ISC Class XII Mathematics

Chapter 9
Differential Equations

143 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

Compiled Answer Key – Step-by-step ISC board-style solutions

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Q. No.	Topic / Statement Summary	Type No.	Kind	Section
9.1 – Order and Degree of Differential Equations				
1	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$\frac{dy}{dx} - \cos x = 0$	1	Numerical	9.1
(ii)	$\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} - \sin^2 y = 0$	1	Numerical	9.1
2	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$y' + y = e^x$	1	Numerical	9.1
(ii)	$\frac{d^2y}{dx^2} = \sin 3x + \cos 3x$	1	Numerical	9.1
3	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$(x^2y - 3x)\,dy + (x^3 - 3y^2)\,dx = 0$	2	Numerical	9.1
(ii)	$\sqrt{1-y^2}\,dx + \sqrt{1-x^2}\,dy = 0$	2	Numerical	9.1

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4	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$\frac{1}{x} \cdot \frac{d^2y}{dx^2} + 5x\frac{dy}{dx} = \sin 2x$	3	Numerical	9.1
(ii)	$\left(\frac{dy}{dx}\right)^4 + 3y\frac{d^2y}{dx^2} = 0$	3	Numerical	9.1
5	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$(y')^2 + y^2 - 1 = 0$	1	Numerical	9.1
(ii)	$y'' + 5x(y')^2 - 6y = \log x$	1	Numerical	9.1
6	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$x^3\left(\frac{d^2y}{dx^2}\right)^2 + x\left(\frac{dy}{dx}\right)^4 = 0$	4	Numerical	9.1
(ii)	$y^{(iv)} + \sin y''' = 0$	4	Numerical	9.1
7	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$\left(\frac{d^2y}{dx^2}\right)^3 + \frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 2x$	4	Numerical	9.1
(ii)	$\left(\frac{d^2y}{dx^2}\right)^3 + 2y\frac{dy}{dx} + \sin y = 5x^{\frac{2}{3}} + \log x$	4	Numerical	9.1
8	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$2x^2\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + y = 0$	4	Numerical	9.1
(ii)	$(y'')^3 + (y')^2 + \sin y' + 1 = 0$	4	Numerical	9.1
9	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$x\frac{d^2y}{dx^2} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^4$	3	Numerical	9.1
(ii)	$\left(\frac{d^4y}{dx^4}\right)^2 = \left(x + \left(\frac{dy}{dx}\right)^2\right)^3$	3	Numerical	9.1
10	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$3x\frac{dy}{dx} + \frac{5}{\frac{dy}{dx}} = y^3$	5	Numerical	9.1

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	$y = x\frac{dy}{dx} + a\sqrt{1+\left(\frac{dy}{dx}\right)^2}$	5	Numerical	9.1
11	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$\sqrt{1+\left(\frac{dy}{dx}\right)^2} = 3x - \frac{dy}{dx}$	5	Numerical	9.1
(ii)	$5\frac{d^2y}{dx^2} = \left(1+\left(\frac{dy}{dx}\right)^2\right)^{\frac{1}{4}}$	5	Numerical	9.1
12	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$y'' + (y')^2 + 2y = 0$	1	Numerical	9.1
(ii)	$y'' + 2y' + \sin y = 0$	1	Numerical	9.1
13	Determine the order and the degree (when defined) of each of the following differential equations:			9.1
(i)	$(y''')^2 + (y'')^3 + (y')^4 + y^5 = 0$	5	Numerical	9.1
(ii)	$(1-(y')^2)^{3/2} = ky''$	5	Numerical	9.1
14	Write the sum of the order and the degree of the following differential equations:			9.1
(i)	$\left(\frac{dy}{dx}\right)^5 + 3xy\left(\frac{d^3y}{dx^3}\right)^2 + y\left(\frac{d^2y}{dx^2}\right)^4 = 0$	6	Numerical	9.1
(ii)	$\left(x+\frac{dy}{dx}\right)^2 = \left(\frac{dy}{dx}\right)^2 + 1$	6	Numerical	9.1
15	Find the product of the order and degree of the following differential equations:			9.1
(i)	$x\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 + y^2 = 0$	7	Numerical	9.1
(ii)	$\left[\frac{d}{dx}(xy^2)\right]\frac{dy}{dx} + y = 0$	7	Numerical	9.1
9.2 – Verifying Solutions of Differential Equations				
1	What is the number of arbitrary constants in the general solution of a differential equation of order 4?	8	Numerical	9.2
2	What is the number of arbitrary constants in a particular solution of a differential equation of order 3?	9	Numerical	9.2
3	Verify that $y = e^{-3x}$ is a solution of the differential equation $\frac{dy}{dx} + 3y = 0$.	10	Proof	9.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
4	In each of the following, show that the given function is a solution of the corresponding differential equation:			9.2
(i)	$y = \cos x + C : \dfrac{dy}{dx} + \sin x = 0$	10	Proof	9.2
(ii)	$y = x^2 + 2x + C : y' - 2x - 2 = 0$	10	Proof	9.2
(iii)	$y = Ae^{-x} : y' + y = 0$	10	Proof	9.2
(iv)	$y = e^x + 1 : y'' - y' = 0$	10	Proof	9.2
5	Show that the function $y = ax + 2a^2$ is a solution of the differential equation $2\left(\dfrac{dy}{dx}\right)^2 + x\dfrac{dy}{dx} - y = 0$.	11	Proof	9.2
6	Verify that $xy = \log y + c$ is a solution of the differential equation $(xy-1)\dfrac{dy}{dx} + y^2 = 0$.	12	Proof	9.2
7	Show that $y = c_1e^x + c_2e^{-x}$ is the general solution of the differential equation $\dfrac{d^2y}{dx^2} - y = 0$.	13	Proof	9.2
8	Show that $y = \sqrt{a^2 - x^2}, x \in (-a, a)$ is a solution of $x + y\dfrac{dy}{dx} = 0$.	14	Proof	9.2
9	Show that $y^2 = 4ax$ is a solution of $y = x\dfrac{dy}{dx} + a\dfrac{dx}{dy}$.	15	Proof	9.2
10	Show that $x + y = \tan^{-1} y$ is a solution of $y^2y' + y^2 + 1 = 0$.	12	Proof	9.2
11	Show that $y = a\cos x + b\sin x$ is a solution of $\dfrac{d^2y}{dx^2} + y = 0$.	16	Proof	9.2
12	Show that $y = x\sin 3x$ is a solution of $\dfrac{d^2y}{dx^2} + 9y - 6\cos 3x = 0$.	17	Proof	9.2
13	Show that $y = e^{-x} + ax + b$ is a solution of the differential equation $e^x\dfrac{d^2y}{dx^2} = 1$.	18	Proof	9.2
14	Show that $y = a\cos(\log x) + b\sin(\log x)$ is a solution of $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.	19	Proof	9.2
15	Show that the differential equation of which $y = 2(x^2 - 1) + ce^{-x^2}$ is a solution is $\dfrac{dy}{dx} + 2xy = 4x^3$.	20	Proof	9.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
16	Show that $y^2 = 4a(x+a)$ is a solution of the differential equation $y(1-y_1^2) = 2xy_1$.	15	Proof	9.2
9.3 – Formation of Differential Equations from a Family of Curves				
1				9.3
(i)	Write the order of the differential equation of the family of circles $x^2 + y^2 = a^2$, where $a(> 0)$ is an arbitrary constant.	21	Numerical	9.3
(ii)	Write the order of the differential equation of the family of parabolas $y^2 = 4ax$, where a is an arbitrary constant.	21	Numerical	9.3
(iii)	Write the order of the differential equation of the family of parabolas $y^2 = 4a(x-b)$, where a, b are arbitrary constants.	21	Numerical	9.3
(iv)	Write the order of the differential equation of the family of ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where a, b are arbitrary constants.	21	Numerical	9.3
2				9.3
(i)	Form the differential equation of the family of straight lines $y = mx$, where m is an arbitrary constant.	22	Numerical	9.3
(ii)	Find the differential equation of the family of straight lines $y = mx + c$, where m, c are arbitrary constants.	22	Numerical	9.3
3	Form the differential equation of the family of concentric circles $x^2 + y^2 = r^2$, where $r(> 0)$ is an arbitrary constant.	23	Numerical	9.3
4	Form the differential equation not containing the arbitrary constant(s) and satisfied by the following equations:			9.3
(i)	$y = ax^3$, a is an arbitrary constant	24	Numerical	9.3
(ii)	$x^2 - y^2 = a^2$, a is an arbitrary constant	24	Numerical	9.3
(iii)	$y = c\sin^{-1} x$, c is an arbitrary constant	24	Numerical	9.3
(iv)	$y^2 = 4ax$, a is an arbitrary constant	24	Numerical	9.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
5	Find the differential equations by eliminating the arbitrary constants and satisfied by the following equations:			9.3
(i)	$y = e^{mx}$, m is an arbitrary constant	25	Numerical	9.3
(ii)	$y = ke^{\tan^{-1} x}$, k is an arbitrary constant	25	Numerical	9.3
(iii)	$y = \tan^{-1} x + ce^{-\tan^{-1} x}$, c is an arbitrary constant	25	Numerical	9.3
6	Find the differential equations by eliminating the arbitrary constants:			9.3
(i)	$x^2 + (y-b)^2 = 1$, b is an arbitrary constant	26	Numerical	9.3
(ii)	$(x-a)^2 - y^2 = 1$, a is an arbitrary constant	26	Numerical	9.3
(iii)	$x^2 + y^2 = ax^3$, a is an arbitrary constant	26	Numerical	9.3
7	Form the differential equation of $\frac{x}{a} + \frac{y}{b} = 1$, where a, b are arbitrary constants.	27	Numerical	9.3
8	Find the differential equation by eliminating arbitrary constants:			9.3
(i)	$y^2 = 4a(x-b)$, a, b are arbitrary constants	28	Numerical	9.3
(ii)	$(y-b)^2 = 4(x-a)$, a, b are arbitrary constants	28	Numerical	9.3
9	Find the differential equations by eliminating a, b or A, B:			9.3
(i)	$y = a\cos(x+b)$, a, b are arbitrary constants	29	Numerical	9.3
(ii)	$y = A\cos 2x + B\sin 2x$, A, B are arbitrary constants	29	Numerical	9.3
10	Find the differential equations by eliminating arbitrary constants:			9.3
(i)	$y = Ae^{2x} + Be^{-2x}$, A, B are arbitrary constants	30	Numerical	9.3
(ii)	$y = ae^{3x} + be^{-2x}$, a, b are arbitrary constants	30	Numerical	9.3
11				9.3
(i)	$y = (\sin^{-1} x)^2 + A\cos^{-1} x + B$, where A, B are arbitrary constants.	31	Numerical	9.3
(ii)	$y = e^x(A\cos x + B\sin x)$, where A, B are arbitrary constants.	31	Numerical	9.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
12	Find the differential equation of the family of curves $(x-h)^2+(y-k)^2=r^2$, where h,k are arbitrary constants and r is a fixed (given) constant.	32	Numerical	9.3
13	Find the differential equation of each of the following family of curves:			9.3
(i)	concentric circles with centre at $(1,2)$ (Exemplar)	33	Numerical	9.3
(ii)	all circles which touch the x-axis at origin	33	Numerical	9.3
(iii)	circles in the first quadrant and touching the coordinate axes	33	Numerical	9.3
(iv)	all parabolas having their vertices at origin and foci on y-axis (NCERT)	33	Numerical	9.3
(v)	all ellipses whose centres are at origin and foci on y-axis (NCERT)	33	Numerical	9.3
(vi)	all hyperbolas having foci on x-axis and centres at the origin (NCERT)	33	Numerical	9.3
(vii)	all non-vertical lines in a plane (Exemplar)	33	Numerical	9.3
14	Form the differential equation of the family of circles having centres on the y-axis and radius 3 units.	34	Numerical	9.3
15	Form the differential equation of simple harmonic motion given by $x=A\cos(nt+\alpha)$, where n is fixed and A,α are parameters.	35	Numerical	9.3
9.4 – Differential Equations with Variables Separable				
1	Solve the following differential equations:			9.4
(i)	$\frac{dy}{dx}=\sin x-x$	36	Numerical	9.4
(ii)	$\frac{dy}{dx}+2x=e^{3x}$	36	Numerical	9.4
(iii)	$(x^2+4)\frac{dy}{dx}=1$	36	Numerical	9.4
(iv)	$\frac{dy}{dx}=y\sin x$	36	Numerical	9.4
(v)	$\frac{dy}{dx}=\csc y$	36	Numerical	9.4
2	Solve the following differential equations:			9.4
(i)	$\frac{dy}{dx}=\sqrt{4-y^2}$	37	Numerical	9.4
(ii)	$\frac{dy}{dx}+y=1$	37	Numerical	9.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
3	Solve the following differential equations:			9.4
(i)	$x^5 \frac{dy}{dx} = -y^5$	38	Numerical	9.4
(ii)	$x(1+y^2)dx + y(1+x^2)dy = 0$	38	Numerical	9.4
4	Solve the following differential equations:			9.4
(i)	$\frac{dy}{dx} = \frac{x+1}{2-y}$	39	Numerical	9.4
(ii)	$e^{y-x} \frac{dy}{dx} = 1$	39	Numerical	9.4
5	Solve the following differential equations:			9.4
(i)	$\frac{dy}{dx} = (e^x + 1)y$	40	Numerical	9.4
(ii)	$\frac{dy}{dx} + \frac{1+y^2}{y} = 0$	40	Numerical	9.4
6	Solve the following differential equations:			9.4
(i)	$\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$	41	Numerical	9.4
(ii)	$\frac{dy}{dx} = 2^{y-x}$	41	Numerical	9.4
7	Solve the following differential equations:			9.4
(i)	$\frac{dy}{dx} = \log x$	42	Numerical	9.4
(ii)	$\sin\left(\frac{dy}{dx}\right) = a$	42	Numerical	9.4
8	Solve the following differential equations:			9.4
(i)	$e^x \frac{dy}{dx} + 1 = x$	43	Numerical	9.4
(ii)	$\frac{1}{x}\frac{dy}{dx} = \tan^{-1} x$	43	Numerical	9.4
9	Solve the following differential equations:			9.4
(i)	$y' = (\cos^2 x - \sin^2 x)\cos^2 y$	44	Numerical	9.4
(ii)	$(xy^2 + x)dx + (x^2y + y)dy = 0$	44	Numerical	9.4
(iii)	$(x^2 - yx^2)dy + (y^2 + xy^2)dx = 0$ (ISC 2017)	44	Numerical	9.4
10	Solve the following differential equations:			9.4
(i)	$(1+x)(1+y^2)dx + (1+y)(1+x^2)dy = 0$	45	Numerical	9.4
(ii)	$x\sqrt{1-y^2}dx + y\sqrt{1-x^2}dy = 0$	45	Numerical	9.4
11	Solve the following differential equations:			9.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	$\cos x \cos y \, dy + \sin x \sin y \, dx = 0$	46	Numerical	9.4
(ii)	$\cos x(1+\cos y)dx - \sin y(1+\sin x)dy = 0$	46	Numerical	9.4
12	Solve the following differential equations:			9.4
(i)	$x(e^{2y}-1)dy + (x^2-1)e^y dx = 0$	47	Numerical	9.4
(ii)	$e^x \tan y \, dx + (1-e^x)\sec^2 y \, dy = 0$	47	Numerical	9.4
13	Solve the following differential equations:			9.4
(i)	$y \, dx - x \, dy = xy \, dx$	48	Numerical	9.4
(ii)	$(1+y^2)\tan^{-1} x \, dx + 2y(1+x^2)dy = 0$ (Exemplar)	48	Numerical	9.4
14	Solve the following differential equations:			9.4
(i)	$(1+x^2)dy = xy \, dx$	49	Numerical	9.4
(ii)	$xyy' = 1 + x + y + xy$	49	Numerical	9.4
15	Solve the following differential equations:			9.4
(i)	$y(1-x^2)\dfrac{dy}{dx} = x(1+y^2)$	50	Numerical	9.4
(ii)	$(y+xy)dx + x(1-y^2)dy = 0$	50	Numerical	9.4
16	Solve the following differential equations:			9.4
(i)	$(x^2 - yx^2)dy + (y^2 + x^2y^2)dx = 0$	51	Numerical	9.4
(ii)	$(e^x+1)y \, dy = (y+1)e^x dx$	51	Numerical	9.4
17	Solve the following differential equations:			9.4
(i)	$\dfrac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$	52	Numerical	9.4
(ii)	$(1+e^{2x})dy + (1+y^2)e^x dx = 0$	52	Numerical	9.4
18	Solve the following differential equations:			9.4
(i)	$\sqrt{a+x}\dfrac{dy}{dx} + xy = 0$	53	Numerical	9.4
(ii)	$(x-1)dy = 2x^3 y \, dx$	53	Numerical	9.4
19	Solve the following differential equations:			9.4
(i)	$x \log x \, dy - y \, dx = 0$	54	Numerical	9.4
(ii)	$\sin^3 x \, dx - \sin y \, dy = 0$	54	Numerical	9.4
20	Solve the following differential equations:			9.4
(i)	$e^{2x-3y}dx + e^{2y-3x}dy = 0$	55	Numerical	9.4
(ii)	$\log\left(\dfrac{dy}{dx}\right) = x - y$	55	Numerical	9.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
21	If $\dfrac{dy}{dx} + \sqrt{\dfrac{1-y^2}{1-x^2}} = 0$, show that $y\sqrt{1-x^2} + x\sqrt{1-y^2} = A$.	56	Proof	9.4
22	Solve the differential equation $\dfrac{dy}{dx} = 1 + x + y^2 + xy^2$, given that $y = 0$ when $x = 0$. (Exemplar)	57	Numerical	9.4
23	Find the particular solution of the differential equation $\dfrac{dy}{dx} = -4xy^2$, given that $y = 1$ when $x = 0$. (NCERT)	58	Numerical	9.4
24	Solve the following differential equations:			9.4
(i)	$2(y+3) - xy\dfrac{dy}{dx} = 0$, given that $y(1) = -2$ (Exemplar)	59	Numerical	9.4
(ii)	$(1+y^2)dx + x\,dy = 0$, given that $y(1) = 1$	59	Numerical	9.4
(iii)	$\dfrac{dy}{dx} = x^2e^{-3y}$, given that $y = 0$ for $x = 0$	59	Numerical	9.4
(iv)	$\dfrac{dy}{dx} = 2y^2$, given that $y(1) = 1$	59	Numerical	9.4
25	Solve the following differential equations:			9.4
(i)	$(1+y^2)(1+\log x)dx + x\,dy = 0$, given that when $x = 1, y = 1$	60	Numerical	9.4
(ii)	$(1+x^2)\dfrac{dy}{dx} + (1+y^2) = 0$, given that $y = 1$ when $x = 0$	60	Numerical	9.4
(iii)	$x(1+y^2)dx - y(1+x^2)dy = 0$, given that $y = 0$ when $x = 1$	60	Numerical	9.4
26	Solve the following differential equations:			9.4
(i)	$y' = y\tan x$, given that $y = 1$ when $x = 0$	61	Numerical	9.4
(ii)	$\dfrac{dy}{dx} = y^2\tan 2x$, given that $y(0) = 2$	61	Numerical	9.4
(iii)	$y' = y\cot 2x$, given that $y\left(\frac{\pi}{4}\right) = 2$	61	Numerical	9.4
27	Find the particular solution of the differential equation $e^x\tan y\,dx + (2-e^x)\sec^2 y\,dy = 0$, given that $y = \frac{\pi}{4}$ when $x = 0$.	62	Numerical	9.4

9.5 – Differential Equations Reducible to Variable Separable Form

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
1	Solve the differential equation: $$(x+y)^2\frac{dy}{dx}=1$$	63	Numerical	9.5
1	Solve the differential equation: $$\frac{dy}{dx}=(4x+y+1)^2$$	64	Numerical	9.5
2	Solve the differential equation: $$(x-y)^2\frac{dy}{dx}=a^2$$	65	Numerical	9.5
2	Solve the differential equation: $$\frac{dy}{dx}=\tan^2(x+y)$$	66	Numerical	9.5
3	Solve the differential equation: $$\frac{dy}{dx}=(3x+y+4)^2$$	64	Numerical	9.5
3	Solve the differential equation: $$\cos(x+y)\,dy=dx$$	67	Numerical	9.5
4	Solve the differential equation: $$\frac{dy}{dx}=\cos(x+y)$$	68	Numerical	9.5
4	Solve the differential equation: $$\cos^2(x-2y)=1-2\frac{dy}{dx}$$	69	Numerical	9.5
5	Solve the differential equation: $$(x+y+1)\frac{dy}{dx}=1$$	70	Numerical	9.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
5	Solve the differential equation: $$\frac{dy}{dx} = \frac{x+y-1}{x+y+1}$$	71	Numerical	9.5
6	Find a particular solution of the differential equation $(x+y+1)^2 dy = dx$, given that $y = 0$ when $x = -1$.	72	Numerical	9.5
9.6 – Homogeneous Differential Equations				
1	Which of the following differential equations are homogeneous?			9.6
(i)	$(x-y)\frac{dy}{dx} = x + 2y$	73	Numerical	9.6
(ii)	$y - x\frac{dy}{dx} = x + y\frac{dy}{dx}$	73	Numerical	9.6
2	Which of the following differential equations are homogeneous? $$(x^2+3xy-4y^2)dx - x(x^2+2y)dy = 0$$	74	Numerical	9.6
3	Which of the following differential equations are homogeneous?			9.6
(i)	$x\frac{dy}{dx} + y = x\tan^{-1}\left(\frac{y}{x}\right)$	73	Numerical	9.6
(ii)	$x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) - 3x^2$	73	Numerical	9.6
4	Which of the following differential equations are homogeneous?			9.6
(i)	$x\frac{dy}{dx} - y = \sqrt{x^2+y^2}$	73	Numerical	9.6
(ii)	$\left(x^2 + y^2\sin\frac{x}{y}\right)\frac{dy}{dx} = 2xye^{x/y}$	73	Numerical	9.6
5	Which of the following differential equations are homogeneous?			9.6
(i)	$2ye^{x/y}dx + (y - 2xe^{x/y})dy = 0$	73	Numerical	9.6
(ii)	$ydx + x\log\left(\frac{y}{x}\right)dy - 2xdy = 0$	73	Numerical	9.6
6	Solve the following differential equations:			9.6
(i)	$\frac{dy}{dx} = \frac{x+y}{x}$	75	Numerical	9.6
(ii)	$\frac{dy}{dx} = \frac{x+y}{x-y}$	75	Numerical	9.6
7	Solve the following differential equations:			9.6
(i)	$(x-y)y' = x+3y$	75	Numerical	9.6

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	$(x+2y)dx-(2x-y)dy=0$	75	Numerical	9.6
8	Solve the following differential equations:			9.6
(i)	$(y^2-2xy)dx=(x^2-2xy)dy$	75	Numerical	9.6
(ii)	$2xydx+(x^2+2y^2)dy=0$	75	Numerical	9.6
9	Solve the following differential equations:			9.6
(i)	$(x^2-y^2)dx+2xydy=0$ (ISC 2023)	75	Numerical	9.6
(ii)	$x^2\frac{dy}{dx}=x^2+xy+y^2$ (Exemplar)	75	Numerical	9.6
(iii)	$\frac{dy}{dx}=\frac{x^2+y^2}{xy}$	75	Numerical	9.6
10	Solve the following differential equations:			9.6
(i)	$x\frac{dy}{dx}+\frac{y^2}{x}=y$	75	Numerical	9.6
(ii)	$3x^2\frac{dy}{dx}-3xy=y^2$	75	Numerical	9.6
11	Solve the following differential equations:			9.6
(i)	$\frac{dy}{dx}=\frac{y}{x}+\frac{\sqrt{x^2-y^2}}{x}, \quad x>0$	75	Numerical	9.6
(ii)	$x\frac{dy}{dx}-y=2\sqrt{y^2-x^2}, \quad x>0$	75	Numerical	9.6
12	Solve the differential equation: $\left(\frac{y}{x}\cos\frac{y}{x}\right)dx-\left(\frac{x}{y}\sin\frac{y}{x}+\cos\frac{y}{x}\right)dy=0$	76	Numerical	9.6
13	Solve the differential equation: $\left(1+e^{y/x}\right)dy+e^{y/x}\left(1-\frac{y}{x}\right)dx=0$	77	Numerical	9.6
14	Solve the following differential equation: $2ye^{x/y}dx+(y-2xe^{x/y})dy=0, \quad$ given $x=0$ and $y=1$ $(ISC 2024)$	78	Numerical	9.6
15	Solve the differential equation $(x+y)dy+(x-y)dx=0$, given that $y=1$ when $x=1$. (NCERT)	79	Numerical	9.6
16	Solve the differential equation $(x^2-y^2)dx+2xydy=0$, given that $y=1$ when $x=1$.	80	Numerical	9.6

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
17	Find the particular solution of the differential equation: $\frac{dy}{dx} = \frac{y^2}{xy - y^2}$, given that $y = 1$ when $x = 1$	81	Numerical (*ISC*2022)	9.6
18	Find particular solutions of the following differential equations:			9.6
(i)	$(x^2 + y^2)dx = 2xydy$, given $y = 0$ when $x = 1$	82	Numerical	9.6
(ii)	$\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$, given $y = 1$ when $x = 0$	82	Numerical	9.6
(iii)	$2xy + y^2 - 2x^2\frac{dy}{dx} = 0$, given $y(1) = 2$	82	Numerical	9.6
(iv)	$(x^2 + 3xy + y^2)dx - x^2dy = 0$, given $y = 0$ when $x = 1$	82	Numerical	9.6
(v)	$\frac{dy}{dx} - \frac{y}{x} + \csc\left(\frac{y}{x}\right) = 0$, given $y = 0$ when $x = 1$	82	Numerical	9.6
(vi)	$x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$, given $y = \frac{\pi}{4}$ at $x = 1$	82	Numerical	9.6
(vii)	$x\frac{dy}{dx} + x\cos^2\left(\frac{y}{x}\right) = y$, given when $x = 1, y = \frac{\pi}{4}$	82	Numerical	9.6
(viii)	$xe^{y/x} - y + x\frac{dy}{dx} = 0$, given $y(e) = 0$	82	Numerical	9.6
19	Find the particular solution of the differential equation $(x - y)\frac{dy}{dx} = x + 2y$, given that $y = 0$, when $x = 1$.	83	Numerical	9.6
20	Solve the differential equation $xdy - ydx = \sqrt{x^2 + y^2}dx$, given that $y = 0$ when $x = 1$.	84	Numerical	9.6
21	Show that the following differential equations are homogeneous and find their particular solutions:			9.6
(i)	$x\frac{dy}{dx}\sin\left(\frac{y}{x}\right) + x - y\sin\left(\frac{y}{x}\right) = 0$, given $x = 1$ when $y = \frac{\pi}{2}$	82	Proof	9.6
(ii)	$\left(xe^{y/x} + y\right)dx = xdy$, given $y = 0$ when $x = 1$	82	Proof	9.6

9.7 – Linear Differential Equations

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
1	Which of the following equations are linear differential equations of first order in y?			9.7
(i)	$\frac{dy}{dx} + 3y = \sin x$	85	Numerical	9.7
(ii)	$x^2 \frac{dy}{dx} - 2xy = x^3 + 2x^2 - 5$	85	Numerical	9.7
(iii)	$x \log x \, dy + (3y - 5 \log x) \, dx = 0$	85	Numerical	9.7
(iv)	$y \frac{dy}{dx} + 5x = y \cos x$	85	Numerical	9.7
2	Which of the following equations are linear differential equations of first order in x?			9.7
(i)	$\frac{dx}{dy} - \frac{2x}{y} = 3y^3 - 5y + 1$	86	Numerical	9.7
(ii)	$(\cos y) \frac{dx}{dy} - 2x \sin y = \cos^3 y$	86	Numerical	9.7
(iii)	$(1 + y^2) \, dx = (\tan^{-1} y - 2x) \, dy$	86	Numerical	9.7
(iv)	$x^2 \frac{dx}{dy} + 2x^3 y = 3y^2 - 2$	86	Numerical	9.7
3	Write an integrating factor of each of the following linear differential equations:			9.7
(i)	$x \frac{dy}{dx} = 2x^2 + y, x > 0$	87	Numerical	9.7
(ii)	$2x \frac{dy}{dx} + y = 6x^3 + 5, x > 0$	87	Numerical	9.7
(iii)	$\frac{dy}{dx} + 2y \tan x = \sin x$	87	Numerical	9.7
(iv)	$x \frac{dy}{dx} + y - xy \cot x = 0$	87	Numerical	9.7
4	Write an integrating factor of each of the following linear differential equations:			9.7
(i)	$y \, dx - (x + 2y^2) \, dy = 0, y > 0$	88	Numerical	9.7
(ii)	$ye^y \, dx = (y^3 + 2xe^y) \, dy, y \neq 0$	88	Numerical	9.7
(iii)	$dx + x \, dy = (e^{-y} \log y) \, dy, y > 0$	88	Numerical	9.7
(iv)	$(1 - y^2) \frac{dx}{dy} + yx = ay, \|y\| < 1$	88	Numerical	9.7
5	Solve the following differential equations:			9.7
(i)	$\frac{dy}{dx} + \frac{y}{x} = x^2$	89	Numerical	9.7
(ii)	$x \frac{dy}{dx} + 2y = x^2$	89	Numerical	9.7
6	Solve the following differential equations:			9.7
(i)	$\frac{dy}{dx} + 2y = 6e^x$	90	Numerical	9.7
(ii)	$\frac{dy}{dx} + 3y = e^{-2x}$	90	Numerical	9.7
7	Solve the following differential equations:			9.7
(i)	$(1 + x^2) \frac{dy}{dx} = 4x^2 - 2xy$	91	Numerical	9.7
(ii)	$\frac{dy}{dx} + \frac{y}{x} = e^{-x}$	91	Numerical	9.7
(iii)	$x \, dy + (y - x^3) \, dx = 0$	91	Numerical	9.7
(iv)	$x \frac{dy}{dx} - y = x^2 e^x$	91	Numerical	9.7
8	Solve the following differential equations:			9.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	$(x^2+1)\frac{dy}{dx}+2xy=\sqrt{x^2+4}$	92	Numerical	9.7
(ii)	$x\frac{dy}{dx}+y=3x\cos 2x, x>0$	92	Numerical	9.7
9	Solve the following differential equations:			9.7
(i)	$x\frac{dy}{dx}-y=(x-1)e^x, x>0$	93	Numerical	9.7
(ii)	$\frac{dx}{dy}+x=\tan y+\sec^2 y$	93	Numerical	9.7
10	Solve the following differential equations:			9.7
(i)	$x\frac{dy}{dx}+2y=x\cos x$	94	Numerical	9.7
(ii)	$(y-\sin^2 x)\,dx+\tan x\,dy=0$	94	Numerical	9.7
(iii)	$\sin x\frac{dy}{dx}-y=\sin x\tan\frac{x}{2}$	94	Numerical	9.7
(iv)	$(y-\sin^2 x)\,dx+(\tan x)\,dy=0$	94	Numerical	9.7
11	Solve the following differential equations:			9.7
(i)	$\frac{dy}{dx}=\frac{x+y\cos x}{1+\sin x}$	95	Numerical	9.7
(ii)	$\frac{dy}{dx}+y\cos x=e^{\sin x}\cos x$	95	Numerical	9.7
12	Solve the following differential equations:			9.7
(i)	$\frac{dy}{dx}-\tan x\cdot y=e^{\sin x}, 0<x<\frac{\pi}{2}$	96	Numerical	9.7
(ii)	$\frac{dy}{dx}+y=\cos x-\sin x$	96	Numerical	9.7
13	Solve the following differential equations:			9.7
(i)	$x\frac{dy}{dx}+y=x\log x$	97	Numerical	9.7
(ii)	$x\frac{dy}{dx}+2y=x^2\log x$	97	Numerical	9.7
14	Solve the following differential equations:			9.7
(i)	$\frac{dy}{dx}+y\cot x=2\cos x$	98	Numerical	9.7
(ii)	$\frac{dy}{dx}-y\tan x=2\sin x$	98	Numerical	9.7
15	Solve the following differential equations:			9.7
(i)	$(\tan x)\frac{dy}{dx}+2y=\sec x$	99	Numerical	9.7
(ii)	$dy=x(x^2-2y)\,dx$	99	Numerical	9.7
16	Solve the following differential equations:			9.7
(i)	$\frac{dy}{dx}-y=\cos x$	100	Numerical	9.7
(ii)	$x\log x\frac{dy}{dx}+y=2\log x, x>1$	100	Numerical	9.7
17	Solve the following differential equations:			9.7
(i)	$y\,dx+(x-y^2)\,dy=0, y>0$	101	Numerical	9.7
(ii)	$(x+3y^2)\,dy=y\,dx$	101	Numerical	9.7
18	Solve the following differential equations:			9.7
(i)	$dx+x\,dy=e^{-y}\log y\,dy, y>0$	102	Numerical	9.7
(ii)	$(x+2y^3)\,dy=y\,dx, y>0$	102	Numerical	9.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
19	Solve the differential equation: $$(1+y^2)\,dx = (\tan^{-1} y - x)\,dy$$	103	Numerical	9.7
20	Solve the differential equation $(1+x^2)\frac{dy}{dx} + 2xy - 4x^2 = 0$ subject to the initial condition $y(0) = 0$.	104	Numerical	9.7
21	Solve the differential equation $\frac{dy}{dx} + y\cot x = 2x + x^2 \cot x$, given that $y = 0$ when $x = \frac{\pi}{2}$.	105	Numerical	9.7
22	If $\frac{dy}{dx} + 2xy = x$, then prove that $2y = 1 + e^{-x^2}$, given that $y = 1$ when $x = 0$.	106	Proof	9.7
23	Solve $(x+2y^2)\frac{dy}{dx} = y$, given that $x = 2$ when $y = 1$.	107	Numerical	9.7
24	Solve the differential equation: $\frac{dy}{dx} + y\sec^2 x = \tan x \sec^2 x; y(0) = 1$.	108	Numerical	9.7
25	Find the particular solutions of the following differential equations:			9.7
(i)	$y' - y = e^x$, given that $y = 1$ when $x = 0$	109	Numerical	9.7
(ii)	$y' + y = e^x$, given that $y = \frac{1}{2}$ when $x = 0$	109	Numerical	9.7
(iii)	$xy' + y = x\log x$, given that $y = \frac{1}{4}$ when $x = 1$	109	Numerical	9.7
25	Solve the differential equation: $$xy' - y = \log x$$ given that $y = 0$ when $x = 1$.	110	Numerical	9.7
25	Solve the differential equation: $$(1+x^2)\frac{dy}{dx} + 2xy = \frac{1}{1+x^2}$$ given that $y = 0$ when $x = 0$.	111	Numerical	9.7
25	Solve the differential equation: $$\frac{dy}{dx} + 2y\tan x = \sin x$$ given that $y = 0$ when $x = \frac{\pi}{3}$.	112	Numerical	9.7
25	Solve the differential equation: $$y' + 2y = e^{-2x}\sin x$$ given that $y = 0$ when $x = 0$.	113	Numerical	9.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
25	Solve the differential equation: $$xy' + y = x\cos x + \sin x$$ given that $y = 1$ when $x = \frac{\pi}{2}$.	114	Numerical	9.7
25	Solve the differential equation: $$x\frac{dy}{dx} + y + \frac{1}{1+x^2} = 0$$ given that $y(1) = 0$.	115	Numerical	9.7
25	Solve the differential equation: $$\frac{dy}{dx} - 2xy = 3x^2e^{x^2}$$ given that $y(0) = 5$.	116	Numerical	9.7
Multiple Choice Question 1	The degree of the differential equation $x^2\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^3$ is: (a) 1; (b) 2; (c) 3; (d) 6	1	Numerical	9.7
Multiple Choice Question 2	The degree of the differential equation $1 + \left(\frac{dy}{dx}\right)^2 = x$ is: (a) 0; (b) 1; (c) 2; (d) 3	1	Numerical	9.7
Multiple Choice Question 3	The degree of the differential equation $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x^2\log\left(\frac{d^2y}{dx^2}\right)$ is: (a) 1; (b) 2; (c) 3; (d) not defined	4	Numerical	9.7
Multiple Choice Question 4	The order and the degree of the differential equation $1 + \left(\frac{dy}{dx}\right)^2 = \frac{d^2y}{dx^2}$ are: (a) 2 and $\frac{3}{2}$; (b) 2 and 3; (c) 3 and 4; (d) 2 and 1	1	Numerical	9.7
Multiple Choice Question 5	The order and the degree of the differential equation $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 3$, respectively, are: (a) 1, 3; (b) 3, 1; (c) 2, 1; (d) 2, 2	1	Numerical	9.7
Multiple Choice Question 6	The order and the degree of the differential equation $\left(3 - \left(\frac{dy}{dx}\right)^2\right)^{5/3} = x^3\frac{d^2y}{dx^2}$, respectively, are: (a) 2, 1; (b) 2, 2; (c) 2, 3; (d) 2, 5	1	Numerical	9.7

Formula & Standard Results Sheet

A. Order and Degree (Ex. 9.1)

Order & Degree Rule

$$\text{Required Value} = \text{Order} + \text{Degree} \quad \text{or} \quad \text{Order} \times \text{Degree}$$

Order = highest-order derivative present; Degree = power of the highest-order derivative, after the equation is made a polynomial in derivatives (only defined when this is possible).

B. Verifying Solutions (Ex. 9.2)

Differentiation of e^{kx}

$$\frac{d}{dx}\left(e^{kx}\right) = k\,e^{kx}$$

Used repeatedly when verifying exponential-form solutions.

Product Rule

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

Needed whenever the proposed solution is a product of two functions of x.

Implicit / Logarithmic Differentiation

$$\frac{d}{dx}(\log y) = \frac{1}{y}\frac{dy}{dx}$$

Used when the given solution relation is implicit in y.

Derivative of $\tan^{-1}$

$$\frac{d}{dx}(\tan^{-1} y) = \frac{1}{1+y^2}\frac{dy}{dx}$$

Used for solutions expressed via inverse trigonometric functions.

Reciprocal Derivative Relation

$$\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx}\right)}$$

Switches between $\frac{dy}{dx}$ and $\frac{dx}{dy}$ when it is easier to differentiate x as a function of y.

C. Forming Differential Equations (Ex. 9.3)

Constant-Elimination Rule

$$\text{No. of arbitrary constants} = \text{order of the resulting differential equation}$$

One arbitrary constant $\implies$ differentiate once; two arbitrary constants $\implies$ differentiate twice, then eliminate both constants between the relations obtained.

D. Variable Separable Method – Standard Integrals (Ex. 9.4)

Logarithmic Form

$$\int \frac{f'(x)}{f(x)}\,dx = \log|f(x)| + C$$

The most frequently used integral in this exercise – recognise the numerator as the derivative of the denominator.

Inverse Tangent Form

$$\int \frac{du}{1+u^2} = \tan^{-1} u + C$$

Standard result for rational integrands with $1 + u^2$ in the denominator.

Inverse Sine Form

$$\int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + C$$

Standard result for radical integrands of this form.

Power Rule (Negative Index)

$$\int u^{-n}\,du = \frac{u^{-n+1}}{-n+1} + C, \quad n \neq 1$$

Used after separating variables into a $y^{-n}dy = \dots dx$ form.

Integration by Parts

$$\int u\,dv = uv - \int v\,du$$

Needed when the separated integral is a product of two different types of functions (e.g. $\log x$).

Tangent / Cotangent Integrals

$$\int \tan x\,dx = \log|\sec x| + C \qquad \int \cot x\,dx = \log|\sin x| + C$$

Standard trigonometric integrals used after separating variables.

E. Reducible to Variable Separable (Ex. 9.5)

Linear Combination Substitution

$$\frac{dy}{dx} = f(ax+by+c) \implies \text{let } v = ax+by+c$$

Reduces an equation that is a function of a linear combination of x, y to a separable one in v and x.

Inverse Tangent Form (in v)

$$\int \frac{dv}{v^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{v}{a}\right) + C$$

Arises after the linear-combination substitution when the resulting expression is a sum of squares.

Log-Ratio Form (in v)

$$\int \frac{dv}{v^2-a^2} = \frac{1}{2a}\log\left|\frac{v-a}{v+a}\right| + C$$

Arises after the linear-combination substitution when the resulting expression is a difference of squares.

F. Homogeneous Differential Equations (Ex. 9.6)

Homogeneity Test

$$f(\lambda x, \lambda y) = \lambda^0 f(x,y) = f(x,y)$$

A function $f(x,y)$ is homogeneous of degree zero – and $\frac{dy}{dx} = f(x,y)$ is a homogeneous differential equation – iff this holds for all λ.

Standard Substitution (y as subject)

$$y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$$

Used when the equation is naturally written as $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$.

Standard Substitution (x as subject)

$$x = vy \implies \frac{dx}{dy} = v + y\frac{dv}{dy}$$

Used instead of the above when $\frac{dx}{dy} = g\left(\frac{x}{y}\right)$ is the more natural homogeneous form (e.g. when the x-terms alone don't simplify).

G. Linear Differential Equations (Ex. 9.7)

Standard Linear Form (in y)

$$\frac{dy}{dx} + P(x)\,y = Q(x)$$

P, Q are functions of x alone (or constants).

Standard Linear Form (in x)

$$\frac{dx}{dy} + P(y)\,x = Q(y)$$

Used when the equation is linear in x as a function of y instead.

Integrating Factor

$$\text{I.F.} = e^{\int P\,dx} \quad (\text{or } e^{\int P(y)\,dy} \text{ for the } x\text{-linear form})$$

Multiplying the linear equation by the I.F. makes the left side an exact derivative $\frac{d}{dx}(y \cdot \text{I.F.})$.

General Solution via I.F.

$$y \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.})\,dx + C$$

Final step after computing the integrating factor – integrate the right side and add the constant of integration.

9.1 – Order and Degree of Differential Equations

Question 1

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $\dfrac{dy}{dx} - \cos x = 0$

(ii) $\left(\dfrac{dy}{dx}\right)^2 + \dfrac{dy}{dx} - \sin^2 y = 0$

TYPE 1 **Order and Degree of Polynomial Differential Equation**

Given: Given differential equations: (i) $\dfrac{dy}{dx} - \cos x = 0$, (ii) $\left(\dfrac{dy}{dx}\right)^2 + \dfrac{dy}{dx} - \sin^2 y = 0$

Solution:

(i) The given differential equation is:

$$\frac{dy}{dx} - \cos x = 0$$

The highest order derivative appearing in this equation is $\dfrac{dy}{dx}$, so:

$$\text{Order} = 1$$

Since the equation is a polynomial in $\dfrac{dy}{dx}$, its degree is the exponent of $\dfrac{dy}{dx}$, which is 1:

$$\text{Degree} = 1$$

(ii) The given differential equation is:

$$\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} - \sin^2 y = 0$$

The highest order derivative present is $\dfrac{dy}{dx}$, so:

$$\text{Order} = 1$$

The equation is expressed as a polynomial in $\dfrac{dy}{dx}$. The highest exponent of $\dfrac{dy}{dx}$ is 2, so:

$$\text{Degree} = 2$$

Final Answer

(i) Order $= 1$, Degree $= 1$

(ii) Order $= 1$, Degree $= 2$

Question 2

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $y' + y = e^x$

(ii) $\dfrac{d^2y}{dx^2} = \sin 3x + \cos 3x$

TYPE 1 **Order and Degree of Polynomial Differential Equation**

Given: Given differential equations: (i) $y' + y = e^x$, (ii) $\dfrac{d^2y}{dx^2} = \sin 3x + \cos 3x$

Solution:

(i) The given differential equation is:

$$y' + y = e^x \quad \text{or} \quad \frac{dy}{dx} + y = e^x$$

The highest order derivative is $y' = \dfrac{dy}{dx}$, so:

$$\text{Order} = 1$$

The exponent of y' is 1, so:

$$\text{Degree} = 1$$

(ii) The given differential equation is:

$$\frac{d^2y}{dx^2} = \sin 3x + \cos 3x$$

The highest order derivative present is $\dfrac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The exponent of $\dfrac{d^2y}{dx^2}$ is 1, so:

$$\text{Degree} = 1$$

Final Answer

(i) Order $= 1$, Degree $= 1$

(ii) Order $= 2$, Degree $= 1$

Question 3

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $(x^2y - 3x)\,dy + (x^3 - 3y^2)\,dx = 0$

(ii) $\sqrt{1-y^2}\,dx + \sqrt{1-x^2}\,dy = 0$

TYPE 2 **Order and Degree of Differential Equation in Differential Form**

Given: Given differential equations in differential form: (i) $(x^2y - 3x)\,dy + (x^3 - 3y^2)\,dx = 0$, (ii) $\sqrt{1-y^2}\,dx + \sqrt{1-x^2}\,dy = 0$

Concept / Formula Used: $\frac{dy}{dx}$

Solution:

(i) Dividing the given differential equation by dx:

$$(x^2y - 3x)\frac{dy}{dx} + (x^3 - 3y^2) = 0$$

The highest order derivative present is $\frac{dy}{dx}$, so:

$$\text{Order} = 1$$

The equation is a polynomial in $\frac{dy}{dx}$ with exponent 1, so:

$$\text{Degree} = 1$$

(ii) Dividing the given equation by dx:

$$\sqrt{1-y^2} + \sqrt{1-x^2}\frac{dy}{dx} = 0$$

The highest order derivative present is $\frac{dy}{dx}$, so:

$$\text{Order} = 1$$

The equation is a polynomial in $\frac{dy}{dx}$ with exponent 1, so:

$$\text{Degree} = 1$$

Final Answer

(i) Order $= 1$, Degree $= 1$

(ii) Order $= 1$, Degree $= 1$

Question 4

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $\frac{1}{x} \cdot \frac{d^2y}{dx^2} + 5x\frac{dy}{dx} = \sin 2x$

(ii) $\left(\frac{dy}{dx}\right)^4 + 3y\frac{d^2y}{dx^2} = 0$

TYPE 3 **Order and Degree Determination for Higher-Order Differential Equations**

Given: Given differential equations: (i) $\frac{1}{x} \cdot \frac{d^2y}{dx^2} + 5x\frac{dy}{dx} = \sin 2x$, (ii) $\left(\frac{dy}{dx}\right)^4 + 3y\frac{d^2y}{dx^2} = 0$

Solution:

(i) Multiplying the equation by x:

$$\frac{d^2y}{dx^2} + 5x^2\frac{dy}{dx} = x \sin 2x$$

The highest order derivative present is $\frac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The power of the highest order derivative $\frac{d^2y}{dx^2}$ is 1, so:

$$\text{Degree} = 1$$

(ii) The given differential equation is:

$$\left(\frac{dy}{dx}\right)^4 + 3y\frac{d^2y}{dx^2} = 0$$

The highest order derivative present is $\frac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The exponent of the highest order derivative $\frac{d^2y}{dx^2}$ is 1 (even though $\frac{dy}{dx}$ has power 4), so:

$$\text{Degree} = 1$$

Final Answer

(i) Order $= 2$, Degree $= 1$

(ii) Order $= 2$, Degree $= 1$

Common Mistake: In subpart (ii), confusing the power 4 on $\frac{dy}{dx}$ as the degree is a common trap. Degree is strictly the exponent of the highest order derivative, which is $\frac{d^2y}{dx^2}$.

Question 5

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $(y')^2 + y^2 - 1 = 0$

(ii) $y'' + 5x(y')^2 - 6y = \log x$

TYPE 1 Order and Degree of Polynomial Differential Equation

Given: Given differential equations: (i) $(y')^2+y^2-1=0$, (ii) $y''+5x(y')^2-6y=\log x$

Solution:

(i) The given equation is:

$$(y')^2 + y^2 - 1 = 0$$

The highest order derivative present is y', so:

$$\text{Order} = 1$$

The exponent of y' is 2, so:

$$\text{Degree} = 2$$

(ii) The given equation is:

$$y'' + 5x(y')^2 - 6y = \log x$$

The highest order derivative present is $y'' = \dfrac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The exponent of y'' is 1, so:

$$\text{Degree} = 1$$

Final Answer

(i) Order $= 1$, Degree $= 2$

(ii) Order $= 2$, Degree $= 1$

Question 6

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $x^3\left(\dfrac{d^2y}{dx^2}\right)^2 + x\left(\dfrac{dy}{dx}\right)^4 = 0$

(ii) $y^{(iv)} + \sin y''' = 0$

TYPE 4 Order and Undefined Degree from Non-Polynomial Derivative Term

Given: Given differential equations: (i) $x^3\left(\frac{d^2y}{dx^2}\right)^2+x\left(\frac{dy}{dx}\right)^4=0$, (ii) $y^{(iv)}+\sin y'''=0$

Concept / Formula Used: $\sin(y'''), e^{y'}, \log(y')$

Solution:

(i) The given equation is:

$$x^3\left(\frac{d^2y}{dx^2}\right)^2+x\left(\frac{dy}{dx}\right)^4=0$$

The highest order derivative present is $\frac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The exponent of $\frac{d^2y}{dx^2}$ is 2, so:

$$\text{Degree} = 2$$

(ii) The given equation is:

$$y^{(iv)}+\sin y'''=0$$

The highest order derivative present is $y^{(iv)}=\frac{d^4y}{dx^4}$, so:

$$\text{Order} = 4$$

Because of the presence of the trigonometric composition $\sin(y''')$, the differential equation cannot be written as a polynomial in derivatives. Therefore:

$$\text{Degree is not defined}$$

Final Answer

(i) Order = 2, Degree = 2

(ii) Order = 4, Degree = Not defined

Common Mistake: A transcendental function of x or y (like $\log x$ or $\sin y$) does not affect the degree, but a transcendental function involving a derivative (like $\sin y'''$) makes the degree undefined.

Question 7

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $\left(\frac{d^2y}{dx^2}\right)^3 + \frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 2x$

(ii) $\left(\frac{d^2y}{dx^2}\right)^3 + 2y\frac{dy}{dx} + \sin y = 5x^{\frac{2}{3}} + \log x$

TYPE 4 Order and Undefined Degree from Non-Polynomial Derivative Term

Given: Given differential equations: (i) $\left(\frac{d^2y}{dx^2}\right)^3 + \frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 2x$, (ii) $\left(\frac{d^2y}{dx^2}\right)^3 + 2y\frac{dy}{dx} + \sin y = 5x^{\frac{2}{3}} + \log x$

Solution:

(i) The given equation is:

$$\left(\frac{d^2y}{dx^2}\right)^3 + \frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 2x$$

The highest order derivative present is $\frac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

Due to the term $\sin\left(\frac{dy}{dx}\right)$, the differential equation cannot be expressed as a polynomial in derivatives. Hence:

$$\text{Degree is not defined}$$

(ii) The given equation is:

$$\left(\frac{d^2y}{dx^2}\right)^3 + 2y\frac{dy}{dx} + \sin y = 5x^{\frac{2}{3}} + \log x$$

The highest order derivative present is $\frac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

Note that $\sin y$, $x^{2/3}$, and $\log x$ do not involve any derivative. The equation is a polynomial in derivatives $\frac{d^2y}{dx^2}$ and $\frac{dy}{dx}$. The exponent of the highest-order derivative $\frac{d^2y}{dx^2}$ is 3, so:

$$\text{Degree} = 3$$

Final Answer

(i) Order = 2, Degree = Not defined

(ii) Order = 2, Degree = 3

Question 8

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $2x^2\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + y = 0$

(ii) $(y'')^3 + (y')^2 + \sin y' + 1 = 0$

TYPE 4 **Order and Undefined Degree from Non-Polynomial Derivative Term**

Given: Given differential equations: (i) $2x^2\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + y = 0$, (ii) $(y'')^3 + (y')^2 + \sin y' + 1 = 0$

Solution:

(i) The given differential equation is:

$$2x^2\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + y = 0$$

The highest order derivative present is $\dfrac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The exponent of $\dfrac{d^2y}{dx^2}$ is 1, so:

$$\text{Degree} = 1$$

(ii) The given differential equation is:

$$(y'')^3 + (y')^2 + \sin y' + 1 = 0$$

The highest order derivative present is y'', so:

$$\text{Order} = 2$$

The equation contains $\sin y'$, so it is not a polynomial in derivatives. Therefore:

$$\text{Degree is not defined}$$

Final Answer

(i) Order $= 2$, Degree $= 1$

(ii) Order $= 2$, Degree $=$ Not defined

Question 9

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $x\dfrac{d^2y}{dx^2} = \left(1 + \left(\dfrac{dy}{dx}\right)^2\right)^4$

(ii) $\left(\dfrac{d^4y}{dx^4}\right)^2 = \left(x + \left(\dfrac{dy}{dx}\right)^2\right)^3$

TYPE 3 Order and Degree Determination for Higher-Order Differential Equations

Given: Given differential equations: (i) $x\dfrac{d^2y}{dx^2} = \left(1 + \left(\dfrac{dy}{dx}\right)^2\right)^4$, (ii) $\left(\dfrac{d^4y}{dx^4}\right)^2 = \left(x + \left(\dfrac{dy}{dx}\right)^2\right)^3$

Solution:

(i) The given differential equation is:

$$x\frac{d^2y}{dx^2} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^4$$

The highest order derivative present is $\dfrac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The equation is free from radicals involving derivatives. The exponent of $\dfrac{d^2y}{dx^2}$ is 1, so:

$$\text{Degree} = 1$$

(ii) The given differential equation is:

$$\left(\frac{d^4y}{dx^4}\right)^2 = \left(x + \left(\frac{dy}{dx}\right)^2\right)^3$$

The highest order derivative present is $\dfrac{d^4y}{dx^4}$, so:

$$\text{Order} = 4$$

The exponent of the highest-order derivative $\dfrac{d^4y}{dx^4}$ is 2, so:

$$\text{Degree} = 2$$

Final Answer

(i) Order $= 2$, Degree $= 1$

(ii) Order $= 4$, Degree $= 2$

Question 10

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $3x\frac{dy}{dx} + \frac{5}{\frac{dy}{dx}} = y^3$

(ii) $y = x\frac{dy}{dx} + a\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

TYPE 5 **Order and Degree after Radical or Rational Derivative Simplification**

Given: Given differential equations: (i) $3x\frac{dy}{dx} + \frac{5}{\frac{dy}{dx}} = y^3$, (ii) $y = x\frac{dy}{dx} + a\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

Solution:

(i) The given equation is:

$$3x\frac{dy}{dx} + \frac{5}{\frac{dy}{dx}} = y^3$$

Multiplying both sides by $\frac{dy}{dx}$ to eliminate the denominator:

$$3x\left(\frac{dy}{dx}\right)^2 + 5 = y^3\frac{dy}{dx}$$

Rearranging in polynomial form:

$$3x\left(\frac{dy}{dx}\right)^2 - y^3\frac{dy}{dx} + 5 = 0$$

The highest order derivative present is $\frac{dy}{dx}$, so:

$$\text{Order} = 1$$

The highest exponent of $\frac{dy}{dx}$ is 2, so:

$$\text{Degree} = 2$$

(ii) The given equation is:

$$y = x\frac{dy}{dx} + a\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Isolating the square root term on one side:

$$y - x\frac{dy}{dx} = a\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Squaring both sides to eliminate the radical:

$$\left(y - x\frac{dy}{dx}\right)^2 = a^2\left(1 + \left(\frac{dy}{dx}\right)^2\right)$$

Expanding the left side:

$$y^2 - 2xy\frac{dy}{dx} + x^2\left(\frac{dy}{dx}\right)^2 = a^2 + a^2\left(\frac{dy}{dx}\right)^2$$

Rearranging terms:

$$(x^2 - a^2)\left(\frac{dy}{dx}\right)^2 - 2xy\frac{dy}{dx} + (y^2 - a^2) = 0$$

The highest order derivative present is $\frac{dy}{dx}$, so:

$$\text{Order} = 1$$

The exponent of $\frac{dy}{dx}$ in polynomial form is 2, so:

$$\text{Degree} = 2$$

Final Answer

(i) Order $= 1$, Degree $= 2$

(ii) Order $= 1$, Degree $= 2$

Common Mistake: Reading the degree of a differential equation before clearing fractions or square roots involving derivatives leads to incorrect answers.

Question 11

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = 3x - \frac{dy}{dx}$

(ii) $5\frac{d^2y}{dx^2} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{\frac{1}{4}}$

TYPE 5 **Order and Degree after Radical or Rational Derivative Simplification**

Given: Given differential equations: (i) $\sqrt{1+\left(\frac{dy}{dx}\right)^2} = 3x - \frac{dy}{dx}$, (ii) $5\frac{d^2y}{dx^2} = \left(1+\left(\frac{dy}{dx}\right)^2\right)^{\frac{1}{4}}$

Solution:

(i) The given differential equation is:

$$\sqrt{1+\left(\frac{dy}{dx}\right)^2} = 3x - \frac{dy}{dx}$$

Squaring both sides to clear the radical:

$$1+\left(\frac{dy}{dx}\right)^2 = \left(3x - \frac{dy}{dx}\right)^2$$

Expanding the right side:

$$1+\left(\frac{dy}{dx}\right)^2 = 9x^2 - 6x\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2$$

Subtracting $\left(\frac{dy}{dx}\right)^2$ from both sides:

$$1 = 9x^2 - 6x\frac{dy}{dx}$$

Rearranging:

$$6x\frac{dy}{dx} = 9x^2 - 1$$

The highest order derivative present is $\frac{dy}{dx}$, so:

$$\text{Order} = 1$$

The exponent of $\frac{dy}{dx}$ is 1, so:

$$\text{Degree} = 1$$

(ii) The given differential equation is:

$$5\frac{d^2y}{dx^2} = \left(1+\left(\frac{dy}{dx}\right)^2\right)^{\frac{1}{4}}$$

Raising both sides to the 4^{th} power:

$$\left(5\frac{d^2y}{dx^2}\right)^4 = 1+\left(\frac{dy}{dx}\right)^2$$

$$625\left(\frac{d^2y}{dx^2}\right)^4 = 1 + \left(\frac{dy}{dx}\right)^2$$

The highest order derivative present is $\frac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The exponent of $\frac{d^2y}{dx^2}$ is 4, so:

$$\text{Degree} = 4$$

Final Answer

(i) Order $= 1$, Degree $= 1$

(ii) Order $= 2$, Degree $= 4$

Question 12

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $y'' + (y')^2 + 2y = 0$

(ii) $y'' + 2y' + \sin y = 0$

TYPE 1 **Order and Degree of Polynomial Differential Equation**

Given: Given differential equations: (i) $y'' + (y')^2 + 2y = 0$, (ii) $y'' + 2y' + \sin y = 0$

Solution:

(i) The given equation is:

$$y'' + (y')^2 + 2y = 0$$

The highest order derivative present is $y'' = \frac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The exponent of y'' is 1, so:

$$\text{Degree} = 1$$

(ii) The given equation is:

$$y'' + 2y' + \sin y = 0$$

The highest order derivative present is $y'' = \frac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

Since $\sin y$ is a function of y and not of a derivative, the equation is a polynomial in derivatives y'' and y'. The exponent of y'' is 1, so:

$$\text{Degree} = 1$$

Final Answer

(i) Order $= 2$, Degree $= 1$

(ii) Order $= 2$, Degree $= 1$

Question 13

Determine the order and the degree (when defined) of each of the following differential equations:

(i) $(y''')^2 + (y'')^3 + (y')^4 + y^5 = 0$

(ii) $(1 - (y')^2)^{3/2} = ky''$

TYPE 5 **Order and Degree after Radical or Rational Derivative Simplification**

Given: Given differential equations: (i) $(y''')^2 + (y'')^3 + (y')^4 + y^5 = 0$, (ii) $(1 - (y')^2)^{3/2} = ky''$

Solution:

(i) The given equation is:

$$(y''')^2 + (y'')^3 + (y')^4 + y^5 = 0$$

The highest order derivative present is $y''' = \dfrac{d^3y}{dx^3}$, so:

$$\text{Order} = 3$$

The exponent of y''' is 2, so:

$$\text{Degree} = 2$$

(ii) The given equation is:

$$\left(1 - (y')^2\right)^{3/2} = ky''$$

Squaring both sides to eliminate the fractional exponent $3/2$:

$$\left[\left(1 - (y')^2\right)^{3/2}\right]^2 = (ky'')^2$$

$$\left(1 - (y')^2\right)^3 = k^2(y'')^2$$

The highest order derivative present is $y'' = \dfrac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The exponent of y'' in this polynomial form is 2, so:

$$\text{Degree} = 2$$

Final Answer

(i) Order $= 3$, Degree $= 2$

(ii) Order $= 2$, Degree $= 2$

Question 14

Write the sum of the order and the degree of the following differential equations:

(i) $\left(\frac{dy}{dx}\right)^5 + 3xy\left(\frac{d^3y}{dx^3}\right)^2 + y\left(\frac{d^2y}{dx^2}\right)^4 = 0$

(ii) $\left(x + \frac{dy}{dx}\right)^2 = \left(\frac{dy}{dx}\right)^2 + 1$

TYPE 6 **Sum of Order and Degree after Algebraic Simplification**

Given: Given equations: (i) $\left(\frac{dy}{dx}\right)^5 + 3xy\left(\frac{d^3y}{dx^3}\right)^2 + y\left(\frac{d^2y}{dx^2}\right)^4 = 0$, (ii) $\left(x + \frac{dy}{dx}\right)^2 = \left(\frac{dy}{dx}\right)^2 + 1$

Concept / Formula Used: Required Value = Order + Degree

Solution:

(i) The given equation is:

$$\left(\frac{dy}{dx}\right)^5 + 3xy\left(\frac{d^3y}{dx^3}\right)^2 + y\left(\frac{d^2y}{dx^2}\right)^4 = 0$$

The highest order derivative present is $\frac{d^3y}{dx^3}$, so:

$$\text{Order} = 3$$

The power of the highest order derivative $\frac{d^3y}{dx^3}$ is 2, so:

$$\text{Degree} = 2$$

Calculating the sum:

$$\text{Sum} = \text{Order} + \text{Degree} = 3 + 2 = 5$$

(ii) The given equation is:

$$\left(x + \frac{dy}{dx}\right)^2 = \left(\frac{dy}{dx}\right)^2 + 1$$

Expanding the left side using $(a+b)^2 = a^2 + 2ab + b^2$:

$$x^2 + 2x\frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2 = \left(\frac{dy}{dx}\right)^2 + 1$$

Subtracting $\left(\frac{dy}{dx}\right)^2$ from both sides:

$$x^2 + 2x\frac{dy}{dx} = 1$$

$$2x\frac{dy}{dx} = 1 - x^2$$

The highest order derivative present is $\frac{dy}{dx}$, so:

$$\text{Order} = 1$$

The exponent of $\frac{dy}{dx}$ in this simplified form is 1, so:

$$\text{Degree} = 1$$

Calculating the sum:

$$\text{Sum} = \text{Order} + \text{Degree} = 1 + 1 = 2$$

Final Answer

(i) Sum $= 5$

(ii) Sum $= 2$

Question 15

Find the product of the order and degree of the following differential equations:

(i) $x\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 + y^2 = 0$

(ii) $\left[\frac{d}{dx}(xy^2)\right]\frac{dy}{dx} + y = 0$

TYPE 7 **Product of Order and Degree with Product Rule Differentiation**

Given: Given equations: (i) $x\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 + y^2 = 0$, (ii) $\left[\frac{d}{dx}(xy^2)\right]\frac{dy}{dx} + y = 0$

Concept / Formula Used: Required Value $=$ Order $\times$ Degree

Solution:

(i) The given equation is:

$$x\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 + y^2 = 0$$

The highest order derivative present is $\frac{d^2y}{dx^2}$, so:

$$\text{Order} = 2$$

The exponent of $\frac{d^2y}{dx^2}$ is 2, so:

$$\text{Degree} = 2$$

Calculating the product:

$$\text{Product} = \text{Order} \times \text{Degree} = 2 \times 2 = 4$$

(ii) The given equation is:

$$\left[\frac{d}{dx}\left(xy^2\right)\right]\frac{dy}{dx}+y=0$$

Differentiating xy^2 with respect to x using the product rule $\frac{d}{dx}(uv)=u\frac{dv}{dx}+v\frac{du}{dx}$:

$$\frac{d}{dx}\left(xy^2\right)=1\cdot y^2+x\cdot 2y\frac{dy}{dx}=y^2+2xy\frac{dy}{dx}$$

Substituting this back into the differential equation:

$$\left(y^2+2xy\frac{dy}{dx}\right)\frac{dy}{dx}+y=0$$

Expanding:

$$y^2\frac{dy}{dx}+2xy\left(\frac{dy}{dx}\right)^2+y=0$$

The highest order derivative present is $\frac{dy}{dx}$, so:

$$\text{Order}=1$$

The highest exponent of $\frac{dy}{dx}$ is 2, so:

$$\text{Degree}=2$$

Calculating the product:

$$\text{Product}=\text{Order}\times\text{Degree}=1\times 2=2$$

Final Answer

(i) Product $=4$

(ii) Product $=2$

9.2 – Verifying Solutions of Differential Equations

Question 1

What is the number of arbitrary constants in the general solution of a differential equation of order 4?

TYPE 8 Number of Arbitrary Constants in General Solution

To Find: The number of arbitrary constants in the general solution of a 4^{th} order differential equation.

Concept / Formula Used: n^{th} n

Solution:

Given the order of the differential equation:

$$n = 4$$

By definition, the general solution of a differential equation of order n contains exactly n independent arbitrary constants.

Therefore, for $n = 4$, the number of arbitrary constants present in the general solution is 4.

Final Answer

The number of arbitrary constants in the general solution is 4.

Question 2

What is the number of arbitrary constants in a particular solution of a differential equation of order 3?

TYPE 9 **Number of Arbitrary Constants in Particular Solution**

To Find: The number of arbitrary constants in a particular solution of a 3^{rd} order differential equation.

Solution:

Given the order of the differential equation:

$$n = 3$$

A particular solution is derived from the general solution by substituting specific values for all the arbitrary constants based on given initial or boundary conditions.

Consequently, no arbitrary constants remain in a particular solution.

Final Answer

The number of arbitrary constants in a particular solution is 0.

Question 3

Verify that $y = e^{-3x}$ is a solution of the differential equation $\dfrac{dy}{dx} + 3y = 0$.

TYPE 10 **Verification of Explicit Solution of First-Order Differential Equation**

To Prove: Show that $y = e^{-3x}$ satisfies $\dfrac{dy}{dx} + 3y = 0$.

Concept / Formula Used: $\dfrac{d}{dx}\left(e^{kx}\right) = ke^{kx}$

Solution:

Given function:

$$y = e^{-3x}$$

Differentiating both sides with respect to x:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left(e^{-3x}\right)\\ &= -3e^{-3x}\end{aligned}$$

Substituting y and $\dfrac{dy}{dx}$ into the Left Hand Side (LHS) of the differential equation:

$$\begin{aligned}\text{LHS} &= \frac{dy}{dx} + 3y\\ &= \left(-3e^{-3x}\right) + 3\left(e^{-3x}\right)\\ &= -3e^{-3x} + 3e^{-3x}\\ &= 0\\ &= \text{RHS}\end{aligned}$$

Since LHS = RHS, $y = e^{-3x}$ is a solution of the differential equation.

Hence Proved

LHS = RHS, hence $y = e^{-3x}$ is a solution of $\dfrac{dy}{dx} + 3y = 0$.

Question 4

In each of the following, show that the given function is a solution of the corresponding differential equation:

(i) $y = \cos x + C : \dfrac{dy}{dx} + \sin x = 0$

(ii) $y = x^2 + 2x + C : y' - 2x - 2 = 0$

(iii) $y = Ae^{-x} : y' + y = 0$

(iv) $y = e^x + 1 : y'' - y' = 0$

TYPE 10 **Verification of Explicit Solution of First-Order Differential Equation**

To Prove: Verify that each given function satisfies its corresponding differential equation.

Concept / Formula Used: $\dfrac{d}{dx}(\cos x) = -\sin x \qquad \dfrac{d}{dx}(x^n) = nx^{n-1} \qquad \dfrac{d}{dx}(e^{kx}) = ke^{kx}$

Solution:

(i) Given function: $y = \cos x + C$.
Differentiating with respect to x:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(\cos x + C)\\ &= -\sin x + 0\\ &= -\sin x\end{aligned}$$

Substitute $\frac{dy}{dx}$ into LHS of the differential equation:

$$\begin{aligned}\text{LHS} &= \frac{dy}{dx} + \sin x\\ &= (-\sin x) + \sin x\\ &= 0 = \text{RHS}\end{aligned}$$

(ii) Given function: $y = x^2 + 2x + C$.
Differentiating with respect to x:

$$\begin{aligned}y' = \frac{dy}{dx} &= \frac{d}{dx}(x^2 + 2x + C)\\ &= 2x + 2\end{aligned}$$

Substitute y' into LHS:

$$\begin{aligned}\text{LHS} &= y' - 2x - 2\\ &= (2x + 2) - 2x - 2\\ &= 0 = \text{RHS}\end{aligned}$$

(iii) Given function: $y = Ae^{-x}$.
Differentiating with respect to x:

$$\begin{aligned}y' = \frac{dy}{dx} &= \frac{d}{dx}\left(Ae^{-x}\right)\\ &= -Ae^{-x}\end{aligned}$$

Substitute y' and y into LHS:

$$\begin{aligned}\text{LHS} &= y' + y\\ &= \left(-Ae^{-x}\right) + \left(Ae^{-x}\right)\\ &= 0 = \text{RHS}\end{aligned}$$

(iv) Given function: $y = e^x + 1$.
First derivative:

$$\begin{aligned}y' &= \frac{d}{dx}(e^x + 1)\\ &= e^x\end{aligned}$$

Second derivative:

$$\begin{aligned} y'' &= \frac{d}{dx}(e^x) \\ &= e^x \end{aligned}$$

Substitute y'' and y' into LHS:

$$\begin{aligned} \text{LHS} &= y'' - y' \\ &= e^x - e^x \\ &= 0 = \text{RHS} \end{aligned}$$

Hence Proved

All given functions satisfy their respective differential equations.

Question 5

Show that the function $y = ax+2a^2$ is a solution of the differential equation $2\left(\frac{dy}{dx}\right)^2 + x\frac{dy}{dx} - y = 0$.

TYPE 11 **Verification of Solution for Non-Linear First-Order Differential Equation**

To Prove: Show that $y = ax + 2a^2$ satisfies $2\left(\frac{dy}{dx}\right)^2 + x\frac{dy}{dx} - y = 0$.

Concept / Formula Used: $\frac{d}{dx}(ax + k) = a$

Solution:

Given function:

$$y = ax + 2a^2$$

Differentiating with respect to x:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(ax + 2a^2\right) \\ &= a + 0 \\ &= a \end{aligned}$$

Substituting $\frac{dy}{dx} = a$ and $y = ax + 2a^2$ into LHS of the differential equation:

$$\begin{aligned} \text{LHS} &= 2\left(\frac{dy}{dx}\right)^2 + x\frac{dy}{dx} - y \\ &= 2(a)^2 + x(a) - (ax + 2a^2) \\ &= 2a^2 + ax - ax - 2a^2 \\ &= 0 \\ &= \text{RHS} \end{aligned}$$

Since LHS = RHS, $y = ax + 2a^2$ is a solution.

Hence Proved

LHS = RHS, hence $y = ax + 2a^2$ is a solution of $2\left(\frac{dy}{dx}\right)^2 + x\frac{dy}{dx} - y = 0$.

Question 6

Verify that $xy = \log y + c$ is a solution of the differential equation $(xy-1)\frac{dy}{dx} + y^2 = 0$.

TYPE 12 Verification of Implicit Solution via Implicit Differentiation

★ Likely Exam Question

To Prove: Show that $xy = \log y + c$ satisfies $(xy-1)\frac{dy}{dx} + y^2 = 0$.

Concept / Formula Used: $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$ $\quad \frac{d}{dx}(\log y) = \frac{1}{y}\frac{dy}{dx}$

Solution:

Given implicit equation:

$$xy = \log y + c$$

Differentiating both sides with respect to x:

$$\begin{aligned} \frac{d}{dx}(xy) &= \frac{d}{dx}(\log y + c) \\ x\frac{dy}{dx} + y\frac{d}{dx}(x) &= \frac{1}{y}\frac{dy}{dx} + 0 \\ x\frac{dy}{dx} + y &= \frac{1}{y}\frac{dy}{dx} \end{aligned}$$

Multiplying the entire equation by y:

$$\begin{aligned} y\left(x\frac{dy}{dx} + y\right) &= y\left(\frac{1}{y}\frac{dy}{dx}\right) \\ xy\frac{dy}{dx} + y^2 &= \frac{dy}{dx} \end{aligned}$$

Rearranging terms to collect $\frac{dy}{dx}$ on one side:

$$\begin{aligned} xy\frac{dy}{dx} - \frac{dy}{dx} + y^2 &= 0 \\ (xy-1)\frac{dy}{dx} + y^2 &= 0 \end{aligned}$$

This is precisely the given differential equation.

Hence Proved

Implicit differentiation yields $(xy-1)\dfrac{dy}{dx}+y^2=0$, hence verified.

Question 7

Show that $y=c_1e^x+c_2e^{-x}$ is the general solution of the differential equation $\dfrac{d^2y}{dx^2}-y=0$.

TYPE 13 Verification of General Solution of Second-Order Differential Equation

To Prove: Show that $y=c_1e^x+c_2e^{-x}$ satisfies $\dfrac{d^2y}{dx^2}-y=0$.

Concept / Formula Used: $\dfrac{d}{dx}(e^x)=e^x \qquad \dfrac{d}{dx}(e^{-x})=-e^{-x}$

Solution:

Given function:

$$y=c_1e^x+c_2e^{-x}$$

Differentiating first time with respect to x:

$$\begin{aligned}\frac{dy}{dx}&=\frac{d}{dx}\left(c_1e^x+c_2e^{-x}\right)\\&=c_1e^x-c_2e^{-x}\end{aligned}$$

Differentiating second time with respect to x:

$$\begin{aligned}\frac{d^2y}{dx^2}&=\frac{d}{dx}\left(c_1e^x-c_2e^{-x}\right)\\&=c_1e^x-c_2(-e^{-x})\\&=c_1e^x+c_2e^{-x}\end{aligned}$$

Substitute $\dfrac{d^2y}{dx^2}$ and y into LHS:

$$\begin{aligned}\text{LHS}&=\frac{d^2y}{dx^2}-y\\&=\left(c_1e^x+c_2e^{-x}\right)-\left(c_1e^x+c_2e^{-x}\right)\\&=0\\&=\text{RHS}\end{aligned}$$

Since the solution satisfies the 2^{nd} order differential equation and contains 2 arbitrary constants (c_1 and c_2), it is the general solution.

Hence Proved

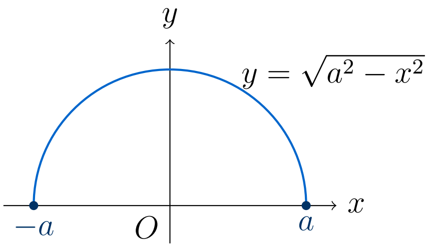
LHS = RHS, hence $y = c_1e^x + c_2e^{-x}$ is the general solution of $\frac{d^2y}{dx^2} - y = 0$.

Question 8

Show that $y = \sqrt{a^2 - x^2}, x \in (-a, a)$ is a solution of $x + y\frac{dy}{dx} = 0$.

TYPE 14 **Verification of Radical Function Solution of First-Order Differential Equation**

★ Likely Exam Question



Semicircle $y = \sqrt{a^2 - x^2}$ *defined on* $x \in (-a, a)$

To Prove: Verify that $y = \sqrt{a^2 - x^2}$ satisfies $x + y\frac{dy}{dx} = 0$.

Concept / Formula Used: $\frac{d}{dx}\sqrt{u} = \frac{1}{2\sqrt{u}}\frac{du}{dx}$

Solution:

Given function:

$$y = \sqrt{a^2 - x^2}$$

Squaring both sides:

$$y^2 = a^2 - x^2$$
$$x^2 + y^2 = a^2$$

Differentiating both sides with respect to x:

$$\frac{d}{dx}\left(x^2 + y^2\right) = \frac{d}{dx}\left(a^2\right)$$
$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0$$
$$2x + 2y\frac{dy}{dx} = 0$$

Dividing both sides by 2:

$$x + y\frac{dy}{dx} = 0$$

Hence Proved

$y = \sqrt{a^2 - x^2}$ satisfies $x + y\frac{dy}{dx} = 0$.

Question 9

Show that $y^2 = 4ax$ is a solution of $y = x\frac{dy}{dx} + a\frac{dx}{dy}$.

TYPE 15 **Verification of Solution by Parameter Elimination in Differential Equation**

To Prove: Verify that $y^2 = 4ax$ satisfies $y = x\frac{dy}{dx} + a\frac{dx}{dy}$.

Concept / Formula Used: $\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx}\right)}$

Solution:

Given relation:

$$y^2 = 4ax$$

Differentiating both sides with respect to x:

$$\begin{aligned}
\frac{d}{dx}\left(y^2\right) &= \frac{d}{dx}(4ax) \\
2y\frac{dy}{dx} &= 4a \\
\frac{dy}{dx} &= \frac{4a}{2y} \\
\frac{dy}{dx} &= \frac{2a}{y}
\end{aligned}$$

From this expression, isolate parameter a:

$$a = \frac{y}{2}\frac{dy}{dx}$$

Taking the reciprocal to find $\frac{dx}{dy}$:

$$\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx}\right)} = \frac{y}{2a}$$

Substitute $x = \frac{y^2}{4a}$ and $\frac{dx}{dy} = \frac{y}{2a}$ into RHS of the differential equation:

$$\begin{aligned}\text{RHS} &= x\frac{dy}{dx} + a\frac{dx}{dy} \\ &= \left(\frac{y^2}{4a}\right)\left(\frac{2a}{y}\right) + a\left(\frac{y}{2a}\right) \\ &= \frac{2ay^2}{4ay} + \frac{ay}{2a} \\ &= \frac{y}{2} + \frac{y}{2} \\ &= y \\ &= \text{LHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence $y^2 = 4ax$ satisfies $y = x\frac{dy}{dx} + a\frac{dx}{dy}$.

Question 10

Show that $x + y = \tan^{-1} y$ is a solution of $y^2y' + y^2 + 1 = 0$.

TYPE 12 Verification of Implicit Solution via Implicit Differentiation

★ Likely Exam Question

To Prove: Show that $x + y = \tan^{-1} y$ satisfies $y^2y' + y^2 + 1 = 0$.

Concept / Formula Used: $\frac{d}{dx}(\tan^{-1} y) = \frac{1}{1+y^2}\frac{dy}{dx}$

Solution:

Given implicit equation:

$$x + y = \tan^{-1} y$$

Differentiating both sides with respect to x (letting $y' = \frac{dy}{dx}$):

$$\begin{aligned}\frac{d}{dx}(x+y) &= \frac{d}{dx}\left(\tan^{-1} y\right) \\ 1 + y' &= \frac{1}{1+y^2} \cdot y'\end{aligned}$$

Multiplying both sides by $(1 + y^2)$:

$$\begin{aligned}(1+y')(1+y^2) &= y' \\ 1(1+y^2) + y'(1+y^2) &= y' \\ 1 + y^2 + y' + y^2y' &= y'\end{aligned}$$

Subtracting y' from both sides:

$$1 + y^2 + y^2y' = 0$$
$$y^2y' + y^2 + 1 = 0$$

Hence Proved

$x + y = \tan^{-1} y$ satisfies $y^2y' + y^2 + 1 = 0$.

Question 11

Show that $y = a\cos x + b\sin x$ is a solution of $\dfrac{d^2y}{dx^2} + y = 0$.

TYPE 16 **Verification of Trigonometric Solution of Second-Order Differential Equation**

To Prove: Show that $y = a\cos x + b\sin x$ satisfies $\dfrac{d^2y}{dx^2} + y = 0$.

Concept / Formula Used: $\dfrac{d}{dx}(\cos x) = -\sin x \qquad \dfrac{d}{dx}(\sin x) = \cos x$

Solution:

Given function:

$$y = a\cos x + b\sin x$$

Differentiating with respect to x:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(a\cos x + b\sin x)\\ &= -a\sin x + b\cos x\end{aligned}$$

Differentiating a second time with respect to x:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx}(-a\sin x + b\cos x)\\ &= -a\cos x - b\sin x\\ &= -(a\cos x + b\sin x)\\ &= -y\end{aligned}$$

Adding y to both sides:

$$\frac{d^2y}{dx^2} + y = 0$$

Hence Proved

$y = a\cos x + b\sin x$ satisfies $\dfrac{d^2y}{dx^2} + y = 0$.

Question 12

Show that $y = x\sin 3x$ is a solution of $\dfrac{d^2y}{dx^2} + 9y - 6\cos 3x = 0$.

TYPE 17 **Verification of Product-Term Solution of Second-Order Differential Equation**

★ Likely Exam Question

To Prove: Verify that $y = x\sin 3x$ satisfies $\dfrac{d^2y}{dx^2} + 9y - 6\cos 3x = 0$.

Concept / Formula Used: $\dfrac{d}{dx}(uv) = u\dfrac{dv}{dx} + v\dfrac{du}{dx}$

Solution:

Given function:

$$y = x\sin 3x$$

First derivative with respect to x:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x\sin 3x)\\ &= x\frac{d}{dx}(\sin 3x) + \sin 3x\frac{d}{dx}(x)\\ &= x(3\cos 3x) + \sin 3x(1)\\ &= 3x\cos 3x + \sin 3x\end{aligned}$$

Second derivative with respect to x:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx}(3x\cos 3x + \sin 3x)\\ &= \frac{d}{dx}(3x\cos 3x) + \frac{d}{dx}(\sin 3x)\\ &= [3x(-3\sin 3x) + 3\cos 3x(1)] + 3\cos 3x\\ &= -9x\sin 3x + 3\cos 3x + 3\cos 3x\\ &= -9x\sin 3x + 6\cos 3x\end{aligned}$$

Substitute $x\sin 3x = y$:

$$\frac{d^2y}{dx^2} = -9y + 6\cos 3x$$

Rearranging all terms to LHS:

$$\frac{d^2y}{dx^2} + 9y - 6\cos 3x = 0$$

Hence Proved

$y = x\sin 3x$ satisfies $\dfrac{d^2y}{dx^2} + 9y - 6\cos 3x = 0$.

Question 13

Show that $y = e^{-x} + ax + b$ is a solution of the differential equation $e^x \dfrac{d^2y}{dx^2} = 1$.

TYPE 18 Verification of Second-Order Differential Equation Solution with Exponential Term

To Prove: Show that $y = e^{-x} + ax + b$ satisfies $e^x \dfrac{d^2y}{dx^2} = 1$.

Concept / Formula Used: $\dfrac{d}{dx}(e^{-x}) = -e^{-x} \qquad \dfrac{d}{dx}(ax+b) = a$

Solution:

Given function:

$$y = e^{-x} + ax + b$$

First derivative with respect to x:

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}\left(e^{-x} + ax + b\right) \\ &= -e^{-x} + a\end{aligned}$$

Second derivative with respect to x:

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx}\left(-e^{-x} + a\right) \\ &= -(-e^{-x}) + 0 \\ &= e^{-x}\end{aligned}$$

Substitute $\dfrac{d^2y}{dx^2} = e^{-x}$ into LHS of the differential equation:

$$\begin{aligned}\text{LHS} &= e^x \frac{d^2y}{dx^2} \\ &= e^x \cdot e^{-x} \\ &= e^{x-x} \\ &= e^0 \\ &= 1 \\ &= \text{RHS}\end{aligned}$$

Hence Proved

LHS = RHS, hence $y = e^{-x} + ax + b$ satisfies $e^x \dfrac{d^2y}{dx^2} = 1$.

Question 14

Show that $y = a\cos(\log x) + b\sin(\log x)$ is a solution of $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

TYPE 19 **Verification of Logarithmic-Trigonometric Solution of Second-Order Differential Equation**

★ **Likely Exam Question**

To Prove: Show that $y = a\cos(\log x) + b\sin(\log x)$ satisfies $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

Concept / Formula Used: $\dfrac{d}{dx}[\cos(\log x)] = -\dfrac{\sin(\log x)}{x}$ $\dfrac{d}{dx}[\sin(\log x)] = \dfrac{\cos(\log x)}{x}$

Solution:

Given function:

$$y = a\cos(\log x) + b\sin(\log x)$$

Differentiating with respect to x:

$$\frac{dy}{dx} = a\left(-\sin(\log x)\cdot\frac{1}{x}\right) + b\left(\cos(\log x)\cdot\frac{1}{x}\right)$$
$$\frac{dy}{dx} = \frac{-a\sin(\log x) + b\cos(\log x)}{x}$$

Multiplying both sides by x:

$$x\frac{dy}{dx} = -a\sin(\log x) + b\cos(\log x)$$

Differentiating again with respect to x using Product Rule on LHS:

$$\frac{d}{dx}\left(x\frac{dy}{dx}\right) = \frac{d}{dx}\left[-a\sin(\log x) + b\cos(\log x)\right]$$
$$x\frac{d^2y}{dx^2} + \frac{dy}{dx}\cdot 1 = -a\cos(\log x)\cdot\frac{1}{x} - b\sin(\log x)\cdot\frac{1}{x}$$
$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} = -\frac{1}{x}\left[a\cos(\log x) + b\sin(\log x)\right]$$

Substituting $y = a\cos(\log x) + b\sin(\log x)$:

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} = -\frac{y}{x}$$

Multiplying the entire equation by x:

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = -y$$
$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$$

Hence Proved

$y = a\cos(\log x) + b\sin(\log x)$ satisfies $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + y = 0$.

Question 15

Show that the differential equation of which $y = 2(x^2 - 1) + ce^{-x^2}$ is a solution is $\dfrac{dy}{dx} + 2xy = 4x^3$.

TYPE 20 **Verification of Solution of First-Order Linear Differential Equation**

To Prove: Show that $y = 2(x^2 - 1) + ce^{-x^2}$ satisfies $\dfrac{dy}{dx} + 2xy = 4x^3$.

Concept / Formula Used: $\dfrac{d}{dx}\left(e^{-x^2}\right) = -2xe^{-x^2}$

Solution:

Given function:

$$\begin{aligned} y &= 2(x^2 - 1) + ce^{-x^2} \\ y &= 2x^2 - 2 + ce^{-x^2} \end{aligned}$$

Differentiating with respect to x:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(2x^2 - 2 + ce^{-x^2}\right) \\ &= 4x - 0 + c\left(e^{-x^2} \cdot (-2x)\right) \\ &= 4x - 2cxe^{-x^2} \end{aligned}$$

Now, evaluate the term $2xy$:

$$\begin{aligned} 2xy &= 2x\left[2(x^2 - 1) + ce^{-x^2}\right] \\ &= 2x(2x^2 - 2 + ce^{-x^2}) \\ &= 4x^3 - 4x + 2cxe^{-x^2} \end{aligned}$$

Substitute $\dfrac{dy}{dx}$ and $2xy$ into LHS of the target differential equation:

$$\begin{aligned} \text{LHS} &= \frac{dy}{dx} + 2xy \\ &= \left(4x - 2cxe^{-x^2}\right) + \left(4x^3 - 4x + 2cxe^{-x^2}\right) \\ &= 4x - 4x - 2cxe^{-x^2} + 2cxe^{-x^2} + 4x^3 \\ &= 4x^3 \\ &= \text{RHS} \end{aligned}$$

Hence Proved

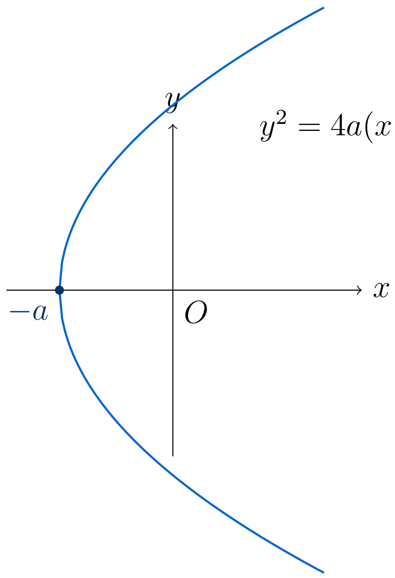
LHS = RHS, hence $y = 2(x^2 - 1) + ce^{-x^2}$ satisfies $\dfrac{dy}{dx} + 2xy = 4x^3$.

Question 16

Show that $y^2 = 4a(x + a)$ is a solution of the differential equation $y\left(1 - y_1^2\right) = 2xy_1$.

TYPE 15 **Verification of Solution by Parameter Elimination in Differential Equation**

★ Likely Exam Question



Parabola $y^2 = 4a(x + a)$ with vertex at $(-a, 0)$

To Prove: Show that $y^2 = 4a(x + a)$ satisfies $y\left(1 - y_1^2\right) = 2xy_1$, where $y_1 = \dfrac{dy}{dx}$.

Concept / Formula Used: a

Solution:

Given equation of parabola:

$$y^2 = 4a(x + a)$$
$$y^2 = 4ax + 4a^2$$

Differentiating both sides with respect to x (writing $y_1 = \frac{dy}{dx}$):

$$\frac{d}{dx}\left(y^2\right) = \frac{d}{dx}\left(4ax + 4a^2\right)$$
$$2yy_1 = 4a$$
$$a = \frac{2yy_1}{4}$$
$$a = \frac{yy_1}{2}$$

Substitute $a = \frac{yy_1}{2}$ back into the original parabola equation $y^2 = 4ax + 4a^2$:

$$y^2 = 4\left(\frac{yy_1}{2}\right)x + 4\left(\frac{yy_1}{2}\right)^2$$
$$y^2 = 2xyy_1 + 4\left(\frac{y^2y_1^2}{4}\right)$$
$$y^2 = 2xyy_1 + y^2y_1^2$$

Dividing both sides by y (assuming $y \neq 0$):

$$y = 2xy_1 + yy_1^2$$

Rearranging terms by subtracting yy_1^2 from both sides:

$$y - yy_1^2 = 2xy_1$$
$$y\left(1 - y_1^2\right) = 2xy_1$$

Hence Proved

$y^2 = 4a(x + a)$ satisfies $y\left(1 - y_1^2\right) = 2xy_1$.

9.3 – Formation of Differential Equations from a Family of Curves

Question 1

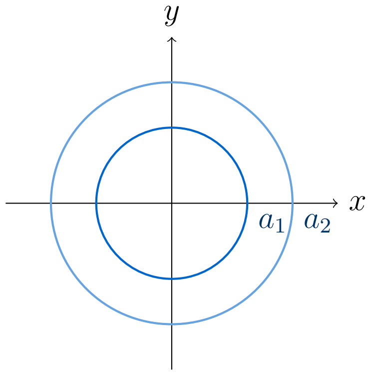
(i) Write the order of the differential equation of the family of circles $x^2 + y^2 = a^2$, where $a(> 0)$ is an arbitrary constant.

(ii) Write the order of the differential equation of the family of parabolas $y^2 = 4ax$, where a is an arbitrary constant.

(iii) Write the order of the differential equation of the family of parabolas $y^2 = 4a(x - b)$, where a, b are arbitrary constants.

(iv) Write the order of the differential equation of the family of ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where a, b are arbitrary constants.

TYPE 21 Order Determination from Number of Arbitrary Constants



Family of geometric curves with varying parameters

Given: Given families of curves with arbitrary constants: (i) $x^2+y^2 = a^2$ (1 constant), (ii) $y^2 = 4ax$ (1 constant), (iii) $y^2 = 4a(x-b)$ (2 constants), (iv) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ (2 constants).

Solution:

(i) The equation $x^2 + y^2 = a^2$ contains 1 essential arbitrary constant (a). Hence, the order of the corresponding differential equation is 1.

(ii) The equation $y^2 = 4ax$ contains 1 essential arbitrary constant (a). Hence, the order of the corresponding differential equation is 1.

(iii) The equation $y^2 = 4a(x-b)$ contains 2 essential arbitrary constants (a and b). Hence, the order of the corresponding differential equation is 2.

(iv) The equation $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ contains 2 essential arbitrary constants (a and b). Hence, the order of the corresponding differential equation is 2.

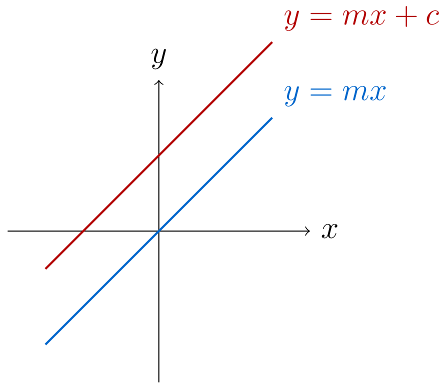
Final Answer

(i) Order $= 1$
(ii) Order $= 1$
(iii) Order $= 2$
(iv) Order $= 2$

Question 2

(i) Form the differential equation of the family of straight lines $y = mx$, where m is an arbitrary constant.
(ii) Find the differential equation of the family of straight lines $y = mx + c$, where m, c are arbitrary constants.

TYPE 22 **Differential Equation Formation for Straight Line Families**



Family of straight lines

Given: Equations of straight lines: (i) $y = mx$ with 1 arbitrary constant m, (ii) $y = mx + c$ with 2 arbitrary constants m, c.

Concept / Formula Used: x

Solution:

(i) Given equation:

$$y = mx \tag{1}$$

Differentiating equation (1) with respect to x:

$$\frac{dy}{dx} = m$$

Substituting the value of m into equation (1):

$$y = \left(\frac{dy}{dx}\right) x$$

$$x\frac{dy}{dx} = y$$

(ii) Given equation:

$$y = mx + c \tag{2}$$

Differentiating equation (2) with respect to x:

$$\frac{dy}{dx} = m$$

Differentiating once more with respect to x to eliminate m:

$$\frac{d^2y}{dx^2} = 0$$

Final Answer

(i) $x\dfrac{dy}{dx} = y$

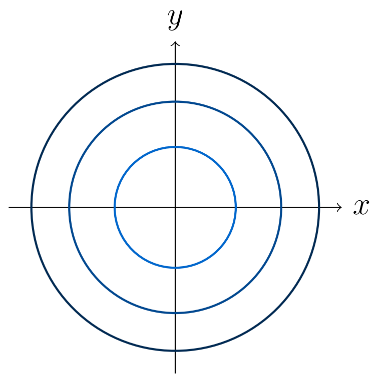
(ii) $\dfrac{d^2y}{dx^2} = 0$

Question 3

Form the differential equation of the family of concentric circles $x^2 + y^2 = r^2$, where $r(> 0)$ is an arbitrary constant.

TYPE 23 Differential Equation Formation for Concentric Circles at Origin

★ Likely Exam Question



Family of concentric circles centered at origin

Given: Equation of concentric circles: $x^2 + y^2 = r^2$, where r is an arbitrary constant.

Concept / Formula Used: $x^2 + y^2 = r^2$ x r

Solution:

Given equation of the family of circles:

$$x^2 + y^2 = r^2$$

Since there is only one arbitrary constant (r), we differentiate once with respect to x:

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(r^2)$$
$$2x + 2y\frac{dy}{dx} = 0$$

Dividing both sides by 2:

$$x + y\frac{dy}{dx} = 0$$

Final Answer

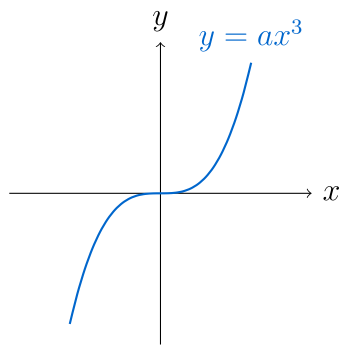
$x + y\dfrac{dy}{dx} = 0$

Question 4

Form the differential equation not containing the arbitrary constant(s) and satisfied by the following equations:

(i) $y = ax^3$, a is an arbitrary constant
(ii) $x^2 - y^2 = a^2$, a is an arbitrary constant
(iii) $y = c\sin^{-1} x$, c is an arbitrary constant
(iv) $y^2 = 4ax$, a is an arbitrary constant

TYPE 24 **Differential Equation Formation by Eliminating Single Multiplicative Constant**



Curves represented by parameter-dependent equations

Given: Given curves with 1 arbitrary constant each: (i) $y = ax^3$, (ii) $x^2 - y^2 = a^2$, (iii) $y = c\sin^{-1} x$, (iv) $y^2 = 4ax$.

Concept / Formula Used: $\dfrac{dy}{dx}$

Solution:

(i) Given equation:

$$y = ax^3 \implies a = \frac{y}{x^3}$$

Differentiating with respect to x:

$$\frac{dy}{dx} = 3ax^2$$

Substituting $a = \dfrac{y}{x^3}$:

$$\begin{aligned} \frac{dy}{dx} &= 3\left(\frac{y}{x^3}\right)x^2 \\ \frac{dy}{dx} &= \frac{3y}{x} \\ xy_1 &= 3y \end{aligned}$$

(ii) Given equation:

$$x^2 - y^2 = a^2$$

Differentiating both sides with respect to x:

$$\begin{aligned} 2x - 2y\frac{dy}{dx} &= 0 \\ 2y\frac{dy}{dx} &= 2x \\ yy_1 &= x \end{aligned}$$

(iii) Given equation:

$$y = c\sin^{-1}x \implies c = \frac{y}{\sin^{-1}x}$$

Differentiating with respect to x:

$$\frac{dy}{dx} = c \cdot \frac{1}{\sqrt{1-x^2}}$$

Substituting $c = \dfrac{y}{\sin^{-1}x}$:

$$\begin{aligned} \frac{dy}{dx} &= \frac{y}{\sqrt{1-x^2}\sin^{-1}x} \\ \sqrt{1-x^2}\sin^{-1}x \cdot \frac{dy}{dx} &= y \end{aligned}$$

Alternatively, writing $\dfrac{dy}{dx} = y_1$:

$$\frac{dy}{dx} = \frac{y}{\sqrt{1-x^2}\sin^{-1}x}$$

(iv) Given equation:

$$y^2 = 4ax$$

Differentiating with respect to x:

$$2y\frac{dy}{dx} = 4a \implies 4a = 2y\frac{dy}{dx}$$

Substituting $4a = 2y\dfrac{dy}{dx}$ into $y^2 = 4ax$:

$$\begin{aligned} y^2 &= \left(2y\frac{dy}{dx}\right)x \\ y^2 &= 2xy\frac{dy}{dx} \\ y &= 2x\frac{dy}{dx} \quad (\text{for } y \neq 0) \\ 2x\frac{dy}{dx} &= y \end{aligned}$$

Final Answer

(i) $xy_1 = 3y$

(ii) $yy_1 = x$

(iii) $\dfrac{dy}{dx} = \dfrac{y}{\sqrt{1-x^2}\sin^{-1}x}$

(iv) $2x\dfrac{dy}{dx} = y$

Question 5

Find the differential equations by eliminating the arbitrary constants and satisfied by the following equations:

(i) $y = e^{mx}$, m is an arbitrary constant

(ii) $y = ke^{\tan^{-1}x}$, k is an arbitrary constant

(iii) $y = \tan^{-1}x + ce^{-\tan^{-1}x}$, c is an arbitrary constant

TYPE 25 Differential Equation Formation by Eliminating Single Exponential Parameter

Given: Equations containing single arbitrary constant: (i) $y = e^{mx}$, (ii) $y = ke^{\tan^{-1}x}$, (iii) $y = \tan^{-1}x + ce^{-\tan^{-1}x}$.

Concept / Formula Used: $\dfrac{d}{dx}(e^u) = e^u\dfrac{du}{dx}$ $\quad \dfrac{d}{dx}(\tan^{-1}x) = \dfrac{1}{1+x^2}$

Solution:

(i) Given equation:

$$y = e^{mx}$$

Taking natural logarithm on both sides:

$$\ln y = mx \implies m = \frac{\ln y}{x}$$

Differentiating $y = e^{mx}$ with respect to x:

$$\frac{dy}{dx} = me^{mx}$$
$$\frac{dy}{dx} = my$$

Substituting $m = \frac{\ln y}{x}$:

$$\frac{dy}{dx} = \left(\frac{\ln y}{x}\right) y$$
$$x\frac{dy}{dx} = y \ln y$$

(ii) Given equation:

$$y = ke^{\tan^{-1} x}$$

Differentiating both sides with respect to x:

$$\frac{dy}{dx} = ke^{\tan^{-1} x} \cdot \frac{d}{dx}(\tan^{-1} x)$$
$$\frac{dy}{dx} = ke^{\tan^{-1} x} \cdot \frac{1}{1+x^2}$$

Since $y = ke^{\tan^{-1} x}$, substituting y:

$$\frac{dy}{dx} = \frac{y}{1+x^2}$$
$$(1+x^2)\frac{dy}{dx} = y$$

(iii) Given equation:

$$y = \tan^{-1} x + ce^{-\tan^{-1} x}$$
$$y - \tan^{-1} x = ce^{-\tan^{-1} x}$$

Differentiating both sides with respect to x:

$$\frac{dy}{dx} - \frac{1}{1+x^2} = ce^{-\tan^{-1} x} \cdot \left(-\frac{1}{1+x^2}\right)$$

Substitute $ce^{-\tan^{-1} x} = y - \tan^{-1} x$:

$$\frac{dy}{dx} - \frac{1}{1+x^2} = -(y - \tan^{-1} x) \cdot \frac{1}{1+x^2}$$

Multiplying the entire equation by $(1+x^2)$:

$$(1+x^2)\frac{dy}{dx} - 1 = -(y - \tan^{-1} x)$$
$$(1+x^2)\frac{dy}{dx} - 1 = -y + \tan^{-1} x$$
$$(1+x^2)\frac{dy}{dx} + y = 1 + \tan^{-1} x$$

Final Answer

(i) $x\dfrac{dy}{dx} = y\ln y$

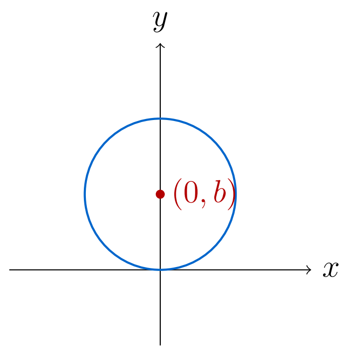
(ii) $(1+x^2)\dfrac{dy}{dx} = y$

(iii) $(1+x^2)\dfrac{dy}{dx} + y = 1 + \tan^{-1} x$

Question 6

Find the differential equations by eliminating the arbitrary constants:
(i) $x^2 + (y-b)^2 = 1$, b is an arbitrary constant
(ii) $(x-a)^2 - y^2 = 1$, a is an arbitrary constant
(iii) $x^2 + y^2 = ax^3$, a is an arbitrary constant

TYPE 26 Differential Equation Formation by Eliminating Single Constant in Shifted Conics



Shifted circle with center on y-axis

Given: Given curves with 1 arbitrary constant each: (i) $x^2 + (y-b)^2 = 1$, (ii) $(x-a)^2 - y^2 = 1$, (iii) $x^2 + y^2 = ax^3$.

Concept / Formula Used: x x, y, y_1

Solution:

(i) Given equation:

$$x^2 + (y-b)^2 = 1$$

Differentiating with respect to x (let $y_1 = \dfrac{dy}{dx}$):

$$\begin{aligned} 2x + 2(y-b)y_1 &= 0 \\ (y-b)y_1 &= -x \\ y - b &= -\frac{x}{y_1} \end{aligned}$$

Substituting $(y-b) = -\dfrac{x}{y_1}$ into $x^2 + (y-b)^2 = 1$:

$$x^2 + \left(-\frac{x}{y_1}\right)^2 = 1$$
$$x^2 + \frac{x^2}{y_1^2} = 1$$
$$x^2\left(1 + \frac{1}{y_1^2}\right) = 1$$
$$x^2\left(\frac{y_1^2+1}{y_1^2}\right) = 1$$
$$x^2(y_1^2+1) = y_1^2$$

(ii) Given equation:

$$(x-a)^2 - y^2 = 1$$

Differentiating with respect to x:

$$2(x-a) - 2yy_1 = 0$$
$$x - a = yy_1$$

Substituting $(x-a) = yy_1$ into $(x-a)^2 - y^2 = 1$:

$$(yy_1)^2 - y^2 = 1$$
$$y^2y_1^2 - y^2 = 1$$
$$y^2(y_1^2 - 1) = 1$$

(iii) Given equation:

$$x^2 + y^2 = ax^3$$

Isolating a:

$$a = \frac{x^2+y^2}{x^3}$$
$$a = x^{-1} + y^2x^{-3}$$

Differentiating with respect to x:

$$0 = -x^{-2} + \left(2yy_1x^{-3} - 3y^2x^{-4}\right)$$
$$0 = -\frac{1}{x^2} + \frac{2yy_1}{x^3} - \frac{3y^2}{x^4}$$

Multiplying through by x^4:

$$0 = -x^2 + 2xyy_1 - 3y^2$$
$$2xyy_1 = x^2 + 3y^2$$

Final Answer

(i) $x^2(y_1^2+1) = y_1^2$

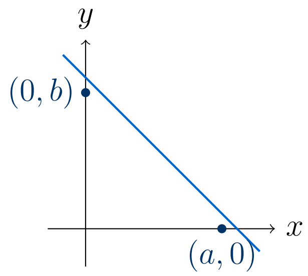
(ii) $y^2(y_1^2-1) = 1$

(iii) $2xyy_1 = x^2 + 3y^2$

Question 7

Form the differential equation of $\dfrac{x}{a} + \dfrac{y}{b} = 1$, where a, b are arbitrary constants.

TYPE 27 **Differential Equation Formation for Two-Intercept Line Family**



Straight line in intercept form

Given: Equation of straight line: $\dfrac{x}{a} + \dfrac{y}{b} = 1$, with 2 arbitrary constants a and b.

Concept / Formula Used: x

Solution:

Given equation:

$$\frac{x}{a} + \frac{y}{b} = 1$$

Differentiating for the first time with respect to x:

$$\begin{aligned} \frac{d}{dx}\left(\frac{x}{a}\right) + \frac{d}{dx}\left(\frac{y}{b}\right) &= \frac{d}{dx}(1) \\ \frac{1}{a} + \frac{1}{b}\frac{dy}{dx} &= 0 \\ \frac{1}{a} + \frac{1}{b}y_1 &= 0 \end{aligned}$$

Differentiating for the second time with respect to x:

$$\begin{aligned} \frac{d}{dx}\left(\frac{1}{a}\right) + \frac{1}{b}\frac{d}{dx}(y_1) &= 0 \\ 0 + \frac{1}{b}y_2 &= 0 \\ \frac{1}{b}y_2 &= 0 \end{aligned}$$

Since $b \neq 0$, we divide by $\dfrac{1}{b}$:

$$y_2 = 0$$

Final Answer

$y_2 = 0 \quad \text{or} \quad \dfrac{d^2y}{dx^2} = 0$

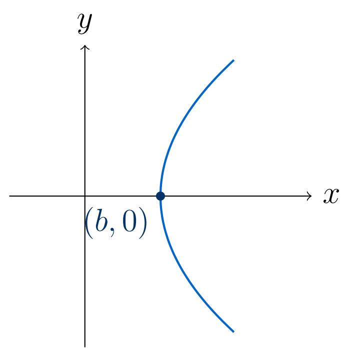
Question 8

Find the differential equation by eliminating arbitrary constants:

(i) $y^2 = 4a(x-b)$, a, b are arbitrary constants

(ii) $(y-b)^2 = 4(x-a)$, a, b are arbitrary constants

TYPE 28 **Differential Equation Formation for Shifted Parabola Families**



Parabola with shifted vertex

Given: Parabola equations with 2 arbitrary constants each: (i) $y^2 = 4a(x-b)$, (ii) $(y-b)^2 = 4(x-a)$.

Solution:

(i) Given equation:

$$y^2 = 4a(x-b)$$

Differentiating the first time with respect to x:

$$2yy_1 = 4a \implies yy_1 = 2a$$

Differentiating the second time with respect to x using the product rule:

$$\begin{aligned} \frac{d}{dx}(yy_1) &= \frac{d}{dx}(2a) \\ y_1 \cdot y_1 + y \cdot y_2 &= 0 \\ y_1^2 + yy_2 &= 0 \\ yy_2 + y_1^2 &= 0 \end{aligned}$$

(ii) Given equation:

$$(y-b)^2 = 4(x-a)$$

Differentiating the first time with respect to x:

$$2(y-b)y_1 = 4$$
$$(y-b)y_1 = 2 \implies y - b = \frac{2}{y_1}$$

Differentiating $(y-b)y_1 = 2$ with respect to x using the product rule:

$$y_1 \cdot y_1 + (y-b)y_2 = 0$$
$$y_1^2 + (y-b)y_2 = 0$$

Substituting $(y-b) = \dfrac{2}{y_1}$:

$$y_1^2 + \left(\frac{2}{y_1}\right) y_2 = 0$$
$$y_1^3 + 2y_2 = 0$$
$$2y_2 + y_1^3 = 0$$

Final Answer

(i) $yy_2 + y_1^2 = 0$

(ii) $2y_2 + y_1^3 = 0$

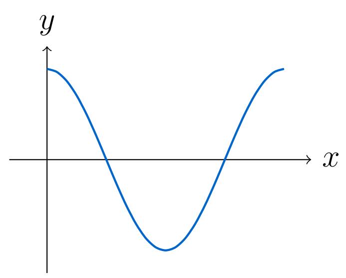
Question 9

Find the differential equations by eliminating a, b or A, B:

(i) $y = a\cos(x+b)$, a, b are arbitrary constants

(ii) $y = A\cos 2x + B\sin 2x$, A, B are arbitrary constants

TYPE 29 **Differential Equation Formation for Harmonic Trigonometric Functions**



Sinusoidal oscillation waveform

Given: Trigonometric family of curves: (i) $y = a\cos(x+b)$, (ii) $y = A\cos 2x + B\sin 2x$.

Concept / Formula Used: y

Solution:

(i) Given equation:

$$y = a\cos(x+b)$$

Differentiating the first time with respect to x:

$$y_1 = -a\sin(x+b)$$

Differentiating the second time with respect to x:

$$y_2 = -a\cos(x+b)$$

Substituting $y = a\cos(x+b)$:

$$y_2 = -y$$
$$y_2 + y = 0$$

(ii) Given equation:

$$y = A\cos 2x + B\sin 2x$$

Differentiating the first time with respect to x:

$$y_1 = -2A\sin 2x + 2B\cos 2x$$

Differentiating the second time with respect to x:

$$y_2 = -4A\cos 2x - 4B\sin 2x$$
$$y_2 = -4(A\cos 2x + B\sin 2x)$$

Substituting $y = A\cos 2x + B\sin 2x$:

$$y_2 = -4y$$
$$y_2 + 4y = 0$$

Final Answer

(i) $y_2 + y = 0$

(ii) $y_2 + 4y = 0$

Question 10

Find the differential equations by eliminating arbitrary constants:

(i) $y = Ae^{2x} + Be^{-2x}$, A, B are arbitrary constants

(ii) $y = ae^{3x} + be^{-2x}$, a, b are arbitrary constants

TYPE 30 Differential Equation Formation for Linear Combination of Exponentials

Given: Exponential equations with 2 arbitrary constants each: (i) $y = Ae^{2x} + Be^{-2x}$, (ii) $y = ae^{3x} + be^{-2x}$.

Solution:

(i) Given equation:

$$y = Ae^{2x} + Be^{-2x}$$

First derivative:

$$y_1 = 2Ae^{2x} - 2Be^{-2x}$$

Second derivative:

$$y_2 = 4Ae^{2x} + 4Be^{-2x}$$
$$y_2 = 4(Ae^{2x} + Be^{-2x})$$

Substituting $y = Ae^{2x} + Be^{-2x}$:

$$y_2 = 4y$$
$$y_2 - 4y = 0$$

(ii) Given equation:

$$y = ae^{3x} + be^{-2x}$$

First derivative:

$$y_1 = 3ae^{3x} - 2be^{-2x}$$

Second derivative:

$$y_2 = 9ae^{3x} + 4be^{-2x}$$

To eliminate a and b, calculate $y_2 - y_1$:

$$y_2 - y_1 = (9ae^{3x} + 4be^{-2x}) - (3ae^{3x} - 2be^{-2x})$$
$$y_2 - y_1 = 6ae^{3x} + 6be^{-2x}$$
$$y_2 - y_1 = 6(ae^{3x} + be^{-2x})$$

Substituting $y = ae^{3x} + be^{-2x}$:

$$y_2 - y_1 = 6y$$
$$y_2 - y_1 - 6y = 0$$

Final Answer

(i) $y_2 - 4y = 0$

(ii) $y_2 - y_1 - 6y = 0$

Question 11

(i) $y = (\sin^{-1} x)^2 + A\cos^{-1} x + B$, where A, B are arbitrary constants.

(ii) $y = e^x(A\cos x + B\sin x)$, where A, B are arbitrary constants.

TYPE 31 Differential Equation Formation for Compound Exponential and Inverse Trig Functions

Given: Equations to eliminate A, B: (i) $y = (\sin^{-1} x)^2 + A\cos^{-1} x + B$, (ii) $y = e^x(A\cos x + B\sin x)$.

Concept / Formula Used: Two arbitrary constants $\implies$ differentiate twice and eliminate A, B between the resulting relations.

Solution:

(i) Given equation:

$$y = (\sin^{-1} x)^2 + A\cos^{-1} x + B$$

Differentiating with respect to x:

$$y_1 = 2\sin^{-1} x \cdot \frac{1}{\sqrt{1-x^2}} + A\left(-\frac{1}{\sqrt{1-x^2}}\right) + 0$$
$$y_1 = \frac{2\sin^{-1} x - A}{\sqrt{1-x^2}}$$

Multiplying both sides by $\sqrt{1-x^2}$:

$$\sqrt{1-x^2}\, y_1 = 2\sin^{-1} x - A$$

This still contains the constant A and the transcendental term $\sin^{-1} x$, so we differentiate once more with respect to x. Using the product rule on the left side, and noting $\frac{d}{dx}(2\sin^{-1} x) = \frac{2}{\sqrt{1-x^2}}$ and $\frac{d}{dx}(A) = 0$ on the right side:

$$\frac{d}{dx}\left[\sqrt{1-x^2}\right] y_1 + \sqrt{1-x^2}\, y_2 = \frac{2}{\sqrt{1-x^2}}$$
$$\left(-\frac{x}{\sqrt{1-x^2}}\right) y_1 + \sqrt{1-x^2}\, y_2 = \frac{2}{\sqrt{1-x^2}}$$

Multiplying throughout by $\sqrt{1-x^2}$ to clear denominators:

$$-x\, y_1 + (1-x^2)\, y_2 = 2$$

which no longer contains A or B.

(ii) Given equation:

$$y = e^x(A\cos x + B\sin x)$$

Differentiating with respect to x (product rule):

$$y_1 = e^x(A\cos x + B\sin x) + e^x(-A\sin x + B\cos x)$$
$$y_1 = y + e^x(B\cos x - A\sin x)$$

So:

$$e^x(B\cos x - A\sin x) = y_1 - y \qquad (*)$$

Differentiating $(*)$ again with respect to x (product rule on the left):

$$e^x(B\cos x - A\sin x) + e^x(-B\sin x - A\cos x) = y_2 - y_1$$

Using $(*)$ to replace the first term on the left:

$$(y_1 - y) + e^x(-B\sin x - A\cos x) = y_2 - y_1$$

Since $e^x(-B\sin x - A\cos x) = -e^x(A\cos x + B\sin x) = -y$:

$$(y_1 - y) - y = y_2 - y_1$$
$$y_2 - 2y_1 + 2y = 0$$

Final Answer

(i) $(1 - x^2)y_2 - xy_1 - 2 = 0$

(ii) $y_2 - 2y_1 + 2y = 0$

Common Mistake: The draft's derivation for part (i) stopped after the first differentiation, leaving the constant A still present in the answer – with two arbitrary constants a second differentiation is mandatory. Part (ii) was missing entirely from the draft and has been solved from scratch. The draft had also appended a block of unrelated circle/parabola/ellipse/hyperbola derivations (matching none of this exercise's indexed problems) inside this question's solution; that contamination has been removed.

Question 12

Find the differential equation of the family of curves $(x - h)^2 + (y - k)^2 = r^2$, where h, k are arbitrary constants and r is a fixed (given) constant.

TYPE 32 **Differential Equation Formation for Circles of Fixed Radius with Variable Center**

Given: Family of circles $(x - h)^2 + (y - k)^2 = r^2$ with fixed radius r; h, k arbitrary.

Concept / Formula Used: Two arbitrary constants $\Longrightarrow$ differentiate twice.

Solution:

Given:

$$(x - h)^2 + (y - k)^2 = r^2 \qquad (1)$$

Differentiating (1) with respect to x:

$$2(x - h) + 2(y - k)y_1 = 0$$
$$(x - h) + (y - k)y_1 = 0 \qquad (2)$$

Differentiating (2) again with respect to x:

$$1 + y_1 \cdot y_1 + (y-k)y_2 = 0$$
$$1 + y_1^2 + (y-k)y_2 = 0$$

So:

$$(y-k) = -\frac{1+y_1^2}{y_2} \tag{3}$$

Substituting (3) back into (2) to find $(x-h)$:

$$(x-h) = -(y-k)y_1 = \frac{(1+y_1^2)y_1}{y_2} \tag{4}$$

Substituting (3) and (4) into the original equation (1):

$$\left(\frac{(1+y_1^2)y_1}{y_2}\right)^2 + \left(-\frac{1+y_1^2}{y_2}\right)^2 = r^2$$
$$\frac{(1+y_1^2)^2y_1^2}{y_2^2} + \frac{(1+y_1^2)^2}{y_2^2} = r^2$$
$$\frac{(1+y_1^2)^2(y_1^2+1)}{y_2^2} = r^2$$
$$(1+y_1^2)^3 = r^2y_2^2$$

Final Answer

$(1+y_1^2)^3 = r^2y_2^2$

Question 13

Find the differential equation of each of the following family of curves:

(i) concentric circles with centre at $(1, 2)$ *(Exemplar)*

(ii) all circles which touch the x-axis at origin

(iii) circles in the first quadrant and touching the coordinate axes

(iv) all parabolas having their vertices at origin and foci on y-axis *(NCERT)*

(v) all ellipses whose centres are at origin and foci on y-axis *(NCERT)*

(vi) all hyperbolas having foci on x-axis and centres at the origin *(NCERT)*

(vii) all non-vertical lines in a plane *(Exemplar)*

TYPE 33 **Differential Equation Formation from Geometric Descriptions of Conic Families**

Given: Seven distinct families of curves, each described geometrically rather than by an explicit equation with named constants.

Concept / Formula Used: Number of arbitrary constants in the family's equation = order of the required differential equation.

Solution:

(i) **Concentric circles with centre $(1, 2)$:** Since the centre is fixed, only the radius r is an arbitrary constant. Equation of the family:

$$(x-1)^2 + (y-2)^2 = r^2$$

One arbitrary constant $\implies$ differentiate once with respect to x:

$$\begin{aligned} 2(x-1) + 2(y-2)y_1 &= 0 \\ (x-1) + (y-2)y_1 &= 0 \end{aligned}$$

This is already free of r, so it is the required differential equation.

$$\boxed{(x-1) + (y-2)y_1 = 0}$$

(ii) **Circles touching the x-axis at the origin:** Such a circle has centre $(0, a)$ and radius $|a|$, for arbitrary constant a. Equation:

$$\begin{aligned} x^2 + (y-a)^2 &= a^2 \\ x^2 + y^2 - 2ay &= 0 \\ x^2 + y^2 &= 2ay \qquad (1) \end{aligned}$$

One arbitrary constant $\implies$ differentiate once:

$$\begin{aligned} 2x + 2yy_1 &= 2ay_1 \\ a &= \frac{x + yy_1}{y_1} \qquad (2) \end{aligned}$$

Substituting (2) into (1):

$$\begin{aligned} x^2 + y^2 &= 2y\left(\frac{x + yy_1}{y_1}\right) \\ (x^2 + y^2)y_1 &= 2xy + 2y^2y_1 \\ x^2y_1 + y^2y_1 - 2y^2y_1 &= 2xy \\ (x^2 - y^2)y_1 &= 2xy \end{aligned}$$

$$\boxed{(x^2 - y^2)y_1 = 2xy}$$

(iii) **Circles in the first quadrant touching both coordinate axes:** Such a circle has centre (r, r) and radius r, for arbitrary constant r. Equation:

$$(x-r)^2 + (y-r)^2 = r^2$$

Expanding:

$$\begin{aligned} x^2 - 2xr + r^2 + y^2 - 2yr + r^2 &= r^2 \\ x^2 + y^2 - 2r(x+y) + r^2 &= 0 \qquad (1) \end{aligned}$$

One arbitrary constant $\implies$ differentiate once:

$$\begin{aligned} 2x + 2yy_1 - 2r(1 + y_1) &= 0 \\ r(1 + y_1) &= x + yy_1 \\ r &= \frac{x + yy_1}{1 + y_1} \qquad (2) \end{aligned}$$

Using (2), compute $(x-r)$ and $(y-r)$ separately:

$$x - r = x - \frac{x+yy_1}{1+y_1} = \frac{x(1+y_1)-(x+yy_1)}{1+y_1} = \frac{(x-y)y_1}{1+y_1}$$
$$y - r = y - \frac{x+yy_1}{1+y_1} = \frac{y(1+y_1)-(x+yy_1)}{1+y_1} = \frac{-(x-y)}{1+y_1}$$

Substituting into $(x-r)^2+(y-r)^2=r^2$:

$$\left(\frac{(x-y)y_1}{1+y_1}\right)^2 + \left(\frac{-(x-y)}{1+y_1}\right)^2 = \left(\frac{x+yy_1}{1+y_1}\right)^2$$

Multiplying throughout by $(1+y_1)^2$:

$$(x-y)^2y_1^2 + (x-y)^2 = (x+yy_1)^2$$
$$(x-y)^2(1+y_1^2) = (x+yy_1)^2$$

$$\boxed{(x-y)^2(1+y_1^2) = (x+yy_1)^2}$$

(iv) **Parabolas with vertex at origin, foci on the y-axis:** Equation: $x^2 = 4ay$, for arbitrary constant a. One constant $\implies$ differentiate once:

$$2x = 4ay_1 \implies 4a = \frac{2x}{y_1} \tag{1}$$

Substituting (1) into $x^2 = 4ay$:

$$x^2 = \left(\frac{2x}{y_1}\right)y$$
$$x^2y_1 = 2xy$$
$$xy_1 = 2y$$

$$\boxed{x\,y_1 = 2y}$$

(v) **Ellipses with centre at origin, foci on the y-axis:** Equation: $\dfrac{x^2}{b^2}+\dfrac{y^2}{a^2}=1$. Let $A=\dfrac{1}{b^2}, B=\dfrac{1}{a^2}$ (two arbitrary constants):

$$Ax^2 + By^2 = 1$$

Two arbitrary constants $\implies$ differentiate twice. First derivative:

$$2Ax + 2Byy_1 = 0$$
$$Ax + Byy_1 = 0 \tag{1}$$

Second derivative:

$$A + B(y_1^2 + yy_2) = 0$$
$$A = -B(y_1^2+yy_2) \tag{2}$$

From (1): $A = -\dfrac{Byy_1}{x}$ (assuming $x \neq 0$) (3) Equating (2) and (3):

$$-B(y_1^2 + yy_2) = -\frac{Byy_1}{x}$$

Dividing throughout by $-B$ ($B \neq 0$):

$$y_1^2 + yy_2 = \frac{yy_1}{x}$$

Multiplying throughout by x:

$$x(yy_2 + y_1^2) = yy_1$$

$$\boxed{xy\,y_2 + x\,y_1^2 - y\,y_1 = 0}$$

(vi) **Hyperbolas with foci on the x-axis, centre at origin:** Equation: $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$.
Let $A = \dfrac{1}{a^2}, B = \dfrac{1}{b^2}$ (two arbitrary constants):

$$Ax^2 - By^2 = 1$$

Two arbitrary constants $\implies$ differentiate twice. First derivative:

$$2Ax - 2Byy_1 = 0$$
$$Ax = Byy_1 \implies A = \frac{Byy_1}{x} \tag{1}$$

Second derivative:

$$A - B(y_1^2 + yy_2) = 0$$
$$A = B(y_1^2 + yy_2) \tag{2}$$

Equating (1) and (2):

$$\frac{Byy_1}{x} = B(y_1^2 + yy_2)$$

Dividing throughout by B ($B \neq 0$) and multiplying by x:

$$yy_1 = x(y_1^2 + yy_2)$$

$$\boxed{xy\,y_2 + x\,y_1^2 - y\,y_1 = 0}$$

Note that this is the same differential equation as for the family of ellipses in part (v) – both are central conics with axes along the coordinate axes, and eliminating two constants from either family produces the same order-2 relation between x, y, y_1, y_2.

(vii) **All non-vertical lines in a plane:** General equation of a non-vertical line (slope defined):

$$y = mx + c$$

where m (slope) and c (intercept) are two arbitrary constants. Two arbitrary constants $\implies$ differentiate twice:

$$y_1 = m \qquad (m \text{ is constant, so differentiating again:})$$
$$y_2 = 0$$

$$\boxed{\frac{d^2y}{dx^2} = 0}$$

Final Answer

(i) $(x-1)+(y-2)y_1 = 0$ (ii) $(x^2-y^2)y_1 = 2xy$ (iii) $(x-y)^2(1+y_1^2) = (x+yy_1)^2$
(iv) $xy_1 = 2y$ (v) $xyy_2 + xy_1^2 - yy_1 = 0$ (vi) $xyy_2 + xy_1^2 - yy_1 = 0$ (vii) $y_2 = 0$

Common Mistake: This question was solved from the source textbook page after the local-model draft lost the original 7-part statement (only a truncated, math-stripped index fragment survived, with the actual solution content garbled/merged into Question 11's block and mislabelled). All seven parts have been independently re-derived here matching the source statement exactly.

Question 14

Form the differential equation of the family of circles having centres on the y-axis and radius 3 units.

TYPE 34 **Differential Equation Formation for Circles with Center on Axis and Fixed Radius**

Given: Family of circles with centre $(0, b)$ on the y-axis, b arbitrary, and fixed radius 3.

Concept / Formula Used: One arbitrary constant $\implies$ differentiate once.

Solution:

Equation of the family of circles, centre $(0, b)$, radius 3:

$$(x-0)^2 + (y-b)^2 = 3^2$$
$$x^2 + (y-b)^2 = 9 \tag{1}$$

Differentiating (1) with respect to x:

$$2x + 2(y-b)y_1 = 0$$
$$(y-b)y_1 = -x$$
$$y - b = -\frac{x}{y_1} \tag{2}$$

Substituting (2) into (1):

$$x^2 + \left(-\frac{x}{y_1}\right)^2 = 9$$
$$x^2 + \frac{x^2}{y_1^2} = 9$$
$$x^2\left(\frac{y_1^2+1}{y_1^2}\right) = 9$$
$$x^2(1+y_1^2) = 9y_1^2$$

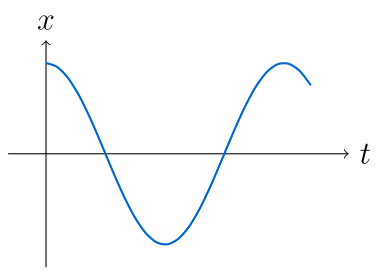
Final Answer

$x^2(1+y_1^2) = 9y_1^2$

Question 15

Form the differential equation of simple harmonic motion given by $x = A\cos(nt+\alpha)$, where n is fixed and A, α are parameters.

TYPE 35 **Differential Equation Formation for Simple Harmonic Motion**



Simple harmonic displacement-time graph

Given: Displacement equation: $x = A\cos(nt+\alpha)$, where n is fixed and A, α are parameters (2 arbitrary constants).

Concept / Formula Used: x t A α

Solution:

Given displacement relation:

$$x = A\cos(nt+\alpha) \tag{3}$$

Differentiating for the first time with respect to t:

$$\frac{dx}{dt} = \frac{d}{dt}\left[A\cos(nt+\alpha)\right]$$
$$\frac{dx}{dt} = -An\sin(nt+\alpha)$$

Differentiating for the second time with respect to t:

$$\frac{d^2x}{dt^2} = \frac{d}{dt}\left[-An\sin(nt+\alpha)\right]$$
$$\frac{d^2x}{dt^2} = -An^2\cos(nt+\alpha)$$
$$\frac{d^2x}{dt^2} = -n^2\left[A\cos(nt+\alpha)\right]$$

Substituting $x = A\cos(nt+\alpha)$:

$$\frac{d^2x}{dt^2} = -n^2x$$
$$\frac{d^2x}{dt^2} + n^2x = 0$$

Final Answer

$\frac{d^2x}{dt^2} + n^2x = 0$

9.4 – Differential Equations with Variables Separable

Question 1

Solve the following differential equations:

(i) $\frac{dy}{dx} = \sin x - x$

(ii) $\frac{dy}{dx} + 2x = e^{3x}$

(iii) $(x^2+4)\frac{dy}{dx} = 1$

(iv) $\frac{dy}{dx} = y\sin x$

(v) $\frac{dy}{dx} = \csc y$

TYPE 36 **General Solution of First-Order Variable Separable Differential Equation**

Given: Given differential equations for subparts (i) through (v) in variable separable form.

Concept / Formula Used: $\int \sin x\,dx = -\cos x + C$ $\quad \int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$ $\quad \int \csc y\,dy = \int \frac{dy}{\csc y} = \int \sin y\,dy$

Solution:

(i) Rewriting the equation by separating variables:

$$dy = (\sin x - x)\,dx$$

Integrating both sides:

$$\int dy = \int (\sin x - x)\,dx$$
$$y = -\cos x - \frac{x^2}{2} + C$$

(ii) Rearranging the terms:

$$\frac{dy}{dx} = e^{3x} - 2x$$
$$dy = (e^{3x} - 2x)\,dx$$

Integrating both sides:

$$\int dy = \int (e^{3x} - 2x)\,dx$$
$$y = \frac{e^{3x}}{3} - x^2 + C$$

(iii) Separating the variables:

$$dy = \frac{dx}{x^2 + 4}$$
$$dy = \frac{dx}{x^2 + 2^2}$$

Integrating both sides:

$$\int dy = \int \frac{dx}{x^2 + 2^2}$$
$$y = \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + C$$

(iv) Separating variables y and x:

$$\frac{dy}{y} = \sin x\,dx$$

Integrating both sides:

$$\int \frac{dy}{y} = \int \sin x\,dx$$
$$\log|y| = -\cos x + C_1$$
$$|y| = e^{-\cos x + C_1}$$
$$y = Ce^{-\cos x} \quad \text{or} \quad \log|y| + \cos x = C$$

(v) Rearranging terms:

$$\frac{dy}{\csc y} = dx$$
$$\sin y\, dy = dx$$

Integrating both sides:

$$\int \sin y\, dy = \int dx$$
$$-\cos y = x + C_1$$
$$x + \cos y = C$$

Final Answer

(i) $y = -\cos x - \frac{x^2}{2} + C$

(ii) $y = \frac{e^{3x}}{3} - x^2 + C$

(iii) $y = \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + C$

(iv) $y = Ce^{-\cos x}$

(v) $x + \cos y = C$

Question 2

Solve the following differential equations:

(i) $\frac{dy}{dx} = \sqrt{4 - y^2}$

(ii) $\frac{dy}{dx} + y = 1$

TYPE 37 **Variable Separable Differential Equation with Radical or Linear Expressions**

Given: Differential equations with variables that can be separated directly.

Concept / Formula Used: $\int \frac{dy}{\sqrt{a^2-y^2}} = \sin^{-1}\left(\frac{y}{a}\right) + C$ $\quad \int \frac{dy}{a-y} = -\log|a-y| + C$

Solution:

(i) Separating variables:

$$\frac{dy}{\sqrt{4 - y^2}} = dx$$
$$\frac{dy}{\sqrt{2^2 - y^2}} = dx$$

Integrating both sides:

$$\begin{aligned}\int \frac{dy}{\sqrt{2^2 - y^2}} &= \int dx \\ \sin^{-1}\left(\frac{y}{2}\right) &= x + C \\ \frac{y}{2} &= \sin(x + C) \\ y &= 2\sin(x + C)\end{aligned}$$

(ii) Rearranging terms:

$$\begin{aligned}\frac{dy}{dx} &= 1 - y \\ \frac{dy}{1 - y} &= dx\end{aligned}$$

Integrating both sides:

$$\begin{aligned}\int \frac{dy}{1 - y} &= \int dx \\ -\log|1 - y| &= x + C_1 \\ \log|1 - y| &= -(x + C_1) \\ 1 - y &= e^{-(x + C_1)} \\ 1 - y &= Ae^{-x} \\ y &= 1 - Ae^{-x}\end{aligned}$$

Final Answer

(i) $y = 2\sin(x + C)$

(ii) $y = 1 - Ae^{-x}$

Question 3

Solve the following differential equations:

(i) $x^5 \dfrac{dy}{dx} = -y^5$

(ii) $x(1 + y^2)dx + y(1 + x^2)dy = 0$

TYPE 38 **Variable Separable Differential Equation with Algebraic Powers and Quadratic Factors**

Given: Differential equations involving powers and quadratic binomial terms.

Concept / Formula Used: $\int y^{-n} dy = \frac{y^{-n+1}}{-n+1} + C$ $\int \frac{2t}{1+t^2} dt = \log(1 + t^2) + C$

Solution:

(i) Separating variables:

$$\frac{dy}{y^5} = -\frac{dx}{x^5}$$
$$y^{-5}dy = -x^{-5}dx$$

Integrating both sides:

$$\int y^{-5}dy = -\int x^{-5}dx$$
$$\frac{y^{-4}}{-4} = -\left(\frac{x^{-4}}{-4}\right) + C_1$$
$$-\frac{1}{4y^4} = \frac{1}{4x^4} + C_1$$
$$\frac{1}{x^4} + \frac{1}{y^4} = C \quad (\text{where } C = -4C_1)$$

(ii) Rearranging terms:

$$y(1+x^2)dy = -x(1+y^2)dx$$
$$\frac{y}{1+y^2}dy = -\frac{x}{1+x^2}dx$$

Multiplying both sides by 2:

$$\frac{2y}{1+y^2}dy = -\frac{2x}{1+x^2}dx$$

Integrating both sides:

$$\int \frac{2y}{1+y^2}dy = -\int \frac{2x}{1+x^2}dx$$
$$\log(1+y^2) = -\log(1+x^2) + \log C$$
$$\log(1+y^2) + \log(1+x^2) = \log C$$
$$\log((1+x^2)(1+y^2)) = \log C$$
$$(1+x^2)(1+y^2) = C$$

Final Answer

(i) $\frac{1}{x^4} + \frac{1}{y^4} = C$

(ii) $(1+x^2)(1+y^2) = C$

Question 4

Solve the following differential equations:

(i) $\frac{dy}{dx} = \frac{x+1}{2-y}$

(ii) $e^{y-x}\frac{dy}{dx} = 1$

TYPE 39 **Variable Separable Differential Equation with Rational and Exponential Terms**

Given: Differential equations with rational functions and exponential terms.

Concept / Formula Used: $e^{y-x} = \frac{e^y}{e^x}$ $\quad \int e^u du = e^u + C$

Solution:

(i) Cross-multiplying to separate variables:

$$(2-y)dy = (x+1)dx$$

Integrating both sides:

$$\int (2-y)dy = \int (x+1)dx$$
$$2y - \frac{y^2}{2} = \frac{x^2}{2} + x + C_1$$

Multiplying by 2:

$$4y - y^2 = x^2 + 2x + 2C_1$$
$$x^2 + y^2 + 2x - 4y = C \quad (\text{where } C = -2C_1)$$

(ii) Expressing e^{y-x} as $\frac{e^y}{e^x}$:

$$\frac{e^y}{e^x}\frac{dy}{dx} = 1$$
$$e^y dy = e^x dx$$

Integrating both sides:

$$\int e^y dy = \int e^x dx$$
$$e^y = e^x + C$$
$$e^y - e^x = C$$

Final Answer

(i) $x^2 + y^2 + 2x - 4y = C$

(ii) $e^y - e^x = C$

Question 5

Solve the following differential equations:

(i) $\dfrac{dy}{dx} = (e^x + 1)y$

(ii) $\dfrac{dy}{dx} + \dfrac{1+y^2}{y} = 0$

TYPE 40 Variable Separable Differential Equation with Product Functions

Given: Differential equations involving product expressions.

Concept / Formula Used: $\int \frac{dy}{y} = \log|y| + C \qquad \int \frac{2y}{1+y^2}dy = \log(1+y^2) + C$

Solution:

(i) Separating variables:

$$\frac{dy}{y} = (e^x + 1)dx$$

Integrating both sides:

$$\int \frac{dy}{y} = \int (e^x + 1)dx$$
$$\log|y| = e^x + x + C$$

(ii) Rearranging terms:

$$\frac{dy}{dx} = -\frac{1+y^2}{y}$$
$$\frac{y}{1+y^2}dy = -dx$$

Multiplying by 2:

$$\frac{2y}{1+y^2}dy = -2dx$$

Integrating both sides:

$$\int \frac{2y}{1+y^2}dy = -2\int dx$$
$$\log(1+y^2) = -2x + C_1$$
$$1+y^2 = e^{-2x+C_1}$$
$$1+y^2 = Ae^{-2x}$$

Final Answer

(i) $\log|y| = e^x + x + C$

(ii) $1 + y^2 = Ae^{-2x}$

Question 6

Solve the following differential equations:

(i) $\dfrac{dy}{dx} = \dfrac{1+y^2}{1+x^2}$

(ii) $\dfrac{dy}{dx} = 2^{y-x}$

TYPE 41 **Variable Separable Differential Equation with Inverse Tangent and Exponential Integrals**

Given: Differential equations with algebraic fractions and exponent rules.

Concept / Formula Used: $\int \frac{du}{1+u^2} = \tan^{-1} u + C \qquad \int a^u du = \frac{a^u}{\log a} + C$

Solution:

(i) Separating variables:

$$\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$$

Integrating both sides:

$$\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$$
$$\tan^{-1} y = \tan^{-1} x + C$$
$$\tan^{-1} y - \tan^{-1} x = C$$

(ii) Writing 2^{y-x} as $\frac{2^y}{2^x}$:

$$\frac{dy}{dx} = \frac{2^y}{2^x}$$
$$2^{-y}dy = 2^{-x}dx$$

Integrating both sides:

$$\int 2^{-y}dy = \int 2^{-x}dx$$
$$\frac{2^{-y}}{-\log 2} = \frac{2^{-x}}{-\log 2} + C_1$$
$$-2^{-y} = -2^{-x} - C_1 \log 2$$
$$2^{-x} - 2^{-y} = C$$

Final Answer

(i) $\tan^{-1} y - \tan^{-1} x = C$

(ii) $2^{-x} - 2^{-y} = C$

Question 7

Solve the following differential equations:

(i) $\dfrac{dy}{dx} = \log x$

(ii) $\sin\left(\dfrac{dy}{dx}\right) = a$

TYPE 42 Variable Separable Differential Equation Involving Logarithmic and Inverse Trigonometric Functions

Given: Differential equations involving logarithmic and inverse trigonometric expressions.

Concept / Formula Used: $\int \log x\, dx = x \log x - x + C$

Solution:

(i) Separating variables:

$$dy = \log x\, dx$$

Integrating both sides using integration by parts on $\int 1 \cdot \log x\, dx$:

$$\begin{aligned} \int dy &= \int \log x\, dx \\ y &= \log x \cdot x - \int \left(\frac{1}{x} \cdot x\right) dx \\ y &= x \log x - \int 1\, dx \\ y &= x \log x - x + C \end{aligned}$$

(ii) Taking inverse sine on both sides:

$$\begin{aligned} \frac{dy}{dx} &= \sin^{-1} a \\ dy &= \sin^{-1} a\, dx \end{aligned}$$

Integrating both sides ($\sin^{-1} a$ is a constant):

$$\begin{aligned} \int dy &= \sin^{-1} a \int dx \\ y &= x \sin^{-1} a + C \end{aligned}$$

Final Answer

(i) $y = x \log x - x + C$

(ii) $y = x \sin^{-1} a + C$

Question 8

Solve the following differential equations:

(i) $e^x \dfrac{dy}{dx} + 1 = x$

(ii) $\dfrac{1}{x}\dfrac{dy}{dx} = \tan^{-1} x$

TYPE 43 **Variable Separable Differential Equation Solved by Integration by Parts**

Given: Differential equations requiring product rule integration on RHS.

Concept / Formula Used: $\int u\,dv = uv - \int v\,du$

Solution:

(i) Rearranging the equation:

$$e^x \frac{dy}{dx} = x - 1$$
$$dy = (x-1)e^{-x}\,dx$$

Integrating both sides:

$$\int dy = \int (x-1)e^{-x}\,dx$$

Applying integration by parts with $u = x - 1 \implies du = dx$ and $dv = e^{-x}dx \implies v = -e^{-x}$:

$$y = (x-1)(-e^{-x}) - \int (-e^{-x})dx$$
$$y = -(x-1)e^{-x} - e^{-x} + C$$
$$y = -xe^{-x} + e^{-x} - e^{-x} + C$$
$$y = -xe^{-x} + C$$

(ii) Separating variables:

$$dy = x\tan^{-1} x\,dx$$

Integrating both sides using parts ($u = \tan^{-1} x, dv = x\,dx \implies du = \frac{dx}{1+x^2}, v = \frac{x^2}{2}$):

$$\begin{aligned}
\int dy &= \int x \tan^{-1} x\,dx \\
y &= \frac{x^2}{2}\tan^{-1} x - \int \frac{x^2}{2(1+x^2)}\,dx \\
y &= \frac{x^2}{2}\tan^{-1} x - \frac{1}{2}\int \frac{(x^2+1)-1}{1+x^2}\,dx \\
y &= \frac{x^2}{2}\tan^{-1} x - \frac{1}{2}\int \left(1 - \frac{1}{1+x^2}\right)dx \\
y &= \frac{x^2}{2}\tan^{-1} x - \frac{1}{2}(x - \tan^{-1} x) + C \\
y &= \frac{1}{2}(x^2+1)\tan^{-1} x - \frac{x}{2} + C
\end{aligned}$$

Final Answer

(i) $y = -xe^{-x} + C$

(ii) $y = \dfrac{1}{2}(x^2+1)\tan^{-1} x - \dfrac{x}{2} + C$

Question 9

Solve the following differential equations:

(i) $y' = (\cos^2 x - \sin^2 x)\cos^2 y$

(ii) $(xy^2 + x)dx + (x^2y + y)dy = 0$

(iii) $(x^2 - yx^2)dy + (y^2 + xy^2)dx = 0$ (ISC 2017)

TYPE 44 Variable Separable Differential Equation with Algebraic Factorization or Trigonometric Identities

Given: Differential equations requiring trigonometric identities and common term factorization.

Concept / Formula Used: $\cos^2 x - \sin^2 x = \cos 2x \qquad \int \sec^2 y\,dy = \tan y + C$

Solution:

(i) Using identity $\cos^2 x - \sin^2 x = \cos 2x$:

$$\begin{aligned}
\frac{dy}{dx} &= \cos 2x \cos^2 y \\
\frac{dy}{\cos^2 y} &= \cos 2x\,dx \\
\sec^2 y\,dy &= \cos 2x\,dx
\end{aligned}$$

Integrating both sides:

$$\int \sec^2 y \, dy = \int \cos 2x \, dx$$
$$\tan y = \frac{1}{2}\sin 2x + C$$

(ii) Factorizing terms:

$$x(y^2+1)dx + y(x^2+1)dy = 0$$
$$y(x^2+1)dy = -x(y^2+1)dx$$
$$\frac{y}{y^2+1}dy = -\frac{x}{x^2+1}dx$$

Multiplying by 2 and integrating:

$$\int \frac{2y}{y^2+1}dy = -\int \frac{2x}{x^2+1}dx$$
$$\log(y^2+1) = -\log(x^2+1) + \log C$$
$$\log((x^2+1)(y^2+1)) = \log C$$
$$(x^2+1)(y^2+1) = C$$

(iii) Factorizing terms:

$$x^2(1-y)dy + y^2(1+x)dx = 0$$
$$x^2(1-y)dy = -y^2(1+x)dx$$
$$\frac{1-y}{y^2}dy = -\frac{1+x}{x^2}dx$$
$$\left(\frac{1}{y^2} - \frac{1}{y}\right)dy = -\left(\frac{1}{x^2} + \frac{1}{x}\right)dx$$

Integrating both sides:

$$\int \left(y^{-2} - \frac{1}{y}\right)dy = \int \left(-x^{-2} - \frac{1}{x}\right)dx$$
$$-\frac{1}{y} - \log|y| = \frac{1}{x} - \log|x| + C_1$$
$$\log|x| - \log|y| - \frac{1}{x} - \frac{1}{y} = C_1$$
$$\log\left|\frac{x}{y}\right| - \frac{1}{x} - \frac{1}{y} = C$$

Final Answer

(i) $\tan y = \frac{1}{2}\sin 2x + C$

(ii) $(x^2+1)(y^2+1) = C$

(iii) $\log\left|\frac{x}{y}\right| - \frac{1}{x} - \frac{1}{y} = C$

Question 10

Solve the following differential equations:

(i) $(1+x)(1+y^2)dx + (1+y)(1+x^2)dy = 0$

(ii) $x\sqrt{1-y^2}dx + y\sqrt{1-x^2}dy = 0$

TYPE 45 Symmetric Variable Separable Differential Equation with Quadratic and Radical Expressions

Given: Symmetric differential equations in x and y.

Concept / Formula Used: $\int \frac{f'(u)}{f(u)}du = \log|f(u)| + C$ $\int \frac{u}{\sqrt{a^2-u^2}}du = -\sqrt{a^2-u^2} + C$

Solution:

(i) Separating variables:

$$
\begin{aligned}
(1+y)(1+x^2)dy &= -(1+x)(1+y^2)dx \\
\frac{1+y}{1+y^2}dy &= -\frac{1+x}{1+x^2}dx \\
\left(\frac{1}{1+y^2} + \frac{y}{1+y^2}\right) dy &= -\left(\frac{1}{1+x^2} + \frac{x}{1+x^2}\right) dx
\end{aligned}
$$

Integrating both sides:

$$
\begin{aligned}
\int \frac{dy}{1+y^2} + \frac{1}{2}\int \frac{2y}{1+y^2}dy &= -\int \frac{dx}{1+x^2} - \frac{1}{2}\int \frac{2x}{1+x^2}dx \\
\tan^{-1} y + \frac{1}{2}\log(1+y^2) &= -\tan^{-1} x - \frac{1}{2}\log(1+x^2) + C
\end{aligned}
$$

$$\tan^{-1} x + \tan^{-1} y + \frac{1}{2}\log((1+x^2)(1+y^2)) = C$$

(ii) Separating variables:

$$
\begin{aligned}
y\sqrt{1-x^2}dy &= -x\sqrt{1-y^2}dx \\
\frac{y}{\sqrt{1-y^2}}dy &= -\frac{x}{\sqrt{1-x^2}}dx
\end{aligned}
$$

Integrating both sides:

$$\int \frac{y}{\sqrt{1-y^2}}dy = -\int \frac{x}{\sqrt{1-x^2}}dx$$

Using substitution $1-y^2 = t \implies -2y\,dy = dt$ and $1-x^2 = u \implies -2x\,dx = du$:

$$
\begin{aligned}
-\sqrt{1-y^2} &= \sqrt{1-x^2} + C_1 \\
\sqrt{1-x^2} + \sqrt{1-y^2} &= C \quad (\text{where } C = -C_1)
\end{aligned}
$$

Final Answer

(i) $\tan^{-1} x + \tan^{-1} y + \frac{1}{2}\log((1+x^2)(1+y^2)) = C$

(ii) $\sqrt{1-x^2} + \sqrt{1-y^2} = C$

Question 11

Solve the following differential equations:

(i) $\cos x \cos y \, dy + \sin x \sin y \, dx = 0$

(ii) $\cos x(1+\cos y)dx - \sin y(1+\sin x)dy = 0$

TYPE 46 Variable Separable Differential Equation with Trigonometric Product Terms

Given: Trigonometric differential expressions.

Concept / Formula Used: $\int \cot y \, dy = \log|\sin y| + C$ $\quad \int \tan x \, dx = \log|\sec x| + C$

Solution:

(i) Separating variables:

$$\begin{aligned} \cos x \cos y \, dy &= -\sin x \sin y \, dx \\ \frac{\cos y}{\sin y} dy &= -\frac{\sin x}{\cos x} dx \\ \cot y \, dy &= -\tan x \, dx \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \int \cot y \, dy &= -\int \tan x \, dx \\ \log|\sin y| &= -\log|\sec x| + \log C \\ \log|\sin y| + \log|\sec x| &= \log C \\ \log|\sin y \sec x| &= \log C \\ \sin y \sec x &= C \end{aligned}$$

(ii) Separating variables:

$$\begin{aligned} \sin y(1+\sin x)dy &= \cos x(1+\cos y)dx \\ \frac{\sin y}{1+\cos y} dy &= \frac{\cos x}{1+\sin x} dx \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \int \frac{\sin y}{1+\cos y} dy &= \int \frac{\cos x}{1+\sin x} dx \\ -\log|1+\cos y| &= \log|1+\sin x| + \log C_1 \\ \log|1+\sin x| + \log|1+\cos y| &= -C_1 = \log C \\ (1+\sin x)(1+\cos y) &= C \end{aligned}$$

Final Answer

(i) $\sin y \sec x = C$

(ii) $(1 + \sin x)(1 + \cos y) = C$

Question 12

Solve the following differential equations:

(i) $x(e^{2y} - 1)dy + (x^2 - 1)e^y dx = 0$

(ii) $e^x \tan y\, dx + (1 - e^x) \sec^2 y\, dy = 0$

TYPE 47 Variable Separable Differential Equation Solved by Logarithmic Derivative Form

Given: Equations containing combined exponential and algebraic/trigonometric expressions.

Concept / Formula Used: $\int \frac{f'(u)}{f(u)} du = \log |f(u)| + C$

Solution:

(i) Separating variables:

$$\begin{aligned} x(e^{2y} - 1)dy &= -(x^2 - 1)e^y dx \\ \frac{e^{2y} - 1}{e^y} dy &= -\frac{x^2 - 1}{x} dx \\ (e^y - e^{-y})dy &= -\left(x - \frac{1}{x}\right) dx \\ (e^y - e^{-y})dy &= \left(\frac{1}{x} - x\right) dx \end{aligned}$$

Integrating both sides:

$$\int (e^y - e^{-y})dy = \int \left(\frac{1}{x} - x\right) dx$$

$$e^y + e^{-y} = \log |x| - \frac{x^2}{2} + C$$

(ii) Separating variables:

$$\begin{aligned} (1 - e^x) \sec^2 y\, dy &= -e^x \tan y\, dx \\ \frac{\sec^2 y}{\tan y} dy &= -\frac{e^x}{1 - e^x} dx \\ \frac{\sec^2 y}{\tan y} dy &= \frac{e^x}{e^x - 1} dx \end{aligned}$$

Integrating both sides:

$$\begin{aligned}\int \frac{\sec^2 y}{\tan y} dy &= \int \frac{e^x}{e^x - 1} dx \\ \log|\tan y| &= \log|e^x - 1| + \log C \\ \log|\tan y| &= \log|C(e^x - 1)| \\ \tan y &= C(e^x - 1)\end{aligned}$$

Final Answer

(i) $e^y + e^{-y} = \log|x| - \dfrac{x^2}{2} + C$

(ii) $\tan y = C(e^x - 1)$

Question 13

Solve the following differential equations:

(i) $y\,dx - x\,dy = xy\,dx$

(ii) $(1 + y^2)\tan^{-1} x\,dx + 2y(1 + x^2)dy = 0$ (Exemplar)

TYPE 48 Variable Separable Differential Equation with Division Transformation or Inverse Trig Substitution

Given: Differential expressions in two variables.

Concept / Formula Used: $\int \frac{g'(x)}{g(x)} dx = \log|g(x)| + C$ $\quad u = \tan^{-1} x$ $\quad du = \frac{dx}{1+x^2}$

Solution:

(i) Dividing throughout by xy:

$$\begin{aligned}\frac{y\,dx - x\,dy}{xy} &= \frac{xy\,dx}{xy} \\ \frac{1}{x}dx - \frac{1}{y}dy &= dx \\ \frac{1}{y}dy &= \left(\frac{1}{x} - 1\right) dx\end{aligned}$$

Integrating both sides:

$$\begin{aligned}\int \frac{1}{y}dy &= \int \left(\frac{1}{x} - 1\right) dx \\ \log|y| &= \log|x| - x + C_1 \\ \log\left|\frac{y}{x}\right| &= -x + C_1 \\ \frac{y}{x} &= e^{-x+C_1} \\ y &= Axe^{-x}\end{aligned}$$

(ii) Separating variables:

$$2y(1+x^2)dy = -(1+y^2)\tan^{-1} x\, dx$$
$$\frac{2y}{1+y^2}dy = -\frac{\tan^{-1} x}{1+x^2}dx$$

Integrating both sides:

$$\int \frac{2y}{1+y^2}dy = -\int \frac{\tan^{-1} x}{1+x^2}dx$$

Let $u = \tan^{-1} x \implies du = \frac{dx}{1+x^2}$:

$$\log(1+y^2) = -\int u\, du$$
$$\log(1+y^2) = -\frac{u^2}{2} + C$$
$$\log(1+y^2) = -\frac{(\tan^{-1} x)^2}{2} + C$$

Final Answer

(i) $y = Axe^{-x}$

(ii) $\log(1+y^2) + \dfrac{(\tan^{-1} x)^2}{2} = C$

Question 14

Solve the following differential equations:

(i) $(1+x^2)dy = xy\, dx$

(ii) $xyy' = 1 + x + y + xy$

TYPE 49 **Variable Separable Differential Equation by Bivariate Polynomial Factorization**

Given: Rational and factorable differential equations.

Concept / Formula Used: $\int \frac{dy}{y} = \log|y| + C \qquad \int \left(1 + \frac{1}{u}\right) du = u + \log|u| + C$

Solution:

(i) Separating variables:

$$\frac{dy}{y} = \frac{x}{1+x^2}dx$$

Multiplying by 2:

$$2\int \frac{dy}{y} = \int \frac{2x}{1+x^2}dx$$
$$2\log|y| = \log(1+x^2) + \log C_1$$
$$\log(y^2) = \log(C_1(1+x^2))$$
$$y^2 = C(1+x^2)$$

(ii) Factoring the RHS:

$$1 + x + y + xy = (1 + x)(1 + y)$$
$$xy\frac{dy}{dx} = (1 + x)(1 + y)$$

Separating variables:

$$\frac{y}{1+y}dy = \frac{1+x}{x}dx$$
$$\left(1 - \frac{1}{1+y}\right)dy = \left(\frac{1}{x} + 1\right)dx$$

Integrating both sides:

$$\int \left(1 - \frac{1}{1+y}\right)dy = \int \left(\frac{1}{x} + 1\right)dx$$
$$y - \log|1 + y| = \log|x| + x + C$$
$$y - x = \log|x| + \log|1 + y| + C$$
$$y - x = \log|x(1 + y)| + C$$

Final Answer

(i) $y^2 = C(1 + x^2)$

(ii) $y - x = \log|x(1 + y)| + C$

Question 15

Solve the following differential equations:

(i) $y(1 - x^2)\frac{dy}{dx} = x(1 + y^2)$

(ii) $(y + xy)dx + x(1 - y^2)dy = 0$

TYPE 50 Variable Separable Differential Equation with Rational Algebraic Functions

Given: Algebraic expressions separable into simple fractions.

Concept / Formula Used: $\int \frac{2u}{1 \pm u^2}du = \pm \log|1 \pm u^2| + C$

Solution:

(i) Separating variables:

$$\frac{y}{1+y^2}dy = \frac{x}{1-x^2}dx$$

Multiplying by 2:

$$\frac{2y}{1+y^2}dy = \frac{2x}{1-x^2}dx$$

Integrating both sides:

$$\int \frac{2y}{1+y^2}dy = -\int \frac{-2x}{1-x^2}dx$$
$$\log(1+y^2) = -\log|1-x^2| + \log C$$
$$\log(1+y^2) + \log|1-x^2| = \log C$$
$$(1+y^2)(1-x^2) = C$$

(ii) Factoring the first term:

$$y(1+x)dx + x(1-y^2)dy = 0$$
$$x(1-y^2)dy = -y(1+x)dx$$
$$\frac{1-y^2}{y}dy = -\frac{1+x}{x}dx$$
$$\left(\frac{1}{y} - y\right)dy = -\left(\frac{1}{x} + 1\right)dx$$

Integrating both sides:

$$\int \left(\frac{1}{y} - y\right)dy = -\int \left(\frac{1}{x} + 1\right)dx$$
$$\log|y| - \frac{y^2}{2} = -\log|x| - x + C$$
$$\log|y| + \log|x| - \frac{y^2}{2} + x = C$$
$$\log|xy| - \frac{y^2}{2} + x = C$$

Final Answer

(i) $(1+y^2)(1-x^2) = C$

(ii) $\log|xy| - \frac{y^2}{2} + x = C$

Question 16

Solve the following differential equations:
(i) $(x^2 - yx^2)dy + (y^2 + x^2y^2)dx = 0$
(ii) $(e^x + 1)y\,dy = (y+1)e^x dx$

TYPE 51 Variable Separable Differential Equation with Common Factor Extraction

Given: Algebraic and exponential variable-separable differential equations.

Concept / Formula Used: $\int u^{-n}du = \frac{u^{-n+1}}{-n+1} + C \qquad \int \frac{f'(x)}{f(x)}dx = \log|f(x)| + C$

Solution:

(i) Factoring common terms:

$$\begin{aligned} x^2(1-y)dy + y^2(1+x^2)dx &= 0 \\ x^2(1-y)dy &= -y^2(1+x^2)dx \\ \frac{1-y}{y^2}dy &= -\frac{1+x^2}{x^2}dx \\ \left(y^{-2} - \frac{1}{y}\right)dy &= -\left(x^{-2}+1\right)dx \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \int \left(y^{-2} - \frac{1}{y}\right)dy &= \int \left(-x^{-2}-1\right)dx \\ -\frac{1}{y} - \log|y| &= \frac{1}{x} - x + C_1 \\ \frac{1}{x} + \frac{1}{y} + \log|y| - x = C \end{aligned}$$

(ii) Separating variables:

$$\begin{aligned} \frac{y}{y+1}dy &= \frac{e^x}{e^x+1}dx \\ \left(1 - \frac{1}{y+1}\right)dy &= \frac{e^x}{e^x+1}dx \end{aligned}$$

Integrating both sides:

$$\begin{aligned} \int \left(1 - \frac{1}{y+1}\right)dy &= \int \frac{e^x}{e^x+1}dx \\ y - \log|y+1| = \log(e^x+1) + C_1 \\ y &= \log|y+1| + \log(e^x+1) + C_1 \\ y &= \log|(y+1)(e^x+1)| + C_1 \end{aligned}$$

Final Answer

(i) $\frac{1}{x} + \frac{1}{y} + \log|y| - x = C$

(ii) $y = \log|(y+1)(e^x+1)| + C$

Question 17

Solve the following differential equations:

(i) $\frac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$

(ii) $(1+e^{2x})dy + (1+y^2)e^x dx = 0$

TYPE 52 Variable Separable Differential Equation with Exponential Factoring and Substitution

Given: Differential equations with exponential power rules.

Concept / Formula Used: $e^{a-b} = e^a e^{-b}$ $\int \frac{du}{1+u^2} = \tan^{-1} u + C$

Solution:

(i) Factoring out e^{-2y} on the RHS:

$$\frac{dy}{dx} = e^{-2y}(e^{3x} + x^2)$$
$$e^{2y}dy = (e^{3x} + x^2)dx$$

Integrating both sides:

$$\int e^{2y}dy = \int (e^{3x} + x^2)dx$$
$$\frac{e^{2y}}{2} = \frac{e^{3x}}{3} + \frac{x^3}{3} + C_1$$

Multiplying throughout by 6:

$$3e^{2y} = 2e^{3x} + 2x^3 + 6C_1$$
$$3e^{2y} - 2e^{3x} - 2x^3 = C$$

(ii) Separating variables:

$$(1 + e^{2x})dy = -(1 + y^2)e^x dx$$
$$\frac{dy}{1+y^2} = -\frac{e^x}{1+e^{2x}}dx$$
$$\frac{dy}{1+y^2} = -\frac{e^x}{1+(e^x)^2}dx$$

Integrating both sides:

$$\int \frac{dy}{1+y^2} = -\int \frac{e^x}{1+(e^x)^2}dx$$

Let $u = e^x \implies du = e^x dx$:

$$\tan^{-1} y = -\tan^{-1} u + C$$
$$\tan^{-1} y = -\tan^{-1}(e^x) + C$$
$$\tan^{-1} y + \tan^{-1}(e^x) = C$$

Final Answer

(i) $3e^{2y} - 2e^{3x} - 2x^3 = C$

(ii) $\tan^{-1} y + \tan^{-1}(e^x) = C$

Question 18

Solve the following differential equations:

(i) $\sqrt{a+x}\dfrac{dy}{dx} + xy = 0$

(ii) $(x-1)dy = 2x^3y\,dx$

TYPE 53 Variable Separable Differential Equation Requiring Radical Substitution or Polynomial Division

Given: Differential equations requiring algebraic transformation for integration.

Concept / Formula Used: $a + x = t^2$

Solution:

(i) Rearranging the terms:

$$\sqrt{a+x}\frac{dy}{dx} = -xy$$
$$\frac{dy}{y} = -\frac{x}{\sqrt{a+x}}dx$$

Integrating both sides:

$$\int \frac{dy}{y} = -\int \frac{x}{\sqrt{a+x}}dx$$

Let $a + x = t^2 \implies x = t^2 - a$ and $dx = 2t\,dt$:

$$\log|y| = -\int \frac{t^2-a}{t}\cdot 2t\,dt$$
$$\log|y| = -2\int (t^2-a)\,dt$$
$$\log|y| = -2\left(\frac{t^3}{3} - at\right) + C$$
$$\log|y| = -2t\left(\frac{t^2-3a}{3}\right) + C$$

Back-substituting $t = \sqrt{a+x}$:

$$\log|y| = -2\sqrt{a+x}\left(\frac{a+x-3a}{3}\right) + C$$
$$\log|y| = \frac{2(2a-x)\sqrt{a+x}}{3} + C$$

(ii) Separating variables:

$$\frac{dy}{y} = \frac{2x^3}{x-1}dx$$

Performing polynomial division on $\frac{2x^3}{x-1}$:

$$\frac{2x^3}{x-1} = 2x^2 + 2x + 2 + \frac{2}{x-1}$$

Integrating both sides:

$$\int \frac{dy}{y} = \int \left(2x^2 + 2x + 2 + \frac{2}{x-1}\right) dx$$
$$\log|y| = \frac{2x^3}{3} + x^2 + 2x + 2\log|x-1| + C$$

Final Answer

(i) $\log|y| = \frac{2(2a-x)\sqrt{a+x}}{3} + C$

(ii) $\log|y| = \frac{2x^3}{3} + x^2 + 2x + 2\log|x-1| + C$

Question 19

Solve the following differential equations:

(i) $x\log x\,dy - y\,dx = 0$

(ii) $\sin^3 x\,dx - \sin y\,dy = 0$

TYPE 54 **Variable Separable Differential Equation with Logarithmic Denominator or Trigonometric Power Integration**

Given: Differential equations containing logarithmic and cubic trigonometric terms.

Concept / Formula Used: $\sin^3 x = (1-\cos^2 x)\sin x \qquad u = \cos x$

Solution:

(i) Separating variables:

$$x\log x\,dy = y\,dx$$
$$\frac{dy}{y} = \frac{dx}{x\log x}$$

Integrating both sides:

$$\int \frac{dy}{y} = \int \frac{1/x}{\log x}dx$$
$$\log|y| = \log|\log x| + \log C$$
$$\log|y| = \log|C\log x|$$
$$y = C\log x$$

(ii) Rearranging the equation:

$$\sin y\,dy = \sin^3 x\,dx$$
$$\sin y\,dy = (1-\cos^2 x)\sin x\,dx$$

Integrating both sides:

$$\int \sin y\,dy = \int (1-\cos^2 x)\sin x\,dx$$

Let $u = \cos x \implies du = -\sin x\,dx$:

$$-\cos y = \int (1-u^2)(-du)$$
$$-\cos y = \int (u^2-1)du$$
$$-\cos y = \frac{u^3}{3} - u + C_1$$
$$\cos y = u - \frac{u^3}{3} - C_1$$

Back-substituting $u = \cos x$:

$$\cos y = \cos x - \frac{\cos^3 x}{3} + C$$

Final Answer

(i) $y = C\log x$

(ii) $\cos y = \cos x - \dfrac{\cos^3 x}{3} + C$

Question 20

Solve the following differential equations:

(i) $e^{2x-3y}dx + e^{2y-3x}dy = 0$

(ii) $\log\left(\dfrac{dy}{dx}\right) = x - y$

TYPE 55 Variable Separable Differential Equation with Logarithmic Derivative or Compound Exponents

Given: Differential equations with logarithmic derivative and exponent laws.

Concept / Formula Used: $\log(A) = B \implies A = e^B \qquad e^{a+b} = e^a \cdot e^b$

Solution:

(i) Expressing exponents separately:

$$\frac{e^{2x}}{e^{3y}}dx + \frac{e^{2y}}{e^{3x}}dy = 0$$

Multiplying throughout by $e^{3x}e^{3y}$:

$$e^{5x}dx + e^{5y}dy = 0$$

Integrating both sides:

$$\int e^{5x}dx + \int e^{5y}dy = 0$$
$$\frac{e^{5x}}{5} + \frac{e^{5y}}{5} = C_1$$
$$e^{5x} + e^{5y} = C$$

(ii) Converting logarithmic form into exponential form:

$$\frac{dy}{dx} = e^{x-y}$$
$$\frac{dy}{dx} = \frac{e^x}{e^y}$$
$$e^y dy = e^x dx$$

Integrating both sides:

$$\int e^y dy = \int e^x dx$$
$$e^y = e^x + C$$
$$e^y - e^x = C$$

Final Answer

(i) $e^{5x} + e^{5y} = C$

(ii) $e^y - e^x = C$

Question 21

If $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$, show that $y\sqrt{1-x^2} + x\sqrt{1-y^2} = A$.

TYPE 56 **Variable Separable Differential Equation Proof Using Compound Angle Trigonometric Identity**

To Prove: $y\sqrt{1-x^2} + x\sqrt{1-y^2} = A$

Concept / Formula Used: $\int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + C$ $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$

Solution:

Given equation:

$$\frac{dy}{dx} = -\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

Separating variables:

$$\frac{dy}{\sqrt{1-y^2}} = -\frac{dx}{\sqrt{1-x^2}}$$

Integrating both sides:

$$\int \frac{dy}{\sqrt{1-y^2}} = -\int \frac{dx}{\sqrt{1-x^2}}$$
$$\sin^{-1} y = -\sin^{-1} x + C$$
$$\sin^{-1} x + \sin^{-1} y = C$$

Taking sine on both sides:

$$\sin(\sin^{-1} x + \sin^{-1} y) = \sin C$$

Using the compound angle identity $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$ where $\alpha = \sin^{-1} x$ and $\beta = \sin^{-1} y$:

$$\sin(\sin^{-1} x)\cos(\sin^{-1} y) + \cos(\sin^{-1} x)\sin(\sin^{-1} y) = \sin C$$
$$x\sqrt{1-y^2} + \sqrt{1-x^2}\,y = A \quad (\text{where } A = \sin C)$$
$$y\sqrt{1-x^2} + x\sqrt{1-y^2} = A$$

Hence Proved

$LHS = RHS. Hence proved.$

Question 22

Solve the differential equation $\frac{dy}{dx} = 1 + x + y^2 + xy^2$, given that $y = 0$ when $x = 0$. (Exemplar)

TYPE 57 **Particular Solution of Variable Separable Differential Equation via Factorization**

Given: Differential equation $\frac{dy}{dx} = 1 + x + y^2 + xy^2$ with initial condition $y(0) = 0$.

Concept / Formula Used: $\int \frac{dy}{1+y^2} = \tan^{-1} y + C$

Solution:

Factoring the RHS:

$$\frac{dy}{dx} = (1+x) + y^2(1+x)$$
$$\frac{dy}{dx} = (1+x)(1+y^2)$$

Separating variables:

$$\frac{dy}{1+y^2} = (1+x)\,dx$$

Integrating both sides:

$$\int \frac{dy}{1+y^2} = \int (1+x)\,dx$$
$$\tan^{-1} y = x + \frac{x^2}{2} + C \quad \text{— (1)}$$

Applying initial condition $x = 0, y = 0$:

$$\tan^{-1}(0) = 0 + \frac{0^2}{2} + C$$
$$0 = C \implies C = 0$$

Substituting $C = 0$ back into (1):

$$\tan^{-1} y = x + \frac{x^2}{2}$$
$$y = \tan\left(x + \frac{x^2}{2}\right)$$

Final Answer

$y = \tan\left(x + \frac{x^2}{2}\right)$

Question 23

Find the particular solution of the differential equation $\frac{dy}{dx} = -4xy^2$, given that $y = 1$ when $x = 0$. (NCERT)

TYPE 58 **Particular Solution of First-Order Variable Separable Differential Equation with Power Terms**

Given: Differential equation $\frac{dy}{dx} = -4xy^2$ with condition $y(0) = 1$.

Concept / Formula Used: $\int y^{-2} dy = -\frac{1}{y} + C$

Solution:

Separating variables:

$$\frac{dy}{y^2} = -4x\,dx$$
$$y^{-2} dy = -4x\,dx$$

Integrating both sides:

$$\int y^{-2}dy = -4\int x\,dx$$
$$-\frac{1}{y} = -4\left(\frac{x^2}{2}\right) + C_1$$
$$-\frac{1}{y} = -2x^2 + C_1$$
$$\frac{1}{y} = 2x^2 + C \quad (\text{where } C = -C_1) \quad — (1)$$

Applying initial condition $x = 0, y = 1$:

$$\frac{1}{1} = 2(0)^2 + C$$
$$1 = C \implies C = 1$$

Substituting $C = 1$ into (1):

$$\frac{1}{y} = 2x^2 + 1$$
$$y = \frac{1}{2x^2 + 1}$$

Final Answer

$y = \dfrac{1}{2x^2 + 1}$

Question 24

Solve the following differential equations:

(i) $2(y+3) - xy\dfrac{dy}{dx} = 0$, given that $y(1) = -2$ (Exemplar)

(ii) $(1+y^2)dx + x\,dy = 0$, given that $y(1) = 1$

(iii) $\dfrac{dy}{dx} = x^2e^{-3y}$, given that $y = 0$ for $x = 0$

(iv) $\dfrac{dy}{dx} = 2y^2$, given that $y(1) = 1$

TYPE 59 Particular Solution of Variable Separable Differential Equation with Initial Boundary Conditions

Given: Differential equations with given initial values.

Concept / Formula Used: $\int \frac{g'(y)}{g(y)}dy = \log|g(y)| + C$

Solution:

(i) Rearranging the equation:

$$\begin{aligned}
xy\frac{dy}{dx} &= 2(y+3)\\
\frac{y}{y+3}dy &= \frac{2}{x}dx\\
\left(1-\frac{3}{y+3}\right)dy &= \frac{2}{x}dx
\end{aligned}$$

Integrating both sides:

$$\begin{aligned}
\int\left(1-\frac{3}{y+3}\right)dy &= 2\int\frac{dx}{x}\\
y-3\log|y+3| &= 2\log|x|+C \quad \text{— (1)}
\end{aligned}$$

Using $x=1, y=-2$:

$$\begin{aligned}
-2-3\log|-2+3| &= 2\log|1|+C\\
-2-3\log(1) &= 2(0)+C\\
-2-0 &= C \implies C=-2
\end{aligned}$$

Substituting $C=-2$ into (1):

$$\begin{aligned}
y-3\log|y+3| &= 2\log|x|-2\\
y+2 &= 2\log|x|+3\log|y+3|\\
y+2 &= \log|x^2(y+3)^3|
\end{aligned}$$

(ii) Separating variables:

$$\begin{aligned}
x\,dy &= -(1+y^2)dx\\
\frac{dy}{1+y^2} &= -\frac{dx}{x}
\end{aligned}$$

Integrating both sides:

$$\begin{aligned}
\int\frac{dy}{1+y^2} &= -\int\frac{dx}{x}\\
\tan^{-1}y &= -\log|x|+C \quad \text{— (2)}
\end{aligned}$$

Using $x=1, y=1$:

$$\begin{aligned}
\tan^{-1}(1) &= -\log|1|+C\\
\frac{\pi}{4} &= 0+C \implies C=\frac{\pi}{4}
\end{aligned}$$

Substituting C into (2):

$$\tan^{-1}y+\log|x| = \frac{\pi}{4}$$

(iii) Separating variables:

$$e^{3y}dy = x^2dx$$

Integrating both sides:

$$\begin{aligned}\int e^{3y}dy &= \int x^2dx \\ \frac{e^{3y}}{3} &= \frac{x^3}{3} + C_1 \\ e^{3y} &= x^3 + C \quad \text{(where } C = 3C_1\text{)} \quad \text{— (3)}\end{aligned}$$

Using $x = 0, y = 0$:

$$\begin{aligned}e^0 &= 0^3 + C \\ 1 &= C \implies C = 1\end{aligned}$$

Substituting $C = 1$ into (3):

$$\begin{aligned}e^{3y} &= x^3 + 1 \\ 3y &= \log(x^3 + 1) \\ y &= \frac{1}{3}\log(x^3 + 1)\end{aligned}$$

(iv) Separating variables:

$$\begin{aligned}\frac{dy}{y^2} &= 2dx \\ y^{-2}dy &= 2dx\end{aligned}$$

Integrating both sides:

$$\begin{aligned}\int y^{-2}dy &= 2\int dx \\ -\frac{1}{y} &= 2x + C \quad \text{— (4)}\end{aligned}$$

Using $x = 1, y = 1$:

$$\begin{aligned}-\frac{1}{1} &= 2(1) + C \\ -1 &= 2 + C \implies C = -3\end{aligned}$$

Substituting $C = -3$ into (4):

$$\begin{aligned}-\frac{1}{y} &= 2x - 3 \\ \frac{1}{y} &= 3 - 2x \\ y &= \frac{1}{3 - 2x}\end{aligned}$$

Final Answer

(i) $y + 2 = \log|x^2(y+3)^3|$

(ii) $\tan^{-1} y + \log|x| = \dfrac{\pi}{4}$

(iii) $y = \dfrac{1}{3}\log(x^3+1)$

(iv) $y = \dfrac{1}{3-2x}$

Question 25

Solve the following differential equations:

(i) $(1+y^2)(1+\log x)dx + x\,dy = 0$, given that when $x = 1, y = 1$

(ii) $(1+x^2)\dfrac{dy}{dx} + (1+y^2) = 0$, given that $y = 1$ when $x = 0$

(iii) $x(1+y^2)dx - y(1+x^2)dy = 0$, given that $y = 0$ when $x = 1$

TYPE 60 **Particular Solution of Variable Separable Differential Equation with Inverse Trig or Logarithmic Integration**

Given: Differential equations with specified boundary values.

Concept / Formula Used: $\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$

Solution:

(i) Separating variables:

$$x\,dy = -(1+y^2)(1+\log x)dx$$
$$\frac{dy}{1+y^2} = -\frac{1+\log x}{x}dx$$

Integrating both sides:

$$\int \frac{dy}{1+y^2} = -\int \frac{1+\log x}{x}dx$$

Let $u = 1 + \log x \implies du = \frac{dx}{x}$:

$$\tan^{-1} y = -\int u\,du$$
$$\tan^{-1} y = -\frac{u^2}{2} + C$$
$$\tan^{-1} y = -\frac{(1+\log x)^2}{2} + C \quad \text{— (1)}$$

Applying $x = 1, y = 1$:

$$\tan^{-1}(1) = -\frac{(1+\log 1)^2}{2} + C$$
$$\frac{\pi}{4} = -\frac{1}{2} + C \implies C = \frac{\pi}{4} + \frac{1}{2}$$

Substituting C into (1):

$$\tan^{-1} y + \frac{(1+\log x)^2}{2} = \frac{\pi}{4} + \frac{1}{2}$$

(ii) Separating variables:

$$(1+x^2)\frac{dy}{dx} = -(1+y^2)$$
$$\frac{dy}{1+y^2} = -\frac{dx}{1+x^2}$$

Integrating both sides:

$$\int \frac{dy}{1+y^2} = -\int \frac{dx}{1+x^2}$$
$$\tan^{-1} y = -\tan^{-1} x + C$$
$$\tan^{-1} x + \tan^{-1} y = C \quad \text{— (2)}$$

Applying $x = 0, y = 1$:

$$\tan^{-1}(0) + \tan^{-1}(1) = C$$
$$0 + \frac{\pi}{4} = C \implies C = \frac{\pi}{4}$$

Substituting $C = \frac{\pi}{4}$ into (2):

$$\tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}$$

Taking tangent on both sides:

$$\tan(\tan^{-1} x + \tan^{-1} y) = \tan\frac{\pi}{4}$$
$$\frac{x+y}{1-xy} = 1$$
$$x + y = 1 - xy$$
$$x + y + xy = 1$$

(iii) Separating variables:

$$y(1+x^2)dy = x(1+y^2)dx$$
$$\frac{y}{1+y^2}dy = \frac{x}{1+x^2}dx$$

Multiplying by 2 and integrating:

$$\int \frac{2y}{1+y^2}dy = \int \frac{2x}{1+x^2}dx$$
$$\log(1+y^2) = \log(1+x^2) + \log C$$
$$1 + y^2 = C(1+x^2) \quad \text{— (3)}$$

Applying $x = 1, y = 0$:

$$1 + 0^2 = C(1 + 1^2)$$
$$1 = 2C \implies C = \frac{1}{2}$$

Substituting $C = \frac{1}{2}$ into (3):

$$1 + y^2 = \frac{1}{2}(1 + x^2)$$
$$2 + 2y^2 = 1 + x^2$$
$$x^2 - 2y^2 = 1$$

Final Answer

(i) $\tan^{-1} y + \frac{(1 + \log x)^2}{2} = \frac{\pi}{4} + \frac{1}{2}$
(ii) $x + y + xy = 1$
(iii) $x^2 - 2y^2 = 1$

Question 26

Solve the following differential equations:

(i) $y' = y \tan x$, given that $y = 1$ when $x = 0$
(ii) $\frac{dy}{dx} = y^2 \tan 2x$, given that $y(0) = 2$
(iii) $y' = y \cot 2x$, given that $y\left(\frac{\pi}{4}\right) = 2$

TYPE 61 Particular Solution of Variable Separable Differential Equation with Trigonometric Functions

Given: Trigonometric differential equations with initial values.

Concept / Formula Used: $\int \tan ax\, dx = \frac{1}{a} \log|\sec ax| + C$ $\int \cot ax\, dx = \frac{1}{a} \log|\sin ax| + C$

Solution:

(i) Separating variables:

$$\frac{dy}{y} = \tan x\, dx$$

Integrating both sides:

$$\int \frac{dy}{y} = \int \tan x\, dx$$
$$\log|y| = \log|\sec x| + \log C$$
$$y = C \sec x \quad \text{— (1)}$$

Applying $x = 0, y = 1$:

$$\begin{aligned}
1 &= C\sec(0) \\
1 &= C(1) \implies C = 1
\end{aligned}$$

Substituting $C = 1$ into (1):

$$y = \sec x$$

(ii) Separating variables:

$$\begin{aligned}
\frac{dy}{y^2} &= \tan 2x\, dx \\
y^{-2}dy &= \tan 2x\, dx
\end{aligned}$$

Integrating both sides:

$$\begin{aligned}
\int y^{-2}dy &= \int \tan 2x\, dx \\
-\frac{1}{y} &= \frac{1}{2}\log|\sec 2x| + C \quad \text{— (2)}
\end{aligned}$$

Applying $x = 0, y = 2$:

$$\begin{aligned}
-\frac{1}{2} &= \frac{1}{2}\log|\sec 0| + C \\
-\frac{1}{2} &= \frac{1}{2}(0) + C \implies C = -\frac{1}{2}
\end{aligned}$$

Substituting $C = -\frac{1}{2}$ into (2):

$$\begin{aligned}
-\frac{1}{y} &= \frac{1}{2}\log|\sec 2x| - \frac{1}{2} \\
\frac{1}{y} &= \frac{1}{2}(1 - \log|\sec 2x|) \\
y &= \frac{2}{1 - \log|\sec 2x|}
\end{aligned}$$

(iii) Separating variables:

$$\frac{dy}{y} = \cot 2x\, dx$$

Integrating both sides:

$$\begin{aligned}
\int \frac{dy}{y} &= \int \cot 2x\, dx \\
\log|y| &= \frac{1}{2}\log|\sin 2x| + \log C \\
\log|y| &= \log(C\sqrt{\sin 2x}) \\
y &= C\sqrt{\sin 2x} \quad \text{— (3)}
\end{aligned}$$

Applying $x = \frac{\pi}{4}, y = 2$:

$$2 = C\sqrt{\sin\left(2 \cdot \frac{\pi}{4}\right)}$$
$$2 = C\sqrt{\sin\frac{\pi}{2}}$$
$$2 = C(1) \implies C = 2$$

Substituting $C = 2$ into (3):

$$y = 2\sqrt{\sin 2x}$$

Final Answer

(i) $y = \sec x$

(ii) $y = \dfrac{2}{1 - \log|\sec 2x|}$

(iii) $y = 2\sqrt{\sin 2x}$

Question 27

Find the particular solution of the differential equation $e^x \tan y\, dx + (2 - e^x)\sec^2 y\, dy = 0$, given that $y = \frac{\pi}{4}$ when $x = 0$.

TYPE 62 Particular Solution of Variable Separable Differential Equation with Combined Exponential and Trigonometric

Given: The differential equation $e^x \tan y\, dx + (2 - e^x)\sec^2 y\, dy = 0$ with initial condition $y = \frac{\pi}{4}$ when $x = 0$.

Concept / Formula Used: $\int \frac{f'(x)}{f(x)}\, dx = \log|f(x)| + C$

Solution:

The given differential equation is:

$$e^x \tan y\, dx + (2 - e^x)\sec^2 y\, dy = 0$$

Rearranging the terms to isolate dy on the left-hand side:

$$(2 - e^x)\sec^2 y\, dy = -e^x \tan y\, dx$$

Dividing both sides by $(2 - e^x)\tan y$:

$$\frac{\sec^2 y}{\tan y}\, dy = \frac{-e^x}{2 - e^x}\, dx$$

Multiplying the numerator and denominator on the right-hand side by -1:

$$\frac{\sec^2 y}{\tan y}\, dy = \frac{e^x}{e^x - 2}\, dx$$

Integrating both sides:

$$\int \frac{\sec^2 y}{\tan y}\, dy = \int \frac{e^x}{e^x - 2}\, dx$$

For the left-hand side integral, let $\tan y = u$.

Differentiating both sides with respect to y:

$$\sec^2 y\, dy = du$$

For the right-hand side integral, let $e^x - 2 = v$.

Differentiating both sides with respect to x:

$$e^x\, dx = dv$$

Substituting these into the integrated expression:

$$\int \frac{du}{u} = \int \frac{dv}{v}$$

Using the standard integration rule $\int \frac{1}{t}\, dt = \log|t| + C$:

$$\log|u| = \log|v| + \log C$$

Applying the logarithmic property $\log A + \log B = \log(AB)$:

$$\log|u| = \log|Cv|$$

Taking antilogarithm on both sides:

$$u = Cv$$

Back-substituting $u = \tan y$ and $v = e^x - 2$:

$$\tan y = C(e^x - 2)$$

Rewriting $C(e^x - 2)$ as $C'(2 - e^x)$ where $C' = -C$:

$$\tan y = C'(2 - e^x)$$

Now, apply the given boundary condition $y = \frac{\pi}{4}$ when $x = 0$:

$$\tan\left(\frac{\pi}{4}\right) = C'(2 - e^0)$$

Since $\tan\left(\frac{\pi}{4}\right) = 1$ and $e^0 = 1$:

$$1 = C'(2 - 1)$$
$$1 = C' \cdot 1$$
$$C' = 1$$

Substituting $C' = 1$ back into the general solution:

$$\tan y = 1 \cdot (2 - e^x)$$
$$\tan y = 2 - e^x$$

Final Answer

$\tan y = 2 - e^x$

Common Mistake: A common slip is taking $e^0 = 0$ instead of 1 during evaluation of the integration constant, or mismanaging the negative sign when transferring terms across sides.

9.5 – Differential Equations Reducible to Variable Separable Form

Question 1 (i)

Solve the differential equation:

$$(x+y)^2\frac{dy}{dx} = 1$$

TYPE 63 **Differential Equation Solved by Linear Substitution v = x + y with Rational Algebraic Integration**

Given: $(x+y)^2\frac{dy}{dx} = 1$

Concept / Formula Used: $x + y = v$

Solution:

Let $x + y = v$.

Differentiating both sides with respect to x:

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$
$$\frac{dy}{dx} = \frac{dv}{dx} - 1$$

Substituting these into the given differential equation:

$$v^2\left(\frac{dv}{dx} - 1\right) = 1$$
$$\frac{dv}{dx} - 1 = \frac{1}{v^2}$$
$$\frac{dv}{dx} = 1 + \frac{1}{v^2}$$
$$\frac{dv}{dx} = \frac{v^2+1}{v^2}$$

Separating the variables:

$$\frac{v^2}{v^2+1}\,dv = dx$$
$$\frac{(v^2+1)-1}{v^2+1}\,dv = dx$$
$$\left(1 - \frac{1}{v^2+1}\right)dv = dx$$

Integrating both sides:

$$\int \left(1 - \frac{1}{v^2+1}\right) dv = \int dx$$

Using the standard result $\int \frac{1}{v^2+1}\, dv = \tan^{-1} v + C$:

$$v - \tan^{-1} v = x + C$$

Now back-substitute $v = x + y$:

$$(x+y) - \tan^{-1}(x+y) = x + C$$
$$y - \tan^{-1}(x+y) = C$$

Final Answer

$y - \tan^{-1}(x+y) = C$

Common Mistake: Forgetting to differentiate the substitution equation with respect to x to express $\frac{dy}{dx}$ in terms of $\frac{dv}{dx}$ leads to incorrect variable separation.

Question 1 (ii)

Solve the differential equation:

$$\frac{dy}{dx} = (4x + y + 1)^2$$

TYPE 64 Differential Equation Reducible to Variable Separable Form of Type dy/dx = (ax + by + c)2

Given: $\frac{dy}{dx} = (4x + y + 1)^2$

Concept / Formula Used: $\frac{dy}{dx} = f(ax + by + c) \qquad ax + by + c = v$

Solution:

Let $4x + y + 1 = v$.

Differentiating both sides with respect to x:

$$4 + \frac{dy}{dx} + 0 = \frac{dv}{dx}$$
$$\frac{dy}{dx} = \frac{dv}{dx} - 4$$

Substituting into the differential equation:

$$\frac{dv}{dx} - 4 = v^2$$
$$\frac{dv}{dx} = v^2 + 4$$

Separating the variables:

$$\frac{dv}{v^2+4} = dx$$

Integrating both sides:

$$\int \frac{dv}{v^2+2^2} = \int dx$$

Using the standard result $\int \frac{dv}{v^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{v}{a}\right) + C_1$:

$$\frac{1}{2}\tan^{-1}\left(\frac{v}{2}\right) = x + C_1$$

$$\tan^{-1}\left(\frac{v}{2}\right) = 2x + 2C_1$$

Let $c = 2C_1$:

$$\tan^{-1}\left(\frac{v}{2}\right) = 2x + c$$

$$\frac{v}{2} = \tan(2x+c)$$

$$v = 2\tan(2x+c)$$

Now back-substitute $v = 4x + y + 1$:

$$4x + y + 1 = 2\tan(2x+c)$$

Final Answer

$4x + y + 1 = 2\tan(2x+c)$

Question 2 (i)

Solve the differential equation:

$$(x-y)^2\frac{dy}{dx} = a^2$$

TYPE 65 **Differential Equation Solved by Linear Substitution v = x - y with Logarithmic Rational Integration**

Given: $(x-y)^2\frac{dy}{dx} = a^2$

Concept / Formula Used: $\int \frac{dv}{v^2-a^2} = \frac{1}{2a}\log\left|\frac{v-a}{v+a}\right| + C$

Solution:

Let $x - y = v$.

Differentiating both sides with respect to x:

$$1 - \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dy}{dx} = 1 - \frac{dv}{dx}$$

Substituting into the differential equation:

$$v^2\left(1-\frac{dv}{dx}\right)=a^2$$

$$1-\frac{dv}{dx}=\frac{a^2}{v^2}$$

$$\frac{dv}{dx}=1-\frac{a^2}{v^2}$$

$$\frac{dv}{dx}=\frac{v^2-a^2}{v^2}$$

Separating variables:

$$\frac{v^2}{v^2-a^2}\,dv=dx$$

$$\frac{(v^2-a^2)+a^2}{v^2-a^2}\,dv=dx$$

$$\left(1+\frac{a^2}{v^2-a^2}\right)dv=dx$$

Integrating both sides:

$$\int\left(1+\frac{a^2}{v^2-a^2}\right)dv=\int dx$$

$$\int 1\,dv+a^2\int\frac{dv}{v^2-a^2}=\int dx$$

Using the standard result $\int\frac{dv}{v^2-a^2}=\frac{1}{2a}\log\left|\frac{v-a}{v+a}\right|+C_1$:

$$v+a^2\cdot\frac{1}{2a}\log\left|\frac{v-a}{v+a}\right|=x+C_1$$

$$v+\frac{a}{2}\log\left|\frac{v-a}{v+a}\right|=x+C_1$$

Now back-substitute $v=x-y$:

$$(x-y)+\frac{a}{2}\log\left|\frac{x-y-a}{x-y+a}\right|=x+C_1$$

$$-y+\frac{a}{2}\log\left|\frac{x-y-a}{x-y+a}\right|=C_1$$

$$y=\frac{a}{2}\log\left|\frac{x-y-a}{x-y+a}\right|-C_1$$

Setting $C=-C_1$:

$$y=\frac{a}{2}\log\left|\frac{x-y-a}{x-y+a}\right|+C$$

Final Answer

$y=\frac{a}{2}\log\left|\frac{x-y-a}{x-y+a}\right|+C$

Question 2 (ii)

Solve the differential equation:

$$\frac{dy}{dx} = \tan^2(x + y)$$

TYPE 66 **Differential Equation Reducible to Variable Separable Form with Tangent Function of Linear Combination**

Given: $\frac{dy}{dx} = \tan^2(x + y)$

Concept / Formula Used: $1 + \tan^2 v = \sec^2 v \qquad \cos^2 v = \frac{1+\cos 2v}{2}$

Solution:

Let $x + y = v$.

Differentiating both sides with respect to x:

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dy}{dx} = \frac{dv}{dx} - 1$$

Substituting into the differential equation:

$$\frac{dv}{dx} - 1 = \tan^2 v$$

$$\frac{dv}{dx} = 1 + \tan^2 v$$

Using identity $1 + \tan^2 v = \sec^2 v$:

$$\frac{dv}{dx} = \sec^2 v$$

Separating variables:

$$\frac{dv}{\sec^2 v} = dx$$

$$\cos^2 v\, dv = dx$$

Using identity $\cos^2 v = \frac{1+\cos 2v}{2}$:

$$\left(\frac{1 + \cos 2v}{2}\right) dv = dx$$

$$(1 + \cos 2v)\, dv = 2\, dx$$

Integrating both sides:

$$\int (1 + \cos 2v)\, dv = 2 \int dx$$

$$v + \frac{\sin 2v}{2} = 2x + A$$

Now back-substitute $v = x + y$:

$$(x+y) + \frac{1}{2}\sin 2(x+y) = 2x + A$$
$$y - x + \frac{1}{2}\sin 2(x+y) = A$$
$$2(y-x) + \sin 2(x+y) = 2A$$

Renaming constant $2A$ as A:

$$2(y-x) + \sin 2(x+y) = A$$

Final Answer

$2(y-x) + \sin 2(x+y) = A$

Question 3 (i)

Solve the differential equation:

$$\frac{dy}{dx} = (3x+y+4)^2$$

TYPE 64 **Differential Equation Reducible to Variable Separable Form of Type dy/dx = (ax + by + c)2**

Given: $\frac{dy}{dx} = (3x+y+4)^2$

Concept / Formula Used: $\int \frac{dv}{v^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{v}{a}\right) + C$

Solution:

Let $3x + y + 4 = v$.

Differentiating both sides with respect to x:

$$3 + \frac{dy}{dx} + 0 = \frac{dv}{dx}$$
$$\frac{dy}{dx} = \frac{dv}{dx} - 3$$

Substituting into the differential equation:

$$\frac{dv}{dx} - 3 = v^2$$
$$\frac{dv}{dx} = v^2 + 3$$

Separating variables:

$$\frac{dv}{v^2+3} = dx$$
$$\frac{dv}{v^2+(\sqrt{3})^2} = dx$$

Integrating both sides:

$$\int \frac{dv}{v^2 + (\sqrt{3})^2} = \int dx$$

Using the standard result $\int \frac{dv}{v^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{v}{a}\right) + C_1$:

$$\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{v}{\sqrt{3}}\right) = x + c$$

$$\tan^{-1}\left(\frac{v}{\sqrt{3}}\right) = \sqrt{3}(x + c)$$

$$\frac{v}{\sqrt{3}} = \tan\left(\sqrt{3}(x + c)\right)$$

$$v = \sqrt{3}\tan\left(\sqrt{3}(x + c)\right)$$

Now back-substitute $v = 3x + y + 4$:

$$3x + y + 4 = \sqrt{3}\tan\left(\sqrt{3}(x + c)\right)$$

Final Answer

$3x + y + 4 = \sqrt{3}\tan\left(\sqrt{3}(x + c)\right)$

Question 3 (ii)

Solve the differential equation:

$$\cos(x + y)\, dy = dx$$

TYPE 67 Differential Equation Reducible to Variable Separable Form with Secant Function of Linear Combination

Given: $\cos(x + y)\, dy = dx$

Concept / Formula Used: $1 + \cos v = 2\cos^2\left(\frac{v}{2}\right)$

Solution:

Rearranging the given equation:

$$\frac{dy}{dx} = \frac{1}{\cos(x + y)} = \sec(x + y)$$

Let $x + y = v$.

Differentiating both sides with respect to x:

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dy}{dx} = \frac{dv}{dx} - 1$$

Substituting into the differential equation:

$$\frac{dv}{dx} - 1 = \sec v$$
$$\frac{dv}{dx} = 1 + \sec v$$
$$\frac{dv}{dx} = 1 + \frac{1}{\cos v}$$
$$\frac{dv}{dx} = \frac{1 + \cos v}{\cos v}$$

Separating variables:

$$\frac{\cos v}{1 + \cos v}\, dv = dx$$
$$\frac{(1 + \cos v) - 1}{1 + \cos v}\, dv = dx$$
$$\left(1 - \frac{1}{1 + \cos v}\right) dv = dx$$

Using identity $1 + \cos v = 2\cos^2\left(\frac{v}{2}\right)$:

$$\left(1 - \frac{1}{2\cos^2\left(\frac{v}{2}\right)}\right) dv = dx$$
$$\left(1 - \frac{1}{2}\sec^2\left(\frac{v}{2}\right)\right) dv = dx$$

Integrating both sides:

$$\int \left(1 - \frac{1}{2}\sec^2\left(\frac{v}{2}\right)\right) dv = \int dx$$
$$v - \frac{1}{2} \cdot \frac{\tan\left(\frac{v}{2}\right)}{\frac{1}{2}} = x + C$$
$$v - \tan\left(\frac{v}{2}\right) = x + C$$

Now back-substitute $v = x + y$:

$$(x + y) - \tan\left(\frac{x + y}{2}\right) = x + C$$
$$y - \tan\left(\frac{x + y}{2}\right) = C$$
$$y = \tan\left(\frac{x + y}{2}\right) + C$$

Final Answer

$y = \tan\left(\frac{x+y}{2}\right) + C$

Question 4 (i)

Solve the differential equation:

$$\frac{dy}{dx} = \cos(x+y)$$

TYPE 68 **Differential Equation Reducible to Variable Separable Form with Cosine Function of Linear Combination**

Given: $\frac{dy}{dx} = \cos(x+y)$

Concept / Formula Used: $1+\cos v = 2\cos^2\left(\frac{v}{2}\right)$ $\int \sec^2(kx)\,dx = \frac{1}{k}\tan(kx)+C$

Solution:

Let $x+y=v$.

Differentiating both sides with respect to x:

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$
$$\frac{dy}{dx} = \frac{dv}{dx} - 1$$

Substituting into the differential equation:

$$\frac{dv}{dx} - 1 = \cos v$$
$$\frac{dv}{dx} = 1 + \cos v$$

Separating variables:

$$\frac{dv}{1+\cos v} = dx$$

Using identity $1+\cos v = 2\cos^2\left(\frac{v}{2}\right)$:

$$\frac{dv}{2\cos^2\left(\frac{v}{2}\right)} = dx$$
$$\frac{1}{2}\sec^2\left(\frac{v}{2}\right)dv = dx$$

Integrating both sides:

$$\frac{1}{2}\int \sec^2\left(\frac{v}{2}\right)dv = \int dx$$

Using standard result $\int \sec^2\left(\frac{v}{2}\right)dv = 2\tan\left(\frac{v}{2}\right)$:

$$\frac{1}{2}\cdot 2\tan\left(\frac{v}{2}\right) = x + C$$
$$\tan\left(\frac{v}{2}\right) = x + C$$

Now back-substitute $v = x + y$:

$$\tan\left(\frac{x+y}{2}\right) = x + C$$

Final Answer

$\tan\left(\frac{x+y}{2}\right) = x + C$

Question 4 (ii)

Solve the differential equation:

$$\cos^2(x-2y) = 1 - 2\frac{dy}{dx}$$

TYPE 69 **Differential Equation Solved by Derivative Substitution v = x - 2y**

Given: $\cos^2(x-2y) = 1 - 2\frac{dy}{dx}$

Concept / Formula Used: $\int \sec^2 v\, dv = \tan v + C$

Solution:

Let $x - 2y = v$.

Differentiating both sides with respect to x:

$$1 - 2\frac{dy}{dx} = \frac{dv}{dx}$$

Substituting this directly into the differential equation:

$$\cos^2 v = \frac{dv}{dx}$$

Separating variables:

$$\frac{dv}{\cos^2 v} = dx$$

$$\sec^2 v\, dv = dx$$

Integrating both sides:

$$\int \sec^2 v\, dv = \int dx$$

Using standard result $\int \sec^2 v\, dv = \tan v$:

$$\tan v = x + C'$$

Now back-substitute $v = x - 2y$:

$$\tan(x-2y) = x + C'$$

$$x = \tan(x-2y) - C'$$

Setting $C = -C'$:

$$x = \tan(x - 2y) + C$$

Final Answer

$x = \tan(x - 2y) + C$

Question 5 (i)

Solve the differential equation:

$$(x + y + 1)\frac{dy}{dx} = 1$$

TYPE 70 **Differential Equation Solved by Linear Substitution v = x + y + 1 with Rational Fraction Integration**

Given: $(x + y + 1)\frac{dy}{dx} = 1$

Concept / Formula Used: $\int \frac{1}{v+1}\, dv = \log|v + 1| + C$

Solution:

Rearranging the given equation:

$$\frac{dy}{dx} = \frac{1}{x + y + 1}$$

Let $x + y + 1 = v$.

Differentiating both sides with respect to x:

$$1 + \frac{dy}{dx} + 0 = \frac{dv}{dx}$$

$$\frac{dy}{dx} = \frac{dv}{dx} - 1$$

Substituting into the equation:

$$\frac{dv}{dx} - 1 = \frac{1}{v}$$

$$\frac{dv}{dx} = 1 + \frac{1}{v}$$

$$\frac{dv}{dx} = \frac{v + 1}{v}$$

Separating variables:

$$\frac{v}{v + 1}\, dv = dx$$

$$\frac{(v + 1) - 1}{v + 1}\, dv = dx$$

$$\left(1 - \frac{1}{v + 1}\right) dv = dx$$

Integrating both sides:

$$\int \left(1 - \frac{1}{v+1}\right) dv = \int dx$$

$$v - \log|v+1| = x + C_1$$

Now back-substitute $v = x + y + 1$:

$$(x+y+1) - \log|(x+y+1)+1| = x + C_1$$
$$x + y + 1 - \log|x+y+2| = x + C_1$$
$$y - \log|x+y+2| = C_1 - 1$$

Setting $C = C_1 - 1$:

$$y - \log|x+y+2| = C$$

Final Answer

$y - \log|x+y+2| = C$

Question 5 (ii)

Solve the differential equation:

$$\frac{dy}{dx} = \frac{x+y-1}{x+y+1}$$

TYPE 71 **Differential Equation with Rational Function of Linear Combination Solved by Substitution**

Given: $\frac{dy}{dx} = \frac{x+y-1}{x+y+1}$

Concept / Formula Used: $\int \frac{1}{v}\, dv = \log|v| + C$

Solution:

Let $x + y = v$.

Differentiating both sides with respect to x:

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$
$$\frac{dy}{dx} = \frac{dv}{dx} - 1$$

Substituting into the differential equation:

$$\frac{dv}{dx} - 1 = \frac{v-1}{v+1}$$
$$\frac{dv}{dx} = \frac{v-1}{v+1} + 1$$
$$\frac{dv}{dx} = \frac{(v-1)+(v+1)}{v+1}$$
$$\frac{dv}{dx} = \frac{2v}{v+1}$$

Separating variables:

$$\frac{v+1}{2v}\,dv = dx$$
$$\left(\frac{v}{2v} + \frac{1}{2v}\right) dv = dx$$
$$\left(\frac{1}{2} + \frac{1}{2v}\right) dv = dx$$

Multiplying by 2:

$$\left(1 + \frac{1}{v}\right) dv = 2\,dx$$

Integrating both sides:

$$\int \left(1 + \frac{1}{v}\right) dv = 2\int dx$$
$$v + \log|v| = 2x + C$$

Now back-substitute $v = x + y$:

$$(x+y) + \log|x+y| = 2x + C$$
$$\log|x+y| + y - x = C$$

Final Answer

$\log|x+y| + y - x = C$

Question 6

Find a particular solution of the differential equation $(x+y+1)^2 dy = dx$, given that $y = 0$ when $x = -1$.

TYPE 72 **Particular Solution of Differential Equation Reducible to Variable Separable Form via Linear Substitution**

Given: $(x+y+1)^2 dy = dx$ with initial condition $y = 0$ when $x = -1$

Solution:

Rearranging the differential equation:

$$\frac{dy}{dx} = \frac{1}{(x+y+1)^2}$$

Let $x + y + 1 = v$.

Differentiating both sides with respect to x:

$$\begin{aligned} 1 + \frac{dy}{dx} + 0 &= \frac{dv}{dx} \\ \frac{dy}{dx} &= \frac{dv}{dx} - 1 \end{aligned}$$

Substituting into the differential equation:

$$\begin{aligned} \frac{dv}{dx} - 1 &= \frac{1}{v^2} \\ \frac{dv}{dx} &= 1 + \frac{1}{v^2} \\ \frac{dv}{dx} &= \frac{v^2+1}{v^2} \end{aligned}$$

Separating variables:

$$\begin{aligned} \frac{v^2}{v^2+1}\,dv &= dx \\ \frac{(v^2+1)-1}{v^2+1}\,dv &= dx \\ \left(1 - \frac{1}{v^2+1}\right) dv &= dx \end{aligned}$$

Integrating both sides:

$$\int \left(1 - \frac{1}{v^2+1}\right) dv = \int dx$$

Using standard result $\int \frac{1}{v^2+1}\,dv = \tan^{-1} v$:

$$v - \tan^{-1} v = x + C$$

Back-substitute $v = x + y + 1$:

$$\begin{aligned} (x+y+1) - \tan^{-1}(x+y+1) &= x + C \\ y + 1 - \tan^{-1}(x+y+1) &= C \end{aligned}$$

Applying the initial condition $y = 0$ when $x = -1$:

$$\begin{aligned} 0 + 1 - \tan^{-1}(-1+0+1) &= C \\ 1 - \tan^{-1}(0) &= C \end{aligned}$$

Since $\tan^{-1}(0) = 0$:

$$1 - 0 = C$$
$$C = 1$$

Substitute $C = 1$ back into the general solution:

$$y + 1 - \tan^{-1}(x + y + 1) = 1$$
$$y - \tan^{-1}(x + y + 1) = 0$$
$$\tan^{-1}(x + y + 1) = y$$
$$x + y + 1 = \tan y$$

Final Answer

$x + y + 1 = \tan y$

Common Mistake: Evaluating $\tan^{-1}(0)$ incorrectly or forgetting to substitute back $v = x + y + 1$ before finding the constant C leads to incorrect particular solutions.

9.6 – Homogeneous Differential Equations

Question 1

Which of the following differential equations are homogeneous?

(i) $(x - y)\dfrac{dy}{dx} = x + 2y$

(ii) $y - x\dfrac{dy}{dx} = x + y\dfrac{dy}{dx}$

TYPE 73 **Homogeneous Differential Equations Verification**

Given: Given differential equations (i) $(x-y)\dfrac{dy}{dx} = x+2y$ and (ii) $y-x\dfrac{dy}{dx} = x+y\dfrac{dy}{dx}$.

Concept / Formula Used: $\dfrac{dy}{dx} = f(x, y) \qquad f(\lambda x, \lambda y) = \lambda^0 f(x, y) = f(x, y)$

Solution:

(i) Expressing the given differential equation for $\dfrac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{x + 2y}{x - y} = f(x, y)$$

Replacing x with λx and y with λy:

$$\begin{aligned} f(\lambda x, \lambda y) &= \frac{\lambda x + 2\lambda y}{\lambda x - \lambda y} \\ &= \frac{\lambda(x+2y)}{\lambda(x-y)} \\ &= \frac{x+2y}{x-y} \\ &= f(x,y) \end{aligned}$$

Therefore, equation (i) is a homogeneous differential equation of degree 0.

(ii) Rearranging the given equation:

$$\begin{aligned} y - x &= x\frac{dy}{dx} + y\frac{dy}{dx} \\ y - x &= (x+y)\frac{dy}{dx} \\ \frac{dy}{dx} &= \frac{y-x}{x+y} = f(x,y) \end{aligned}$$

Replacing x with λx and y with λy:

$$\begin{aligned} f(\lambda x, \lambda y) &= \frac{\lambda y - \lambda x}{\lambda x + \lambda y} \\ &= \frac{\lambda(y-x)}{\lambda(x+y)} \\ &= \frac{y-x}{x+y} \\ &= f(x,y) \end{aligned}$$

Therefore, equation (ii) is also a homogeneous differential equation of degree 0.

Final Answer

Both (i) and (ii) are homogeneous differential equations.

Question 2

Which of the following differential equations are homogeneous?

$$(x^2 + 3xy - 4y^2)dx - x(x^2 + 2y)dy = 0$$

TYPE 74 **Homogeneity Verification for Non-Homogeneous Polynomial Differential Equation**

Given: Given equation $(x^2 + 3xy - 4y^2)dx - x(x^2 + 2y)dy = 0$.

Concept / Formula Used: $M(x,y) \qquad N(x,y) \qquad Mdx + Ndy = 0$

Solution:

Rearranging the given equation to express $\frac{dy}{dx}$:

$$x(x^2 + 2y)dy = (x^2 + 3xy - 4y^2)dx$$
$$\frac{dy}{dx} = \frac{x^2 + 3xy - 4y^2}{x^3 + 2xy} = f(x, y)$$

Let us check $f(\lambda x, \lambda y)$:

$$\begin{aligned} f(\lambda x, \lambda y) &= \frac{(\lambda x)^2 + 3(\lambda x)(\lambda y) - 4(\lambda y)^2}{(\lambda x)^3 + 2(\lambda x)(\lambda y)} \\ &= \frac{\lambda^2 x^2 + 3\lambda^2 xy - 4\lambda^2 y^2}{\lambda^3 x^3 + 2\lambda^2 xy} \\ &= \frac{\lambda^2(x^2 + 3xy - 4y^2)}{\lambda^2(\lambda x^3 + 2xy)} \\ &= \frac{x^2 + 3xy - 4y^2}{\lambda x^3 + 2xy} \neq f(x, y) \end{aligned}$$

Since the degree of the numerator is 2 while the denominator contains terms of degree 3 and 2, $f(x, y)$ cannot be expressed as a function of degree 0.

Final Answer

The given differential equation is NOT homogeneous.

Question 3

Which of the following differential equations are homogeneous?

(i) $x\frac{dy}{dx} + y = x\tan^{-1}\left(\frac{y}{x}\right)$

(ii) $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) - 3x^2$

TYPE 73 **Homogeneous Differential Equations Verification**

Given: Differential equations (i) $x\frac{dy}{dx} + y = x\tan^{-1}\left(\frac{y}{x}\right)$ and (ii) $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) - 3x^2$.

Concept / Formula Used: $\frac{dy}{dx} = f(x, y)$ $\quad f(x, y) \quad \left(\frac{y}{x}\right)$

Solution:

(i) Rearranging equation (i):

$$x\frac{dy}{dx} = x\tan^{-1}\left(\frac{y}{x}\right) - y$$
$$\frac{dy}{dx} = \tan^{-1}\left(\frac{y}{x}\right) - \frac{y}{x} = f(x, y)$$

Here $f(x, y)$ is purely a function of $\left(\frac{y}{x}\right)$. Thus, $f(\lambda x, \lambda y) = f(x, y)$. Hence, equation (i) is a homogeneous differential equation.

(ii) Rearranging equation (ii):

$$\frac{dy}{dx} = \frac{y\cos\left(\frac{y}{x}\right) - 3x^2}{x\cos\left(\frac{y}{x}\right)}$$
$$= \frac{y}{x} - \frac{3x}{\cos\left(\frac{y}{x}\right)} = f(x, y)$$

Testing with $f(\lambda x, \lambda y)$:

$$f(\lambda x, \lambda y) = \frac{\lambda y}{\lambda x} - \frac{3\lambda x}{\cos\left(\frac{\lambda y}{\lambda x}\right)}$$
$$= \frac{y}{x} - \lambda \cdot \frac{3x}{\cos\left(\frac{y}{x}\right)} \neq f(x, y)$$

Hence, equation (ii) is NOT a homogeneous differential equation.

Final Answer

(i) is homogeneous, but (ii) is not homogeneous.

Question 4

Which of the following differential equations are homogeneous?

(i) $x\frac{dy}{dx} - y = \sqrt{x^2 + y^2}$

(ii) $\left(x^2 + y^2 \sin\frac{x}{y}\right)\frac{dy}{dx} = 2xye^{x/y}$

TYPE 73 **Homogeneous Differential Equations Verification**

Given: Differential equations (i) $x\frac{dy}{dx} - y = \sqrt{x^2 + y^2}$ and (ii) $\left(x^2 + y^2 \sin\frac{x}{y}\right)\frac{dy}{dx} = 2xye^{x/y}$.

Concept / Formula Used: $f(\lambda x, \lambda y) = f(x, y)$

Solution:

(i) Rearranging equation (i):

$$x\frac{dy}{dx} = y + \sqrt{x^2 + y^2}$$
$$\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} = \frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2} = f(x, y)$$

Since $f(x, y)$ depends only on $\left(\frac{y}{x}\right)$, it is homogeneous of degree 0.

(ii) Rearranging equation (ii):

$$\frac{dy}{dx} = \frac{2xye^{x/y}}{x^2 + y^2 \sin\left(\frac{x}{y}\right)}$$

Dividing numerator and denominator by y^2:

$$\frac{dy}{dx} = \frac{2\left(\frac{x}{y}\right)e^{x/y}}{\left(\frac{x}{y}\right)^2 + \sin\left(\frac{x}{y}\right)} = f(x, y)$$

Since $f(x, y)$ depends entirely on $\left(\frac{x}{y}\right)$, it is a homogeneous differential equation.

Final Answer

Both (i) and (ii) are homogeneous differential equations.

Question 5

Which of the following differential equations are homogeneous?

(i) $2ye^{x/y}dx + (y - 2xe^{x/y})dy = 0$

(ii) $ydx + x\log\left(\frac{y}{x}\right)dy - 2xdy = 0$

TYPE 73 Homogeneous Differential Equations Verification

Given: Differential equations (i) $2ye^{x/y}dx + (y - 2xe^{x/y})dy = 0$ and (ii) $ydx + x\log\left(\frac{y}{x}\right)dy - 2xdy = 0$.

Concept / Formula Used: $\frac{dx}{dy} = g\left(\frac{x}{y}\right)$

Solution:

(i) Rearranging equation (i) for $\frac{dx}{dy}$:

$$\begin{aligned} 2ye^{x/y}dx &= (2xe^{x/y} - y)dy \\ \frac{dx}{dy} &= \frac{2xe^{x/y} - y}{2ye^{x/y}} \\ &= \frac{x}{y} - \frac{1}{2e^{x/y}} = g(x, y) \end{aligned}$$

$g(x, y)$ is a function purely of $\left(\frac{x}{y}\right)$, so it is homogeneous.

(ii) Rearranging equation (ii) for $\frac{dx}{dy}$:

$$\begin{aligned} ydx &= \left(2x - x\log\left(\frac{y}{x}\right)\right)dy \\ \frac{dx}{dy} &= \frac{2x - x\log\left(\frac{y}{x}\right)}{y} \\ &= \frac{x}{y}\left[2 - \log\left(\frac{y}{x}\right)\right] = g(x,y) \end{aligned}$$

Since $g(x,y)$ depends only on the ratio $\left(\frac{y}{x}\right)$ or $\left(\frac{x}{y}\right)$, it is homogeneous.

Final Answer

Both (i) and (ii) are homogeneous differential equations.

Question 6

Solve the following differential equations:

(i) $\frac{dy}{dx} = \frac{x+y}{x}$

(ii) $\frac{dy}{dx} = \frac{x+y}{x-y}$

TYPE 75 Homogeneous Differential Equation Substitution

Given: Differential equations (i) $\frac{dy}{dx} = \frac{x+y}{x}$ and (ii) $\frac{dy}{dx} = \frac{x+y}{x-y}$.

Concept / Formula Used: $\frac{dy}{dx} = f\left(\frac{y}{x}\right) \qquad y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

Solution:

(i) Rewrite equation (i):

$$\frac{dy}{dx} = 1 + \frac{y}{x}$$

Let $y = vx$. Differentiating with respect to x:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

Substituting into the differential equation:

$$\begin{aligned} v + x\frac{dv}{dx} &= 1 + v \\ x\frac{dv}{dx} &= 1 \\ dv &= \frac{dx}{x} \end{aligned}$$

Integrating both sides:

$$\int dv = \int \frac{dx}{x}$$
$$v = \log|x| + C$$

Back-substituting $v = \frac{y}{x}$:

$$\frac{y}{x} = \log|x| + C$$
$$y = x\log|x| + Cx$$

(ii) Equation (ii):

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

Let $y = vx$. Differentiating with respect to x:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

Substituting into the differential equation:

$$\begin{aligned} v + x\frac{dv}{dx} &= \frac{x+vx}{x-vx} \\ v + x\frac{dv}{dx} &= \frac{1+v}{1-v} \\ x\frac{dv}{dx} &= \frac{1+v}{1-v} - v \\ x\frac{dv}{dx} &= \frac{1+v-v(1-v)}{1-v} \\ x\frac{dv}{dx} &= \frac{1+v^2}{1-v} \end{aligned}$$

Separating variables:

$$\begin{aligned} \frac{1-v}{1+v^2}dv &= \frac{dx}{x} \\ \int \frac{1}{1+v^2}dv - \int \frac{v}{1+v^2}dv &= \int \frac{dx}{x} \end{aligned}$$

Using standard integration results:

$$\tan^{-1} v - \frac{1}{2}\log(1+v^2) = \log|x| + C$$
$$\begin{aligned} \tan^{-1} v &= \log|x| + \frac{1}{2}\log(1+v^2) + C \\ \tan^{-1} v &= \frac{1}{2}\log(x^2) + \frac{1}{2}\log(1+v^2) + C \\ \tan^{-1} v &= \frac{1}{2}\log(x^2(1+v^2)) + C \end{aligned}$$

Back-substituting $v = \dfrac{y}{x}$:

$$\tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\log\left(x^2\left(1+\frac{y^2}{x^2}\right)\right) + C$$
$$\tan^{-1}\left(\frac{y}{x}\right) = \frac{1}{2}\log(x^2+y^2) + C$$

Final Answer

(i) $y = x\log|x| + Cx$

(ii) $\tan^{-1}\left(\dfrac{y}{x}\right) = \dfrac{1}{2}\log(x^2+y^2) + C$

Question 7

Solve the following differential equations:

(i) $(x-y)y' = x+3y$

(ii) $(x+2y)dx - (2x-y)dy = 0$

TYPE 75 **Homogeneous Differential Equation Substitution**

Given: Differential equations (i) $(x-y)y' = x+3y$ and (ii) $(x+2y)dx-(2x-y)dy = 0$.

Concept / Formula Used: $y = vx \implies y' = v + x\dfrac{dv}{dx}$

Solution:

(i) Rearranging equation (i):

$$\frac{dy}{dx} = \frac{x+3y}{x-y}$$

Let $y = vx$. Differentiating with respect to x:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

Substituting:

$$v + x\frac{dv}{dx} = \frac{1+3v}{1-v}$$
$$x\frac{dv}{dx} = \frac{1+3v-v(1-v)}{1-v}$$
$$x\frac{dv}{dx} = \frac{v^2+2v+1}{1-v} = \frac{(v+1)^2}{1-v}$$

Separating variables:

$$\begin{aligned}\frac{1-v}{(v+1)^2}dv &= \frac{dx}{x}\\ \frac{-(v+1)+2}{(v+1)^2}dv &= \frac{dx}{x}\\ \left[\frac{2}{(v+1)^2}-\frac{1}{v+1}\right]dv &= \frac{dx}{x}\end{aligned}$$

Integrating both sides:

$$\begin{aligned}-\frac{2}{v+1}-\log|v+1| &= \log|x|+C\\ -\frac{2}{v+1} &= \log|x(v+1)|+C\end{aligned}$$

Back-substituting $v=\frac{y}{x}$:

$$\begin{aligned}-\frac{2}{\frac{y}{x}+1} &= \log\left|x\left(\frac{y}{x}+1\right)\right|+C\\ -\frac{2x}{x+y} &= \log|x+y|+C\\ \log|x+y| &= -\frac{2x}{x+y}+C'\end{aligned}$$

(ii) Rearranging equation (ii):

$$\begin{aligned}(2x-y)dy &= (x+2y)dx\\ \frac{dy}{dx} &= \frac{x+2y}{2x-y}\end{aligned}$$

Let $y=vx \implies \frac{dy}{dx}=v+x\frac{dv}{dx}$. Substituting:

$$\begin{aligned}v+x\frac{dv}{dx} &= \frac{1+2v}{2-v}\\ x\frac{dv}{dx} &= \frac{1+2v-v(2-v)}{2-v}\\ x\frac{dv}{dx} &= \frac{1+v^2}{2-v}\end{aligned}$$

Separating variables:

$$\begin{aligned}\frac{2-v}{1+v^2}dv &= \frac{dx}{x}\\ 2\int\frac{dv}{1+v^2}-\frac{1}{2}\int\frac{2v\,dv}{1+v^2} &= \int\frac{dx}{x}\\ 2\tan^{-1}v-\frac{1}{2}\log(1+v^2) &= \log|x|+C_1\\ 4\tan^{-1}v-\log(1+v^2) &= 2\log|x|+2C_1\\ 4\tan^{-1}v &= \log(x^2(1+v^2))+C\end{aligned}$$

Back-substituting $v = \frac{y}{x}$:

$$4\tan^{-1}\left(\frac{y}{x}\right) = \log\left(x^2\left(1 + \frac{y^2}{x^2}\right)\right) + C$$
$$4\tan^{-1}\left(\frac{y}{x}\right) = \log(x^2 + y^2) + C$$
$$x^2 + y^2 = C_2 e^{4\tan^{-1}(y/x)}$$

Final Answer

(i) $\log|x+y| = -\frac{2x}{x+y} + C$

(ii) $x^2 + y^2 = Ce^{4\tan^{-1}(y/x)}$

Question 8

Solve the following differential equations:

(i) $(y^2 - 2xy)dx = (x^2 - 2xy)dy$

(ii) $2xydx + (x^2 + 2y^2)dy = 0$

TYPE 75 Homogeneous Differential Equation Substitution

Given: Differential equations (i) $(y^2 - 2xy)dx = (x^2 - 2xy)dy$ and (ii) $2xydx + (x^2 + 2y^2)dy = 0$.

Concept / Formula Used: $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

Solution:

(i) Rearranging equation (i):

$$\frac{dy}{dx} = \frac{y^2 - 2xy}{x^2 - 2xy}$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$v + x\frac{dv}{dx} = \frac{v^2x^2 - 2vx^2}{x^2 - 2vx^2}$$
$$v + x\frac{dv}{dx} = \frac{v^2 - 2v}{1 - 2v}$$
$$x\frac{dv}{dx} = \frac{v^2 - 2v - v(1-2v)}{1 - 2v}$$
$$x\frac{dv}{dx} = \frac{3v^2 - 3v}{1 - 2v}$$

Separating variables:

$$\frac{1-2v}{3v(v-1)}dv = \frac{dx}{x}$$

Decomposing $\frac{1-2v}{v(v-1)}$ using partial fractions:

$$\frac{1-2v}{v(v-1)} = \frac{A}{v} + \frac{B}{v-1}$$
$$1-2v = A(v-1) + Bv$$

Setting $v = 0 \implies A = -1$.
Setting $v = 1 \implies B = -1$. So:

$$\frac{1}{3}\int\left(-\frac{1}{v} - \frac{1}{v-1}\right)dv = \int\frac{dx}{x}$$
$$-\log|v| - \log|v-1| = 3\log|x| + C'$$
$$\log|v(v-1)| + \log|x^3| = -C'$$
$$\log|x^3v(v-1)| = -C'$$
$$x^3v(v-1) = A$$

Back-substituting $v = \frac{y}{x}$:

$$x^3\left(\frac{y}{x}\right)\left(\frac{y}{x} - 1\right) = A$$
$$x^3\left(\frac{y}{x}\right)\left(\frac{y-x}{x}\right) = A$$
$$xy(y-x) = A$$
$$xy^2 - x^2y = A$$

(ii) Rearranging equation (ii):

$$\frac{dy}{dx} = -\frac{2xy}{x^2+2y^2}$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$v + x\frac{dv}{dx} = -\frac{2v}{1+2v^2}$$
$$x\frac{dv}{dx} = -\frac{2v}{1+2v^2} - v = -\frac{3v+2v^3}{1+2v^2}$$

Separating variables:

$$\frac{1+2v^2}{v(3+2v^2)}dv = -\frac{dx}{x}$$

Using partial fractions:

$$\frac{1+2v^2}{v(3+2v^2)} = \frac{1/3}{v} + \frac{4/3v}{3+2v^2}$$

Integrating:

$$\begin{aligned}\frac{1}{3}\log|v| + \frac{1}{3}\log(3+2v^2) &= -\log|x| + C_1\\ \log|v(3+2v^2)| &= -3\log|x| + 3C_1\\ \log|x^3v(3+2v^2)| &= C_2\\ x^3v(3+2v^2) &= A\end{aligned}$$

Back-substituting $v = \frac{y}{x}$:

$$\begin{aligned}x^3\left(\frac{y}{x}\right)\left(3+2\frac{y^2}{x^2}\right) &= A\\ x^2y\left(\frac{3x^2+2y^2}{x^2}\right) &= A\\ 3x^2y+2y^3 &= A\end{aligned}$$

Final Answer

(i) $xy^2 - x^2y = A$

(ii) $3x^2y + 2y^3 = A$

Question 9

Solve the following differential equations:

(i) $(x^2-y^2)dx + 2xydy = 0$ **(ISC 2023)**

(ii) $x^2\frac{dy}{dx} = x^2 + xy + y^2$ **(Exemplar)**

(iii) $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$

TYPE 75 **Homogeneous Differential Equation Substitution**

Given: Differential equations (i) $(x^2-y^2)dx + 2xydy = 0$, (ii) $x^2\frac{dy}{dx} = x^2+xy+y^2$, and (iii) $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$.

Concept / Formula Used: $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

Solution:

(i) Rearranging equation (i):

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$\begin{aligned} v + x\frac{dv}{dx} &= \frac{v^2 - 1}{2v} \\ x\frac{dv}{dx} &= \frac{v^2 - 1 - 2v^2}{2v} = -\frac{1 + v^2}{2v} \end{aligned}$$

Separating variables:

$$\frac{2v}{1 + v^2}dv = -\frac{dx}{x}$$

Integrating both sides:

$$\begin{aligned} \log(1 + v^2) &= -\log|x| + \log|C| \\ \log|x(1 + v^2)| &= \log|C| \\ x(1 + v^2) &= C \end{aligned}$$

Back-substituting $v = \frac{y}{x}$:

$$\begin{aligned} x\left(1 + \frac{y^2}{x^2}\right) &= C \\ \frac{x^2 + y^2}{x} &= C \\ x^2 + y^2 &= Cx \end{aligned}$$

(ii) Equation (ii):

$$\frac{dy}{dx} = 1 + \frac{y}{x} + \left(\frac{y}{x}\right)^2$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$\begin{aligned} v + x\frac{dv}{dx} &= 1 + v + v^2 \\ x\frac{dv}{dx} &= 1 + v^2 \end{aligned}$$

Separating variables:

$$\frac{dv}{1 + v^2} = \frac{dx}{x}$$

Integrating both sides:

$$\tan^{-1} v = \log|x| + C$$

Back-substituting $v = \frac{y}{x}$:

$$\tan^{-1}\left(\frac{y}{x}\right) = \log|x| + C$$

(iii) Equation (iii):

$$\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$v + x\frac{dv}{dx} = \frac{1}{v} + v$$
$$x\frac{dv}{dx} = \frac{1}{v}$$

Separating variables:

$$vdv = \frac{dx}{x}$$

Integrating both sides:

$$\frac{v^2}{2} = \log|x| + C_1$$
$$v^2 = 2\log|x| + C$$

Back-substituting $v = \frac{y}{x}$:

$$\frac{y^2}{x^2} = 2\log|x| + C$$
$$y^2 = 2x^2\log|x| + Cx^2$$

Final Answer

(i) $x^2 + y^2 = Cx$

(ii) $\tan^{-1}\left(\frac{y}{x}\right) = \log|x| + C$

(iii) $y^2 = 2x^2\log|x| + Cx^2$

Question 10

Solve the following differential equations:

(i) $x\frac{dy}{dx} + \frac{y^2}{x} = y$

(ii) $3x^2\frac{dy}{dx} - 3xy = y^2$

TYPE 75 **Homogeneous Differential Equation Substitution**

Given: Differential equations (i) $x\frac{dy}{dx} + \frac{y^2}{x} = y$ and (ii) $3x^2\frac{dy}{dx} - 3xy = y^2$.

Concept / Formula Used: $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

Solution:

(i) Rearranging equation (i):

$$x\frac{dy}{dx} = y - \frac{y^2}{x}$$
$$\frac{dy}{dx} = \frac{y}{x} - \left(\frac{y}{x}\right)^2$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$v + x\frac{dv}{dx} = v - v^2$$
$$x\frac{dv}{dx} = -v^2$$

Separating variables:

$$-\frac{dv}{v^2} = \frac{dx}{x}$$

Integrating both sides:

$$\frac{1}{v} = \log|x| + C'$$
$$\log|x| = \frac{1}{v} - C'$$
$$x = Ae^{1/v}$$

Back-substituting $v = \frac{y}{x}$:

$$x = Ae^{x/y}$$

(ii) Rearranging equation (ii):

$$3x^2\frac{dy}{dx} = 3xy + y^2$$
$$\frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{3x^2}$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$v + x\frac{dv}{dx} = v + \frac{v^2}{3}$$
$$x\frac{dv}{dx} = \frac{v^2}{3}$$

Separating variables:

$$\frac{3}{v^2}dv = \frac{dx}{x}$$

Integrating both sides:

$$-\frac{3}{v} = \log|x| + C_1$$
$$\log|x| = -C_1 - \frac{3}{v}$$
$$x = Ce^{-3/v}$$

Back-substituting $v = \frac{y}{x}$:

$$x = Ce^{-3x/y}$$

This is the required solution in implicit form; it cannot be solved explicitly for y in terms of elementary functions.

Final Answer

(i) $x = Ae^{x/y}$

(ii) $x = Ce^{-3x/y}$

Common Mistake: The draft's final answer $y = xe^{Cx}$ for part (ii) does not follow from the derived integration steps and has been corrected to the implicit form $x = Ce^{-3x/y}$, which is what the separation and back-substitution actually give.

Question 11

Solve the following differential equations:

(i) $\frac{dy}{dx} = \frac{y}{x} + \frac{\sqrt{x^2 - y^2}}{x}, \quad x > 0$

(ii) $x\frac{dy}{dx} - y = 2\sqrt{y^2 - x^2}, \quad x > 0$

TYPE 75 **Homogeneous Differential Equation Substitution**

Given: Differential equations (i) $\frac{dy}{dx} = \frac{y}{x} + \frac{\sqrt{x^2 - y^2}}{x}$ and (ii) $x\frac{dy}{dx} - y = 2\sqrt{y^2 - x^2}$.

Concept / Formula Used: $x > 0$ $\quad \frac{\sqrt{x^2 - y^2}}{x} = \sqrt{1 - \left(\frac{y}{x}\right)^2} \quad y = vx$

Solution:

(i) Rewriting equation (i) for $x > 0$:

$$\frac{dy}{dx} = \frac{y}{x} + \sqrt{1 - \left(\frac{y}{x}\right)^2}$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$
\begin{aligned}
v + x\frac{dv}{dx} &= v + \sqrt{1 - v^2} \\
x\frac{dv}{dx} &= \sqrt{1 - v^2}
\end{aligned}
$$

Separating variables:

$$\frac{dv}{\sqrt{1 - v^2}} = \frac{dx}{x}$$

Integrating both sides:

$$\sin^{-1} v = \log|x| + C$$

Back-substituting $v = \frac{y}{x}$:

$$\sin^{-1}\left(\frac{y}{x}\right) = \log|x| + C$$

(ii) Rearranging equation (ii) for $x > 0$:

$$\frac{dy}{dx} = \frac{y}{x} + 2\sqrt{\frac{y^2}{x^2} - 1}$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$
\begin{aligned}
v + x\frac{dv}{dx} &= v + 2\sqrt{v^2 - 1} \\
x\frac{dv}{dx} &= 2\sqrt{v^2 - 1}
\end{aligned}
$$

Separating variables:

$$\frac{dv}{\sqrt{v^2 - 1}} = 2\frac{dx}{x}$$

Integrating both sides using $\int \frac{dv}{\sqrt{v^2-1}} = \log|v + \sqrt{v^2 - 1}|$:

$$
\begin{aligned}
\log|v + \sqrt{v^2 - 1}| &= 2\log|x| + \log|A| \\
\log|v + \sqrt{v^2 - 1}| &= \log|Ax^2| \\
v + \sqrt{v^2 - 1} &= Ax^2
\end{aligned}
$$

Back-substituting $v = \frac{y}{x}$:

$$
\begin{aligned}
\frac{y}{x} + \sqrt{\frac{y^2}{x^2} - 1} &= Ax^2 \\
\frac{y + \sqrt{y^2 - x^2}}{x} &= Ax^2 \\
y + \sqrt{y^2 - x^2} &= Ax^3
\end{aligned}
$$

Final Answer

(i) $\sin^{-1}\left(\frac{y}{x}\right) = \log|x| + C$

(ii) $y + \sqrt{y^2 - x^2} = Ax^3$

Question 12

Solve the differential equation:

$$\left(\frac{y}{x}\cos\frac{y}{x}\right)dx - \left(\frac{x}{y}\sin\frac{y}{x} + \cos\frac{y}{x}\right)dy = 0$$

TYPE 76 General Solution of Homogeneous Differential Equation with Trigonometric Ratio Terms

Given: Differential equation $\left(\frac{y}{x}\cos\frac{y}{x}\right)dx - \left(\frac{x}{y}\sin\frac{y}{x} + \cos\frac{y}{x}\right)dy = 0.$

Concept / Formula Used: $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

Solution:

Rearranging the equation to solve for $\frac{dy}{dx}$:

$$\left(\frac{x}{y}\sin\frac{y}{x} + \cos\frac{y}{x}\right)dy = \left(\frac{y}{x}\cos\frac{y}{x}\right)dx$$

$$\frac{dy}{dx} = \frac{\frac{y}{x}\cos\frac{y}{x}}{\frac{x}{y}\sin\frac{y}{x} + \cos\frac{y}{x}}$$

Let $y = vx$. Differentiating with respect to x:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

Substituting $y/x = v$ and $x/y = 1/v$:

$$v + x\frac{dv}{dx} = \frac{v\cos v}{\frac{1}{v}\sin v + \cos v}$$

$$v + x\frac{dv}{dx} = \frac{v^2\cos v}{\sin v + v\cos v}$$

$$x\frac{dv}{dx} = \frac{v^2\cos v}{\sin v + v\cos v} - v$$

$$x\frac{dv}{dx} = \frac{v^2\cos v - v(\sin v + v\cos v)}{\sin v + v\cos v}$$

$$x\frac{dv}{dx} = \frac{v^2\cos v - v\sin v - v^2\cos v}{\sin v + v\cos v}$$

$$x\frac{dv}{dx} = \frac{-v\sin v}{\sin v + v\cos v}$$

Separating variables:

$$\frac{\sin v + v\cos v}{v\sin v}dv = -\frac{dx}{x}$$
$$\left(\frac{1}{v} + \frac{\cos v}{\sin v}\right) dv = -\frac{dx}{x}$$

Integrating both sides:

$$\int \frac{1}{v}dv + \int \frac{\cos v}{\sin v}dv = -\int \frac{dx}{x}$$
$$\log|v| + \log|\sin v| = -\log|x| + \log|A|$$
$$\log|v\sin v| + \log|x| = \log|A|$$
$$\log|xv\sin v| = \log|A|$$
$$xv\sin v = A$$

Back-substituting $v = \dfrac{y}{x}$:

$$x\left(\frac{y}{x}\right)\sin\left(\frac{y}{x}\right) = A$$
$$y\sin\left(\frac{y}{x}\right) = A$$

Final Answer

$y\sin\left(\frac{y}{x}\right) = A$

Question 13

Solve the differential equation:

$$\left(1 + e^{y/x}\right) dy + e^{y/x}\left(1 - \frac{y}{x}\right) dx = 0$$

TYPE 77 **General Solution of Homogeneous Differential Equation with Exponential Ratio Terms**

Given: Differential equation $\left(1 + e^{y/x}\right) dy + e^{y/x}\left(1 - \frac{y}{x}\right) dx = 0$.

Concept / Formula Used: $y = vx \implies \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

Solution:

Rearranging the given equation:

$$\left(1 + e^{y/x}\right) dy = -e^{y/x}\left(1 - \frac{y}{x}\right) dx$$
$$\frac{dy}{dx} = \frac{e^{y/x}\left(\frac{y}{x} - 1\right)}{1 + e^{y/x}}$$

Let $y = vx \implies \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$. Substituting into the equation:

$$\begin{aligned} v + x\frac{dv}{dx} &= \frac{e^v(v-1)}{1+e^v} \\ x\frac{dv}{dx} &= \frac{ve^v - e^v}{1+e^v} - v \\ x\frac{dv}{dx} &= \frac{ve^v - e^v - v(1+e^v)}{1+e^v} \\ x\frac{dv}{dx} &= \frac{ve^v - e^v - v - ve^v}{1+e^v} \\ x\frac{dv}{dx} &= -\frac{v+e^v}{1+e^v} \end{aligned}$$

Separating variables:

$$\frac{1+e^v}{v+e^v}dv = -\frac{dx}{x}$$

Integrating both sides:

$$\int \frac{1+e^v}{v+e^v}dv = -\int \frac{dx}{x}$$

Noting that $\dfrac{d}{dv}(v+e^v) = 1+e^v$:

$$\begin{aligned} \log|v+e^v| &= -\log|x| + \log|C| \\ \log|v+e^v| + \log|x| &= \log|C| \\ \log|x(v+e^v)| &= \log|C| \\ x(v+e^v) &= C \end{aligned}$$

Back-substituting $v = \dfrac{y}{x}$:

$$\begin{aligned} x\left(\frac{y}{x} + e^{y/x}\right) &= C \\ y + xe^{y/x} &= C \end{aligned}$$

Final Answer

$y + xe^{y/x} = C$

Question 14

Solve the following differential equation:

$$2ye^{x/y}dx + (y - 2xe^{x/y})dy = 0, \quad \text{given } x = 0 \text{ and } y = 1$$ **(ISC 2024)**

TYPE 78 Particular Solution of Homogeneous Differential Equation in dx/dy Form with Exponential Ratio

Given: Differential equation $2ye^{x/y}dx + (y - 2xe^{x/y})dy = 0$ with initial condition $x = 0$ when $y = 1$.

Concept / Formula Used: $e^{x/y}$ $\quad \frac{dx}{dy} = g\left(\frac{x}{y}\right) \quad x = vy \implies \frac{dx}{dy} = v + y\frac{dv}{dy}$

Solution:

Rearranging the differential equation to solve for $\frac{dx}{dy}$:

$$2ye^{x/y}dx = (2xe^{x/y} - y)dy$$
$$\frac{dx}{dy} = \frac{2xe^{x/y} - y}{2ye^{x/y}}$$

Let $x = vy$. Differentiating with respect to y:

$$\frac{dx}{dy} = v + y\frac{dv}{dy}$$

Substituting into the differential equation:

$$v + y\frac{dv}{dy} = \frac{2vye^{v} - y}{2ye^{v}}$$
$$v + y\frac{dv}{dy} = \frac{2ve^{v} - 1}{2e^{v}}$$
$$v + y\frac{dv}{dy} = v - \frac{1}{2e^{v}}$$
$$y\frac{dv}{dy} = -\frac{1}{2e^{v}}$$

Separating variables:

$$2e^{v}dv = -\frac{dy}{y}$$

Integrating both sides:

$$2\int e^{v}dv = -\int \frac{dy}{y}$$
$$2e^{v} = -\log|y| + C$$

Back-substituting $v = \frac{x}{y}$:

$$2e^{x/y} + \log|y| = C$$

Using the initial condition $x = 0$ when $y = 1$:

$$2e^{0/1} + \log|1| = C$$
$$2(1) + 0 = C$$
$$C = 2$$

Substituting $C = 2$ back into the general solution gives:

$$2e^{x/y} + \log|y| = 2$$

Final Answer

$2e^{x/y} + \log|y| = 2$

Question 15

Solve the differential equation $(x+y)dy+(x-y)dx = 0$, given that $y = 1$ when $x = 1$. **(NCERT)**

TYPE 79 **Particular Solution of Homogeneous Differential Equation Reducible to Inverse Tangent and Logarithmic Integ**

Given: Differential equation $(x + y)dy + (x - y)dx = 0$ with initial condition $y = 1$ when $x = 1$.

Concept / Formula Used: $y = vx \implies \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

Solution:

Rearranging the differential equation:

$$(x + y)dy = -(x - y)dx$$
$$\frac{dy}{dx} = \frac{y - x}{x + y}$$

Let $y = vx \implies \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$. Substituting into the equation:

$$v + x\frac{dv}{dx} = \frac{vx - x}{x + vx}$$
$$v + x\frac{dv}{dx} = \frac{v - 1}{1 + v}$$
$$x\frac{dv}{dx} = \frac{v - 1 - v(1 + v)}{1 + v}$$
$$x\frac{dv}{dx} = \frac{v - 1 - v - v^2}{1 + v} = -\frac{1 + v^2}{1 + v}$$

Separating variables:

$$\frac{1 + v}{1 + v^2}dv = -\frac{dx}{x}$$
$$\int \frac{1}{1 + v^2}dv + \int \frac{v}{1 + v^2}dv = -\int \frac{dx}{x}$$
$$\tan^{-1} v + \frac{1}{2}\log(1 + v^2) = -\log|x| + C_1$$
$$\tan^{-1} v + \frac{1}{2}\log(1 + v^2) + \frac{1}{2}\log(x^2) = C_1$$
$$\tan^{-1} v + \frac{1}{2}\log(x^2(1 + v^2)) = C_1$$

Back-substituting $v = \dfrac{y}{x}$:

$$\tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{2}\log\left(x^2\left(1 + \frac{y^2}{x^2}\right)\right) = C_1$$
$$\tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{2}\log(x^2 + y^2) = C_1$$

Applying initial condition $x = 1, y = 1$:

$$\tan^{-1}(1) + \frac{1}{2}\log(1^2 + 1^2) = C_1$$
$$\frac{\pi}{4} + \frac{1}{2}\log 2 = C_1$$

Substituting C_1 into the general solution:

$$\tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{2}\log(x^2 + y^2) = \frac{\pi}{4} + \frac{1}{2}\log 2$$

Final Answer

$\tan^{-1}\left(\frac{y}{x}\right) + \frac{1}{2}\log(x^2 + y^2) = \frac{\pi}{4} + \frac{1}{2}\log 2$

Question 16

Solve the differential equation $(x^2 - y^2)dx + 2xydy = 0$, given that $y = 1$ when $x = 1$.

TYPE 80 Particular Solution of Homogeneous Differential Equation Reducible to Logarithmic Form

Given: Differential equation $(x^2 - y^2)dx + 2xydy = 0$ with condition $y = 1$ when $x = 1$.

Concept / Formula Used: $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

Solution:

Rearranging the equation:

$$2xydy = -(x^2 - y^2)dx$$
$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$v + x\frac{dv}{dx} = \frac{v^2 - 1}{2v}$$
$$x\frac{dv}{dx} = \frac{v^2 - 1 - 2v^2}{2v} = -\frac{1 + v^2}{2v}$$

Separating variables:

$$\frac{2v}{1 + v^2}dv = -\frac{dx}{x}$$

Integrating both sides:

$$\log(1 + v^2) = -\log|x| + \log|C|$$
$$\log|x(1 + v^2)| = \log|C|$$
$$x(1 + v^2) = C$$

Back-substituting $v = \dfrac{y}{x}$:

$$x\left(1 + \frac{y^2}{x^2}\right) = C$$
$$\frac{x^2 + y^2}{x} = C$$
$$x^2 + y^2 = Cx$$

Applying condition $x = 1, y = 1$:

$$1^2 + 1^2 = C(1)$$
$$C = 2$$

Therefore, the particular solution is:

$$x^2 + y^2 = 2x$$

Final Answer

$x^2 + y^2 = 2x$

Question 17

Find the particular solution of the differential equation:

$$\frac{dy}{dx} = \frac{y^2}{xy - y^2}, \quad \text{given that } y = 1 \text{ when } x = 1$$ **(ISC 2022)**

TYPE 81 **Particular Solution of Homogeneous Differential Equation with Rational Ratio Terms**

Given: Differential equation $\dfrac{dy}{dx} = \dfrac{y^2}{xy - y^2}$ with condition $y = 1$ when $x = 1$.

Concept / Formula Used: $y = vx \implies \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

Solution:

Given:

$$\frac{dy}{dx} = \frac{y^2}{xy - y^2}$$

Let $y = vx$. Differentiating with respect to x:

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

Substituting into the equation:

$$\begin{aligned}
v + x\frac{dv}{dx} &= \frac{v^2x^2}{x \cdot vx - v^2x^2} \\
v + x\frac{dv}{dx} &= \frac{v^2x^2}{vx^2 - v^2x^2} \\
v + x\frac{dv}{dx} &= \frac{v^2}{v - v^2} \\
v + x\frac{dv}{dx} &= \frac{v}{1 - v} \\
x\frac{dv}{dx} &= \frac{v}{1 - v} - v \\
x\frac{dv}{dx} &= \frac{v - v(1 - v)}{1 - v} \\
x\frac{dv}{dx} &= \frac{v^2}{1 - v}
\end{aligned}$$

Separating variables:

$$\begin{aligned}
\frac{1 - v}{v^2}dv &= \frac{dx}{x} \\
\left(\frac{1}{v^2} - \frac{1}{v}\right) dv &= \frac{dx}{x}
\end{aligned}$$

Integrating both sides:

$$\begin{aligned}
-\frac{1}{v} - \log|v| &= \log|x| + C \\
-\frac{1}{v} - \log|v| - \log|x| &= C \\
-\frac{1}{v} - \log|vx| &= C
\end{aligned}$$

Back-substituting $v = \frac{y}{x}$ (so $vx = y$):

$$-\frac{x}{y} - \log|y| = C$$

Using condition $x = 1, y = 1$:

$$\begin{aligned}
-\frac{1}{1} - \log(1) &= C \\
-1 - 0 = C &\implies C = -1
\end{aligned}$$

Substitute $C = -1$ into the general solution:

$$\begin{aligned}
-\frac{x}{y} - \log|y| &= -1 \\
\frac{x}{y} + \log|y| &= 1 \\
\frac{x}{y} &= 1 - \log|y| \\
x &= y\left(1 - \log|y|\right)
\end{aligned}$$

Final Answer

$x = y(1 - \log|y|)$

Common Mistake: The differential equation's denominator was mistyped in the draft as $xy - x^2$; the correct board-question denominator is $xy - y^2$ (verified: with $xy - x^2$, the equation is singular exactly at the given initial point $(1, 1)$, which cannot be right). All subsequent algebra has been re-derived from the corrected equation.

Question 18

Find particular solutions of the following differential equations:

(i) $(x^2 + y^2)dx = 2xydy$, given $y = 0$ when $x = 1$

(ii) $\dfrac{dy}{dx} = \dfrac{xy}{x^2 + y^2}$, given $y = 1$ when $x = 0$

(iii) $2xy + y^2 - 2x^2\dfrac{dy}{dx} = 0$, given $y(1) = 2$

(iv) $(x^2 + 3xy + y^2)dx - x^2dy = 0$, given $y = 0$ when $x = 1$

(v) $\dfrac{dy}{dx} - \dfrac{y}{x} + \csc\left(\dfrac{y}{x}\right) = 0$, given $y = 0$ when $x = 1$

(vi) $x\dfrac{dy}{dx} = y - x\tan\left(\dfrac{y}{x}\right)$, given $y = \dfrac{\pi}{4}$ at $x = 1$

(vii) $x\dfrac{dy}{dx} + x\cos^2\left(\dfrac{y}{x}\right) = y$, given when $x = 1, y = \dfrac{\pi}{4}$

(viii) $xe^{y/x} - y + x\dfrac{dy}{dx} = 0$, given $y(e) = 0$

TYPE 82 **Particular Solution of Homogeneous Differential Equation**

Given: Homogeneous differential equations (i) to (viii) with their respective initial conditions.

Concept / Formula Used: $y = vx \implies \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

Solution:

(i) $\dfrac{dy}{dx} = \dfrac{x^2 + y^2}{2xy}$. Let $y = vx \implies v + x\dfrac{dv}{dx} = \dfrac{1 + v^2}{2v}$.

$$x\frac{dv}{dx} = \frac{1 - v^2}{2v} \implies \frac{2v}{1 - v^2}dv = \frac{dx}{x}$$
$$-\log|1 - v^2| = \log|x| + C_1 \implies x(1 - v^2) = C \implies x^2 - y^2 = Cx$$

Condition $x = 1, y = 0 \implies 1^2 - 0 = C(1) \implies C = 1$. So $x^2 - y^2 = x$.

(ii) $\dfrac{dy}{dx} = \dfrac{xy}{x^2+y^2}$. Let $y = vx \implies v + x\dfrac{dv}{dx} = \dfrac{v}{1+v^2}$.

$$x\frac{dv}{dx} = \frac{-v^3}{1+v^2} \implies \left(\frac{1}{v^3}+\frac{1}{v}\right)dv = -\frac{dx}{x}$$

$$-\frac{1}{2v^2} + \log|v| = -\log|x| + C_1 \implies \log|vx| - \frac{x^2}{2y^2} = C_1$$

$$\log|y| - \frac{x^2}{2y^2} = C_1$$

Condition $x = 0, y = 1 \implies \log(1) - 0 = C_1 \implies C_1 = 0$. So $x^2 = 2y^2 \log|y|$.

(iii) $2x^2\dfrac{dy}{dx} = 2xy + y^2 \implies \dfrac{dy}{dx} = \dfrac{y}{x} + \dfrac{y^2}{2x^2}$. Let $y = vx$.

$$v + x\frac{dv}{dx} = v + \frac{v^2}{2} \implies \frac{2}{v^2}dv = \frac{dx}{x}$$

$$-\frac{2}{v} = \log|x| + C \implies -\frac{2x}{y} = \log|x| + C$$

Condition $x = 1, y = 2 \implies -\frac{2(1)}{2} = \log(1) + C \implies C = -1$. So $-\dfrac{2x}{y} = \log|x| - 1 \implies y = \dfrac{2x}{1-\log|x|}$.

(iv) $\dfrac{dy}{dx} = 1 + 3\left(\dfrac{y}{x}\right) + \left(\dfrac{y}{x}\right)^2$. Let $y = vx$.

$$v + x\frac{dv}{dx} = 1 + 3v + v^2 \implies \frac{dv}{(v+1)^2} = \frac{dx}{x}$$

$$-\frac{1}{v+1} = \log|x| + C \implies -\frac{x}{x+y} = \log|x| + C$$

Condition $x = 1, y = 0 \implies -1 = 0 + C \implies C = -1$. So $x + y = \dfrac{x}{1-\log|x|}$.

(v) $\dfrac{dy}{dx} = \dfrac{y}{x} - \csc\left(\dfrac{y}{x}\right)$. Let $y = vx$.

$$v + x\frac{dv}{dx} = v - \csc v \implies \sin v\,dv = -\frac{dx}{x}$$

$$-\cos v = -\log|x| + C' \implies \cos\left(\frac{y}{x}\right) = \log|x| + C$$

Condition $x = 1, y = 0 \implies \cos(0) = \log(1) + C \implies C = 1$. So $\cos\left(\dfrac{y}{x}\right) = 1 + \log|x|$.

(vi) $\dfrac{dy}{dx} = \dfrac{y}{x} - \tan\left(\dfrac{y}{x}\right)$. Let $y = vx$.

$$v + x\frac{dv}{dx} = v - \tan v \implies \cot v\,dv = -\frac{dx}{x}$$

$$\log|\sin v| = -\log|x| + \log C \implies x\sin\left(\frac{y}{x}\right) = C$$

Condition $x = 1, y = \pi/4 \implies 1\cdot\sin(\pi/4) = C \implies C = \dfrac{1}{\sqrt{2}}$. So $x\sqrt{2}\sin\left(\dfrac{y}{x}\right) = 1$.

(vii) $\dfrac{dy}{dx} = \dfrac{y}{x} - \cos^2\left(\dfrac{y}{x}\right)$. Let $y = vx$.

$$v + x\frac{dv}{dx} = v - \cos^2 v \implies \sec^2 v dv = -\frac{dx}{x}$$
$$\tan v = -\log|x| + C \implies \tan\left(\frac{y}{x}\right) = -\log|x| + C$$

Condition $x = 1, y = \pi/4 \implies \tan(\pi/4) = -\log(1) + C \implies C = 1$. So $\tan\left(\dfrac{y}{x}\right) + \log|x| = 1$.

(viii) $\dfrac{dy}{dx} = \dfrac{y}{x} - e^{y/x}$. Let $y = vx$.

$$v + x\frac{dv}{dx} = v - e^v \implies e^{-v}dv = -\frac{dx}{x}$$
$$-e^{-v} = -\log|x| + C' \implies e^{-y/x} = \log|x| + C$$

Condition $x = e, y = 0 \implies e^0 = \log(e) + C \implies 1 = 1 + C \implies C = 0$. So $e^{-y/x} = \log|x| \implies y = -x\log(\log|x|)$.

Final Answer

(i) $x^2 - y^2 = x$
(ii) $x^2 = 2y^2 \log|y|$
(iii) $y = \dfrac{2x}{1 - \log|x|}$
(iv) $x + y = \dfrac{x}{1 - \log|x|}$
(v) $\cos\left(\dfrac{y}{x}\right) = 1 + \log|x|$
(vi) $x\sqrt{2}\sin\left(\dfrac{y}{x}\right) = 1$
(vii) $\tan\left(\dfrac{y}{x}\right) + \log|x| = 1$
(viii) $y = -x\log(\log|x|)$

Question 19

Find the particular solution of the differential equation $(x - y)\dfrac{dy}{dx} = x + 2y$, given that $y = 0$, when $x = 1$.

TYPE 83 **Particular Solution of Homogeneous Differential Equation Reducible to Quadratic Denominator Integration**

Given: Differential equation $(x - y)\dfrac{dy}{dx} = x + 2y$ with condition $y = 0$ when $x = 1$.

Concept / Formula Used: $y = vx \implies \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

Solution:

Expressing $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{x+2y}{x-y}$$

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$\begin{aligned} v + x\frac{dv}{dx} &= \frac{1+2v}{1-v} \\ x\frac{dv}{dx} &= \frac{1+2v-v(1-v)}{1-v} = \frac{v^2+v+1}{1-v} \end{aligned}$$

Separating variables:

$$\frac{1-v}{v^2+v+1}dv = \frac{dx}{x}$$

Rewriting the numerator as $1 - v = -\frac{1}{2}(2v+1) + \frac{3}{2}$:

$$\begin{aligned} \int \frac{-\frac{1}{2}(2v+1)+\frac{3}{2}}{v^2+v+1}dv &= \int \frac{dx}{x} \\ -\frac{1}{2}\log(v^2+v+1) + \frac{3}{2}\int \frac{dv}{\left(v+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} &= \log|x| + C_1 \\ -\frac{1}{2}\log(v^2+v+1) + \sqrt{3}\tan^{-1}\left(\frac{2v+1}{\sqrt{3}}\right) &= \log|x| + C_1 \end{aligned}$$

Multiplying by 2 throughout:

$$\begin{aligned} 2\sqrt{3}\tan^{-1}\left(\frac{2v+1}{\sqrt{3}}\right) &= 2\log|x| + \log(v^2+v+1) + 2C_1 \\ 2\sqrt{3}\tan^{-1}\left(\frac{2v+1}{\sqrt{3}}\right) &= \log(x^2(v^2+v+1)) + C \end{aligned}$$

Back-substituting $v = \frac{y}{x}$:

$$2\sqrt{3}\tan^{-1}\left(\frac{2y+x}{\sqrt{3}x}\right) = \log(x^2+xy+y^2) + C$$

Using condition $x = 1, y = 0$:

$$\begin{aligned} 2\sqrt{3}\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) &= \log(1) + C \\ 2\sqrt{3}\left(\frac{\pi}{6}\right) &= 0 + C \implies C = \frac{\pi}{\sqrt{3}} \end{aligned}$$

Substituting C:

$$2\sqrt{3}\tan^{-1}\left(\frac{2y+x}{\sqrt{3}x}\right) = \log(x^2+xy+y^2) + \frac{\pi}{\sqrt{3}}$$

Final Answer

$$2\sqrt{3}\tan^{-1}\left(\frac{2y+x}{\sqrt{3}x}\right) = \log(x^2+xy+y^2) + \frac{\pi}{\sqrt{3}}$$

Question 20

Solve the differential equation $xdy - ydx = \sqrt{x^2 + y^2}dx$, given that $y = 0$ when $x = 1$.

TYPE 84 Particular Solution of Homogeneous Differential Equation Reducible to Logarithmic Radical Integrals

Given: Differential equation $xdy - ydx = \sqrt{x^2 + y^2}dx$ with condition $y = 0$ when $x = 1$.

Concept / Formula Used: $y = vx \implies \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

Solution:

Rearranging the given equation for $x > 0$:

$$xdy = \left(y + \sqrt{x^2 + y^2}\right) dx$$
$$\frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} = \frac{y}{x} + \sqrt{1 + \left(\frac{y}{x}\right)^2}$$

Let $y = vx \implies \dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$. Substituting into the equation:

$$v + x\frac{dv}{dx} = v + \sqrt{1 + v^2}$$
$$x\frac{dv}{dx} = \sqrt{1 + v^2}$$

Separating variables:

$$\frac{dv}{\sqrt{1 + v^2}} = \frac{dx}{x}$$

Integrating both sides using $\int \frac{dv}{\sqrt{1+v^2}} = \log|v + \sqrt{1 + v^2}|$:

$$\log|v + \sqrt{1 + v^2}| = \log|x| + \log|C|$$
$$v + \sqrt{1 + v^2} = Cx$$

Back-substituting $v = \dfrac{y}{x}$:

$$\frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} = Cx$$
$$\frac{y + \sqrt{x^2 + y^2}}{x} = Cx$$
$$y + \sqrt{x^2 + y^2} = Cx^2$$

Applying initial condition $x = 1, y = 0$:

$$0 + \sqrt{1^2 + 0} = C(1)^2$$
$$C = 1$$

Therefore:

$$y + \sqrt{x^2 + y^2} = x^2$$

Final Answer

$y + \sqrt{x^2 + y^2} = x^2$

Question 21

Show that the following differential equations are homogeneous and find their particular solutions:

(i) $x\frac{dy}{dx}\sin\left(\frac{y}{x}\right) + x - y\sin\left(\frac{y}{x}\right) = 0$, given $x = 1$ when $y = \frac{\pi}{2}$

(ii) $\left(xe^{y/x} + y\right)dx = xdy$, given $y = 0$ when $x = 1$

TYPE 82 Particular Solution of Homogeneous Differential Equation

To Prove: Show that differential equations (i) and (ii) are homogeneous and evaluate their particular solutions.

Concept / Formula Used: $f(\lambda x, \lambda y) = f(x, y)$ $\quad y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$

Solution:

(i) Rearranging equation (i) for $\frac{dy}{dx}$:

$$x\sin\left(\frac{y}{x}\right)\frac{dy}{dx} = y\sin\left(\frac{y}{x}\right) - x$$

$$\frac{dy}{dx} = \frac{y\sin\left(\frac{y}{x}\right) - x}{x\sin\left(\frac{y}{x}\right)} = \frac{y}{x} - \csc\left(\frac{y}{x}\right) = f(x, y)$$

Since $f(\lambda x, \lambda y) = \frac{\lambda y}{\lambda x} - \csc\left(\frac{\lambda y}{\lambda x}\right) = f(x, y)$, the DE is homogeneous.

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$v + x\frac{dv}{dx} = v - \csc v$$

$$x\frac{dv}{dx} = -\csc v$$

$$\sin v dv = -\frac{dx}{x}$$

Integrating:

$$-\cos v = -\log|x| + C$$

$$\cos\left(\frac{y}{x}\right) = \log|x| + C$$

Given $x = 1$ when $y = \pi/2$:

$$\cos\left(\frac{\pi}{2}\right) = \log(1) + C \implies 0 = 0 + C \implies C = 0$$

Particular solution:

$$\cos\left(\frac{y}{x}\right) = \log|x|$$

(ii) Rearranging equation (ii) for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = \frac{xe^{y/x} + y}{x} = e^{y/x} + \frac{y}{x} = f(x, y)$$

$f(\lambda x, \lambda y) = e^{\lambda y/\lambda x} + \frac{\lambda y}{\lambda x} = f(x, y)$, so it is homogeneous.

Let $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$. Substituting:

$$v + x\frac{dv}{dx} = e^v + v$$
$$x\frac{dv}{dx} = e^v$$
$$e^{-v}dv = \frac{dx}{x}$$

Integrating:

$$-e^{-v} = \log|x| + C_1$$
$$e^{-y/x} = C - \log|x|$$

Given $x = 1, y = 0$:

$$e^0 = C - \log(1) \implies 1 = C - 0 \implies C = 1$$

Particular solution:

$$e^{-y/x} = 1 - \log|x|$$

Hence Proved

Both differential equations are shown to be homogeneous, and their particular solutions are derived as $\cos\left(\frac{y}{x}\right) = \log|x|$ for (i) and $e^{-y/x} = 1 - \log|x|$ for (ii).

9.7 – Linear Differential Equations

Question 1

Which of the following equations are linear differential equations of first order in y?
(i) $\frac{dy}{dx} + 3y = \sin x$
(ii) $x^2\frac{dy}{dx} - 2xy = x^3 + 2x^2 - 5$
(iii) $x \log x\, dy + (3y - 5\log x)\, dx = 0$
(iv) $y\frac{dy}{dx} + 5x = y\cos x$

TYPE 85 **Identification of First-Order Linear Differential Equations in y**

Concept / Formula Used: $\frac{dy}{dx} + P(x)y = Q(x)$ y $P(x)$ $Q(x)$ x

Solution:

Let us analyze each equation:

(i) $\frac{dy}{dx} + 3y = \sin x$
This is already in the standard form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = 3$ and $Q(x) = \sin x$. Thus, it is a linear differential equation of first order in y.

(ii) $x^2\frac{dy}{dx} - 2xy = x^3 + 2x^2 - 5$
Dividing both sides by x^2:

$$\frac{dy}{dx} - \frac{2}{x}y = x + 2 - \frac{5}{x^2}$$

This is in the standard form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = -\frac{2}{x}$ and $Q(x) = x + 2 - \frac{5}{x^2}$. Thus, it is a linear differential equation of first order in y.

(iii) $x \log x\, dy + (3y - 5\log x)\, dx = 0$
Dividing by dx:

$$x \log x \frac{dy}{dx} + 3y - 5\log x = 0$$

$$x \log x \frac{dy}{dx} + 3y = 5\log x$$

Dividing by $x \log x$:

$$\frac{dy}{dx} + \frac{3}{x\log x}y = \frac{5}{x}$$

This is in the standard form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = \frac{3}{x\log x}$ and $Q(x) = \frac{5}{x}$. Thus, it is a linear differential equation of first order in y.

(iv) $y\frac{dy}{dx} + 5x = y\cos x$
Dividing by y:

$$\frac{dy}{dx} + \frac{5x}{y} = \cos x$$

Here, the term $\frac{5x}{y}$ contains y in the denominator, which is non-linear in y. Alternatively, the coefficient of $\frac{dy}{dx}$ is y, which is a function of y, making it non-linear in y. Thus, it is not a linear differential equation of first order in y.

Final Answer

The linear differential equations of first order in y are (i), (ii), and (iii).

Question 2

Which of the following equations are linear differential equations of first order in x?
(i) $\frac{dx}{dy} - \frac{2x}{y} = 3y^3 - 5y + 1$
(ii) $(\cos y)\frac{dx}{dy} - 2x \sin y = \cos^3 y$
(iii) $(1 + y^2)\, dx = (\tan^{-1} y - 2x)\, dy$
(iv) $x^2 \frac{dx}{dy} + 2x^3 y = 3y^2 - 2$

TYPE 86 **Identification of First-Order Linear Differential Equations in x**

Concept / Formula Used: $\frac{dx}{dy} + P(y)x = Q(y)$ x $P(y)$ $Q(y)$ y

Solution:

Let us analyze each equation:

(i) $\frac{dx}{dy} - \frac{2x}{y} = 3y^3 - 5y + 1$
This is in the standard form $\frac{dx}{dy} + P(y)x = Q(y)$, where $P(y) = -\frac{2}{y}$ and $Q(y) = 3y^3 - 5y + 1$. Thus, it is a linear differential equation of first order in x.

(ii) $(\cos y)\frac{dx}{dy} - 2x \sin y = \cos^3 y$
Dividing both sides by $\cos y$:

$$\frac{dx}{dy} - 2x \tan y = \cos^2 y$$

This is in the standard form $\frac{dx}{dy} + P(y)x = Q(y)$, where $P(y) = -2\tan y$ and $Q(y) = \cos^2 y$. Thus, it is a linear differential equation of first order in x.

(iii) $(1 + y^2)\, dx = (\tan^{-1} y - 2x)\, dy$
Dividing by dy:

$$(1 + y^2)\frac{dx}{dy} = \tan^{-1} y - 2x$$

$$(1 + y^2)\frac{dx}{dy} + 2x = \tan^{-1} y$$

Dividing by $1 + y^2$:

$$\frac{dx}{dy} + \frac{2}{1 + y^2}x = \frac{\tan^{-1} y}{1 + y^2}$$

This is in the standard form $\frac{dx}{dy} + P(y)x = Q(y)$, where $P(y) = \frac{2}{1+y^2}$ and $Q(y) = \frac{\tan^{-1} y}{1+y^2}$. Thus, it is a linear differential equation of first order in x.

(iv) $x^2 \frac{dx}{dy} + 2x^3 y = 3y^2 - 2$
Dividing by x^2:

$$\frac{dx}{dy} + 2xy = \frac{3y^2 - 2}{x^2}$$

Here, the term $2xy$ has x to the power 1, but the right-hand side has x^2 in the denominator, which is non-linear in x. Thus, it is not a linear differential equation of first order in x.

Final Answer

The linear differential equations of first order in x are (i), (ii), and (iii).

Question 3

Write an integrating factor of each of the following linear differential equations:
(i) $x\frac{dy}{dx} = 2x^2 + y, x > 0$
(ii) $2x\frac{dy}{dx} + y = 6x^3 + 5, x > 0$
(iii) $\frac{dy}{dx} + 2y\tan x = \sin x$
(iv) $x\frac{dy}{dx} + y - xy\cot x = 0$

TYPE 87 **Integrating Factor Determination for Linear Differential Equations in y**

Concept / Formula Used: I.F. $= e^{\int P(x)\,dx}$ $\quad y \quad$ $\frac{dy}{dx} + P(x)y = Q(x)$

Solution:

(i) $x\frac{dy}{dx} = 2x^2 + y$
Rearranging to standard form:

$$x\frac{dy}{dx} - y = 2x^2$$

Dividing by x:

$$\frac{dy}{dx} - \frac{1}{x}y = 2x$$

Here, $P(x) = -\frac{1}{x}$. The Integrating Factor is:

$$\text{I.F.} = e^{\int -\frac{1}{x}\,dx} = e^{-\log x} = e^{\log(x^{-1})} = \frac{1}{x}$$

(ii) $2x\frac{dy}{dx} + y = 6x^3 + 5$
Dividing by $2x$:

$$\frac{dy}{dx} + \frac{1}{2x}y = 3x^2 + \frac{5}{2x}$$

Here, $P(x) = \frac{1}{2x}$. The Integrating Factor is:

$$\text{I.F.} = e^{\int \frac{1}{2x}\,dx} = e^{\frac{1}{2}\log x} = e^{\log(x^{1/2})} = \sqrt{x}$$

(iii) $\frac{dy}{dx} + 2y\tan x = \sin x$
This is already in standard form with $P(x) = 2\tan x$. The Integrating Factor is:

$$\text{I.F.} = e^{\int 2\tan x\,dx} = e^{2\log|\sec x|} = e^{\log(\sec^2 x)} = \sec^2 x$$

(iv) $x\frac{dy}{dx} + y - xy\cot x = 0$
Rearranging:

$$x\frac{dy}{dx} + (1 - x\cot x)y = 0$$

Dividing by x:

$$\frac{dy}{dx} + \left(\frac{1}{x} - \cot x\right)y = 0$$

Here, $P(x) = \frac{1}{x} - \cot x$. The Integrating Factor is:

$$\text{I.F.} = e^{\int \left(\frac{1}{x} - \cot x\right) dx} = e^{\log x - \log|\sin x|} = e^{\log\left(\frac{x}{\sin x}\right)} = \frac{x}{\sin x} = x \csc x$$

Final Answer

(i) I.F. $= \frac{1}{x}$
(ii) I.F. $= \sqrt{x}$
(iii) I.F. $= \sec^2 x$
(iv) I.F. $= x \csc x$

Question 4

Write an integrating factor of each of the following linear differential equations:

(i) $y\,dx - (x + 2y^2)\,dy = 0, y > 0$

(ii) $ye^y\,dx = (y^3 + 2xe^y)\,dy, y \neq 0$

(iii) $dx + x\,dy = (e^{-y}\log y)\,dy, y > 0$

(iv) $(1 - y^2)\frac{dx}{dy} + yx = ay, |y| < 1$

TYPE 88 **Integrating Factor Determination for Linear Differential Equations in x**

Concept / Formula Used: I.F. $= e^{\int P(y)\,dy}$ x $\frac{dx}{dy} + P(y)x = Q(y)$

Solution:

(i) $y\,dx - (x + 2y^2)\,dy = 0$
Dividing by dy:

$$y\frac{dx}{dy} - (x + 2y^2) = 0$$

$$y\frac{dx}{dy} - x = 2y^2$$

Dividing by y:

$$\frac{dx}{dy} - \frac{1}{y}x = 2y$$

Here, $P(y) = -\frac{1}{y}$. The Integrating Factor is:

$$\text{I.F.} = e^{\int -\frac{1}{y}\,dy} = e^{-\log y} = \frac{1}{y}$$

(ii) $ye^y\,dx = (y^3 + 2xe^y)\,dy$
Dividing by dy:

$$ye^y\frac{dx}{dy} = y^3 + 2xe^y$$

$$ye^y\frac{dx}{dy} - 2e^y x = y^3$$

Dividing by ye^y:

$$\frac{dx}{dy} - \frac{2}{y}x = y^2e^{-y}$$

Here, $P(y) = -\frac{2}{y}$. The Integrating Factor is:

$$\text{I.F.} = e^{\int -\frac{2}{y}\,dy} = e^{-2\log y} = \frac{1}{y^2}$$

(iii) $dx + x\,dy = (e^{-y}\log y)\,dy$
Dividing by dy:

$$\frac{dx}{dy} + x = e^{-y}\log y$$

Here, $P(y) = 1$. The Integrating Factor is:

$$\text{I.F.} = e^{\int 1\,dy} = e^y$$

(iv) $(1 - y^2)\frac{dx}{dy} + yx = ay$
Dividing by $1 - y^2$:

$$\frac{dx}{dy} + \frac{y}{1 - y^2}x = \frac{ay}{1 - y^2}$$

Here, $P(y) = \frac{y}{1-y^2}$. The Integrating Factor is:

$$\text{I.F.} = e^{\int \frac{y}{1-y^2}\,dy}$$

Let $1 - y^2 = u$. Differentiating: $-2y\,dy = du \implies y\,dy = -\frac{1}{2}du$. The integral becomes:

$$\int \frac{y}{1 - y^2}\,dy = \int -\frac{1}{2u}\,du = -\frac{1}{2}\log|u| = \log\left(\frac{1}{\sqrt{1 - y^2}}\right)$$

Thus:

$$\text{I.F.} = e^{\log(1-y^2)^{-1/2}} = \frac{1}{\sqrt{1 - y^2}}$$

Final Answer

(i) I.F. $= \frac{1}{y}$
(ii) I.F. $= \frac{1}{y^2}$
(iii) I.F. $= e^y$
(iv) I.F. $= \frac{1}{\sqrt{1-y^2}}$

Question 5

Solve the following differential equations:

(i) $\frac{dy}{dx} + \frac{y}{x} = x^2$

(ii) $x\frac{dy}{dx} + 2y = x^2$

TYPE 89 **General Solution of First-Order Linear Differential Equation in y with Polynomial Terms**

Concept / Formula Used: $y \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.})\, dx + C \qquad \frac{dy}{dx} + P(x)y = Q(x)$

Solution:

(i) $\frac{dy}{dx} + \frac{y}{x} = x^2$

This is in standard form with $P(x) = \frac{1}{x}$ and $Q(x) = x^2$. Integrating Factor:

$$\text{I.F.} = e^{\int \frac{1}{x}\, dx} = e^{\log x} = x$$

The general solution is:

$$\begin{aligned} y \cdot x &= \int x^2 \cdot x\, dx + C \\ xy &= \int x^3\, dx + C \\ xy &= \frac{x^4}{4} + C \\ y &= \frac{x^3}{4} + \frac{C}{x} \end{aligned}$$

(ii) $x\frac{dy}{dx} + 2y = x^2$

Dividing by x:

$$\frac{dy}{dx} + \frac{2}{x}y = x$$

Here, $P(x) = \frac{2}{x}$ and $Q(x) = x$. Integrating Factor:

$$\text{I.F.} = e^{\int \frac{2}{x}\, dx} = e^{2\log x} = x^2$$

The general solution is:

$$\begin{aligned} y \cdot x^2 &= \int x \cdot x^2\, dx + C \\ yx^2 &= \int x^3\, dx + C \\ yx^2 &= \frac{x^4}{4} + C \\ y &= \frac{x^2}{4} + \frac{C}{x^2} \end{aligned}$$

Final Answer

(i) $y = \frac{x^3}{4} + \frac{C}{x}$

(ii) $y = \frac{x^2}{4} + \frac{C}{x^2}$

Question 6

Solve the following differential equations:

(i) $\frac{dy}{dx} + 2y = 6e^x$

(ii) $\frac{dy}{dx} + 3y = e^{-2x}$

TYPE 90 **General Solution of First-Order Linear Differential Equation with Exponential Terms**

Concept / Formula Used: $y \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.})\, dx + C \qquad \frac{dy}{dx} + P(x)y = Q(x)$

Solution:

(i) $\frac{dy}{dx} + 2y = 6e^x$

Here, $P(x) = 2$ and $Q(x) = 6e^x$. Integrating Factor:

$$\text{I.F.} = e^{\int 2\, dx} = e^{2x}$$

The general solution is:

$$\begin{aligned} y \cdot e^{2x} &= \int 6e^x \cdot e^{2x}\, dx + C \\ ye^{2x} &= 6 \int e^{3x}\, dx + C \\ ye^{2x} &= 6 \cdot \frac{e^{3x}}{3} + C \\ ye^{2x} &= 2e^{3x} + C \\ y &= 2e^x + Ce^{-2x} \end{aligned}$$

(ii) $\frac{dy}{dx} + 3y = e^{-2x}$

Here, $P(x) = 3$ and $Q(x) = e^{-2x}$. Integrating Factor:

$$\text{I.F.} = e^{\int 3\, dx} = e^{3x}$$

The general solution is:

$$\begin{aligned} y \cdot e^{3x} &= \int e^{-2x} \cdot e^{3x}\, dx + C \\ ye^{3x} &= \int e^x\, dx + C \\ ye^{3x} &= e^x + C \\ y &= e^{-2x} + Ce^{-3x} \end{aligned}$$

Final Answer

(i) $y = 2e^x + Ce^{-2x}$

(ii) $y = e^{-2x} + Ce^{-3x}$

Question 7

Solve the following differential equations:

(i) $(1+x^2)\frac{dy}{dx} = 4x^2 - 2xy$

(ii) $\frac{dy}{dx} + \frac{y}{x} = e^{-x}$

(iii) $x\,dy + (y - x^3)\,dx = 0$

(iv) $x\frac{dy}{dx} - y = x^2e^x$

TYPE 91 **General Solution of First-Order Linear Differential Equation with Polynomial and Exponential Terms**

Concept / Formula Used: $y \cdot (\text{I.F.}) = \int Q(x) \cdot (\text{I.F.})\,dx + C \qquad \frac{dy}{dx} + P(x)y = Q(x)$

Solution:

(i) $(1+x^2)\frac{dy}{dx} = 4x^2 - 2xy$
Rearranging:

$$(1+x^2)\frac{dy}{dx} + 2xy = 4x^2$$

Dividing by $1+x^2$:

$$\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{4x^2}{1+x^2}$$

Here, $P(x) = \frac{2x}{1+x^2}$ and $Q(x) = \frac{4x^2}{1+x^2}$. Integrating Factor:

$$\text{I.F.} = e^{\int \frac{2x}{1+x^2}\,dx} = e^{\log(1+x^2)} = 1+x^2$$

The general solution is:

$$y \cdot (1+x^2) = \int \frac{4x^2}{1+x^2} \cdot (1+x^2)\,dx + C$$
$$y(1+x^2) = \int 4x^2\,dx + C$$
$$y(1+x^2) = \frac{4x^3}{3} + C$$

(ii) $\frac{dy}{dx} + \frac{y}{x} = e^{-x}$
Here, $P(x) = \frac{1}{x}$ and $Q(x) = e^{-x}$. Integrating Factor:

$$\text{I.F.} = e^{\int \frac{1}{x}\,dx} = e^{\log x} = x$$

The general solution is:

$$y \cdot x = \int xe^{-x}\,dx + C$$

Using integration by parts for $\int xe^{-x}\,dx$: Let $u = x \implies du = dx$, and $dv =$

$e^{-x}\,dx \implies v = -e^{-x}$.

$$\begin{aligned}
xy &= -xe^{-x} - \int (-e^{-x})\,dx + C \\
xy &= -xe^{-x} - e^{-x} + C \\
xy &= -(x+1)e^{-x} + C \\
y &= -\left(1 + \frac{1}{x}\right)e^{-x} + \frac{C}{x}
\end{aligned}$$

(iii) $x\,dy + (y - x^3)\,dx = 0$
Dividing by dx:

$$x\frac{dy}{dx} + y - x^3 = 0 \implies x\frac{dy}{dx} + y = x^3$$

Dividing by x:

$$\frac{dy}{dx} + \frac{1}{x}y = x^2$$

Integrating Factor:

$$\text{I.F.} = e^{\int \frac{1}{x}\,dx} = x$$

The general solution is:

$$\begin{aligned}
y \cdot x &= \int x^2 \cdot x\,dx + C \\
xy &= \int x^3\,dx + C \\
xy &= \frac{x^4}{4} + C \\
y &= \frac{x^3}{4} + \frac{C}{x}
\end{aligned}$$

(iv) $x\frac{dy}{dx} - y = x^2e^x$
Dividing by x:

$$\frac{dy}{dx} - \frac{1}{x}y = xe^x$$

Integrating Factor:

$$\text{I.F.} = e^{\int -\frac{1}{x}\,dx} = e^{-\log x} = \frac{1}{x}$$

The general solution is:

$$\begin{aligned}
y \cdot \frac{1}{x} &= \int xe^x \cdot \frac{1}{x}\,dx + C \\
\frac{y}{x} &= \int e^x\,dx + C \\
\frac{y}{x} &= e^x + C \\
y &= xe^x + Cx
\end{aligned}$$

Final Answer

(i) $y(1+x^2) = \frac{4x^3}{3} + C$
(ii) $y = -\left(1 + \frac{1}{x}\right)e^{-x} + \frac{C}{x}$
(iii) $y = \frac{x^3}{4} + \frac{C}{x}$
(iv) $y = xe^x + Cx$

Question 8

Solve the following differential equations:

(i) $(x^2+1)\frac{dy}{dx}+2xy=\sqrt{x^2+4}$

(ii) $x\frac{dy}{dx}+y=3x\cos 2x, x>0$

TYPE 92 **General Solution of First-Order Linear Differential Equation Requiring Integration by Parts or Radical Integra**

Concept / Formula Used: $y\cdot(\text{I.F.})=\int Q(x)\cdot(\text{I.F.})\,dx+C \qquad \frac{dy}{dx}+P(x)y=Q(x)$

Solution:

(i) $(x^2+1)\frac{dy}{dx}+2xy=\sqrt{x^2+4}$
Dividing by x^2+1:

$$\frac{dy}{dx}+\frac{2x}{x^2+1}y=\frac{\sqrt{x^2+4}}{x^2+1}$$

Integrating Factor:

$$\text{I.F.}=e^{\int\frac{2x}{x^2+1}\,dx}=e^{\log(x^2+1)}=x^2+1$$

The general solution is:

$$y\cdot(x^2+1)=\int\frac{\sqrt{x^2+4}}{x^2+1}\cdot(x^2+1)\,dx+C$$
$$y(x^2+1)=\int\sqrt{x^2+4}\,dx+C$$

Using the standard result $\int\sqrt{x^2+a^2}\,dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\log|x+\sqrt{x^2+a^2}|$:

$$y(x^2+1)=\frac{x}{2}\sqrt{x^2+4}+2\log|x+\sqrt{x^2+4}|+C$$

(ii) $x\frac{dy}{dx}+y=3x\cos 2x$
Dividing by x:

$$\frac{dy}{dx}+\frac{1}{x}y=3\cos 2x$$

Integrating Factor:

$$\text{I.F.}=e^{\int\frac{1}{x}\,dx}=x$$

The general solution is:

$$y\cdot x=\int 3\cos 2x\cdot x\,dx+C$$

$$xy=3\int x\cos 2x\,dx+C$$

Using integration by parts for $\int x\cos 2x\,dx$: Let $u = x \implies du = dx$, and $dv = \cos 2x\,dx \implies v = \frac{\sin 2x}{2}$.

$$xy = 3\left[\frac{x\sin 2x}{2} - \int \frac{\sin 2x}{2}\,dx\right] + C$$

$$xy = 3\left[\frac{x\sin 2x}{2} + \frac{\cos 2x}{4}\right] + C$$

$$xy = \frac{3x\sin 2x}{2} + \frac{3\cos 2x}{4} + C$$

Final Answer

(i) $y(x^2+1) = \frac{x}{2}\sqrt{x^2+4} + 2\log|x+\sqrt{x^2+4}| + C$
(ii) $xy = \frac{3x\sin 2x}{2} + \frac{3\cos 2x}{4} + C$

Question 9

Solve the following differential equations:
(i) $x\frac{dy}{dx} - y = (x-1)e^x, x > 0$
(ii) $\frac{dx}{dy} + x = \tan y + \sec^2 y$

TYPE 93 **General Solution of First-Order Linear Differential Equation with ex (f(x)+f'(x)) Form**

Concept / Formula Used: $x \cdot (\text{I.F.}) = \int Q(y) \cdot (\text{I.F.})\, dy + C \qquad \frac{dx}{dy} + P(y)x = Q(y)$

Solution:

(i) $x\frac{dy}{dx} - y = (x-1)e^x$
Dividing by x:

$$\frac{dy}{dx} - \frac{1}{x}y = \left(1 - \frac{1}{x}\right)e^x$$

Integrating Factor:

$$\text{I.F.} = e^{\int -\frac{1}{x}\,dx} = \frac{1}{x}$$

The general solution is:

$$y \cdot \frac{1}{x} = \int \left(1 - \frac{1}{x}\right)e^x \cdot \frac{1}{x}\,dx + C$$
$$\frac{y}{x} = \int \left(\frac{1}{x} - \frac{1}{x^2}\right)e^x\,dx + C$$

Using the standard result $\int e^x(f(x) + f'(x))\,dx = e^x f(x) + C$ with $f(x) = \frac{1}{x}$:

$$\frac{y}{x} = \frac{e^x}{x} + C$$
$$y = e^x + Cx$$

(ii) $\frac{dx}{dy} + x = \tan y + \sec^2 y$
This is a linear differential equation in x with $P(y) = 1$ and $Q(y) = \tan y + \sec^2 y$.
Integrating Factor:

$$\text{I.F.} = e^{\int 1\,dy} = e^y$$

The general solution is:

$$x \cdot e^y = \int (\tan y + \sec^2 y)e^y\,dy + C$$

Using the standard result $\int e^y(f(y) + f'(y))\,dy = e^y f(y) + C$ with $f(y) = \tan y$:

$$xe^y = e^y \tan y + C$$
$$x = \tan y + Ce^{-y}$$

Final Answer

(i) $y = e^x + Cx$
(ii) $x = \tan y + Ce^{-y}$

Question 10

Solve the following differential equations:

(i) $x\frac{dy}{dx} + 2y = x\cos x$

(ii) $(y - \sin^2 x)\,dx + \tan x\,dy = 0$

(iii) $\sin x\frac{dy}{dx} - y = \sin x \tan\frac{x}{2}$

(iv) $(y - \sin^2 x)\,dx + (\tan x)\,dy = 0$

TYPE 94 General Solution of First-Order Linear Differential Equation with Trigonometric Terms

Solution:

(i) $x\frac{dy}{dx} + 2y = x\cos x$
Dividing by x:

$$\frac{dy}{dx} + \frac{2}{x}y = \cos x$$

Integrating Factor:

$$\text{I.F.} = e^{\int \frac{2}{x}\,dx} = x^2$$

The general solution is:

$$y \cdot x^2 = \int x^2 \cos x\,dx + C$$

Using integration by parts for $\int x^2\cos x\,dx$: Let $u = x^2 \implies du = 2x\,dx$, and $dv = \cos x\,dx \implies v = \sin x$.

$$\int x^2 \cos x\,dx = x^2 \sin x - \int 2x \sin x\,dx$$

Now, for $\int x\sin x\,dx$: Let $u = x \implies du = dx$, and $dv = \sin x\,dx \implies v = -\cos x$.

$$\int x\sin x\,dx = -x\cos x - \int(-\cos x)\,dx = -x\cos x + \sin x$$

Thus:

$$yx^2 = x^2\sin x - 2(-x\cos x + \sin x) + C$$
$$yx^2 = x^2\sin x + 2x\cos x - 2\sin x + C$$
$$y = \sin x + \frac{2\cos x}{x} - \frac{2\sin x}{x^2} + \frac{C}{x^2}$$

(ii) and (iv) $(y - \sin^2 x)\,dx + \tan x\,dy = 0$
Dividing by dx:

$$\tan x\frac{dy}{dx} + y = \sin^2 x$$

Dividing by $\tan x$:

$$\frac{dy}{dx} + \cot x \cdot y = \sin x\cos x$$

Integrating Factor:

$$\text{I.F.} = e^{\int \cot x\,dx} = e^{\log|\sin x|} = \sin x$$

The general solution is:

$$y \cdot \sin x = \int \sin x \cos x \cdot \sin x \, dx + C = \int \sin^2 x \cos x \, dx + C$$

Let $\sin x = t$. Differentiating: $\cos x \, dx = dt$.

$$\begin{aligned} y \sin x &= \int t^2 \, dt + C \\ y \sin x &= \frac{t^3}{3} + C \\ y \sin x &= \frac{\sin^3 x}{3} + C \\ y &= \frac{\sin^2 x}{3} + C \csc x \end{aligned}$$

(iii) $\sin x \frac{dy}{dx} - y = \sin x \tan \frac{x}{2}$
Dividing by $\sin x$:

$$\frac{dy}{dx} - (\csc x)y = \tan \frac{x}{2}$$

Integrating Factor:

$$\text{I.F.} = e^{\int - \csc x \, dx} = e^{-\log|\csc x - \cot x|} = \frac{1}{\csc x - \cot x}$$

Since $\csc x - \cot x = \frac{1 - \cos x}{\sin x} = \tan \frac{x}{2}$:

$$\text{I.F.} = \frac{1}{\tan \frac{x}{2}} = \cot \frac{x}{2}$$

The general solution is:

$$\begin{aligned} y \cdot \cot \frac{x}{2} &= \int \tan \frac{x}{2} \cdot \cot \frac{x}{2} \, dx + C \\ y \cot \frac{x}{2} &= \int 1 \, dx + C \\ y \cot \frac{x}{2} &= x + C \\ y &= (x + C) \tan \frac{x}{2} \end{aligned}$$

Final Answer

(i) $y = \sin x + \frac{2 \cos x}{x} - \frac{2 \sin x}{x^2} + \frac{C}{x^2}$
(ii) and (iv) $y = \frac{\sin^2 x}{3} + C \csc x$
(iii) $y = (x + C) \tan \frac{x}{2}$

Question 11

Solve the following differential equations:

(i) $\frac{dy}{dx} = \frac{x+y\cos x}{1+\sin x}$

(ii) $\frac{dy}{dx} + y\cos x = e^{\sin x}\cos x$

TYPE 95 **General Solution of First-Order Linear Differential Equation with Trigonometric Substitution or Exponential T**

Solution:

(i) $\frac{dy}{dx} = \frac{x+y\cos x}{1+\sin x}$

Rearranging:

$$\frac{dy}{dx} - \frac{\cos x}{1+\sin x}y = \frac{x}{1+\sin x}$$

Integrating Factor:

$$\text{I.F.} = e^{\int -\frac{\cos x}{1+\sin x}\,dx} = e^{-\log(1+\sin x)} = \frac{1}{1+\sin x}$$

The general solution is:

$$\frac{y}{1+\sin x} = \int \frac{x}{(1+\sin x)^2}\,dx + C$$

To evaluate $\int \frac{x}{(1+\sin x)^2}\,dx$, we use the substitution $x = 2\theta - \frac{\pi}{2} \implies dx = 2\,d\theta$. Then $1+\sin x = 1-\cos 2\theta = 2\sin^2\theta$. The integral becomes:

$$\int \frac{2\theta - \pi/2}{4\sin^4\theta}\cdot 2\,d\theta = \int \left(2\theta - \frac{\pi}{2}\right)\csc^4\theta\,d\theta$$

Using integration by parts with $u = 2\theta - \frac{\pi}{2} \implies du = 2\,d\theta$, and $dv = \csc^4\theta\,d\theta \implies v = -\cot\theta - \frac{\cot^3\theta}{3}$:

$$\begin{aligned}
&= \left(2\theta - \frac{\pi}{2}\right)\left(-\cot\theta - \frac{\cot^3\theta}{3}\right) - \int 2\left(-\cot\theta - \frac{\cot^3\theta}{3}\right)d\theta \\
&= -x\left(\cot\theta + \frac{\cot^3\theta}{3}\right) + 2\log|\sin\theta| + \frac{2}{3}\left(-\frac{\cot^2\theta}{2} - \log|\sin\theta|\right) \\
&= -x\left(\cot\theta + \frac{\cot^3\theta}{3}\right) + \frac{4}{3}\log|\sin\theta| - \frac{\cot^2\theta}{3}
\end{aligned}$$

where $\theta = \frac{x}{2} + \frac{\pi}{4}$.

(ii) $\frac{dy}{dx} + y\cos x = e^{\sin x}\cos x$

Integrating Factor:

$$\text{I.F.} = e^{\int \cos x\,dx} = e^{\sin x}$$

The general solution is:

$$y\cdot e^{\sin x} = \int e^{\sin x}\cos x\cdot e^{\sin x}\,dx + C = \int e^{2\sin x}\cos x\,dx + C$$

Let $\sin x = t$. Differentiating: $\cos x\, dx = dt$.

$$ye^{\sin x} = \int e^{2t}\, dt + C$$
$$ye^{\sin x} = \frac{e^{2t}}{2} + C$$
$$ye^{\sin x} = \frac{e^{2\sin x}}{2} + C$$
$$y = \frac{e^{\sin x}}{2} + Ce^{-\sin x}$$

Final Answer

(i) $y = (1 + \sin x)\left[\int \frac{x}{(1+\sin x)^2}\, dx + C\right]$

(ii) $y = \frac{e^{\sin x}}{2} + Ce^{-\sin x}$

Question 12

Solve the following differential equations:

(i) $\frac{dy}{dx} - \tan x \cdot y = e^{\sin x}, 0 < x < \frac{\pi}{2}$

(ii) $\frac{dy}{dx} + y = \cos x - \sin x$

TYPE 96 General Solution of First-Order Linear Differential Equation with Exponential and Trigonometric Product Inte

Solution:

(i) $\frac{dy}{dx} - \tan x \cdot y = e^{\sin x}$

Integrating Factor:

$$\text{I.F.} = e^{\int -\tan x\, dx} = e^{-\log|\sec x|} = \cos x \quad \left(\text{since } 0 < x < \frac{\pi}{2}\right)$$

The general solution is:

$$y \cdot \cos x = \int e^{\sin x} \cos x\, dx + C$$

Let $\sin x = t$. Differentiating: $\cos x\, dx = dt$.

$$y \cos x = \int e^t\, dt + C$$
$$y \cos x = e^t + C$$
$$y \cos x = e^{\sin x} + C$$
$$y = e^{\sin x} \sec x + C \sec x$$

(ii) $\frac{dy}{dx} + y = \cos x - \sin x$

Integrating Factor:

$$\text{I.F.} = e^{\int 1\, dx} = e^x$$

The general solution is:

$$y \cdot e^x = \int e^x(\cos x - \sin x)\, dx + C$$

Using the standard result $\int e^x(f(x) + f'(x))\, dx = e^x f(x) + C$ with $f(x) = \cos x$:

$$ye^x = e^x \cos x + C$$
$$y = \cos x + Ce^{-x}$$

Final Answer

(i) $y = e^{\sin x} \sec x + C \sec x$

(ii) $y = \cos x + Ce^{-x}$

Question 13

Solve the following differential equations:

(i) $x\frac{dy}{dx} + y = x \log x$

(ii) $x\frac{dy}{dx} + 2y = x^2 \log x$

TYPE 97 **General Solution of First-Order Linear Differential Equation with Logarithmic Terms**

Solution:

(i) $x\frac{dy}{dx} + y = x \log x$

Dividing by x:

$$\frac{dy}{dx} + \frac{1}{x}y = \log x$$

Integrating Factor:

$$\text{I.F.} = e^{\int \frac{1}{x}\,dx} = x$$

The general solution is:

$$y \cdot x = \int x \log x\,dx + C$$

Using integration by parts for $\int x \log x\,dx$: Let $u = \log x \implies du = \frac{1}{x}\,dx$, and $dv = x\,dx \implies v = \frac{x^2}{2}$.

$$xy = \frac{x^2}{2}\log x - \int \frac{x^2}{2} \cdot \frac{1}{x}\,dx + C$$

$$xy = \frac{x^2}{2}\log x - \frac{x^2}{4} + C$$

$$y = \frac{x}{2}\log x - \frac{x}{4} + \frac{C}{x}$$

(ii) $x\frac{dy}{dx} + 2y = x^2 \log x$

Dividing by x:

$$\frac{dy}{dx} + \frac{2}{x}y = x \log x$$

Integrating Factor:

$$\text{I.F.} = e^{\int \frac{2}{x}\,dx} = x^2$$

The general solution is:

$$y \cdot x^2 = \int x^2 \cdot x \log x\,dx + C = \int x^3 \log x\,dx + C$$

Using integration by parts for $\int x^3 \log x\,dx$: Let $u = \log x \implies du = \frac{1}{x}\,dx$, and $dv = x^3\,dx \implies v = \frac{x^4}{4}$.

$$yx^2 = \frac{x^4}{4}\log x - \int \frac{x^4}{4} \cdot \frac{1}{x}\,dx + C$$

$$yx^2 = \frac{x^4}{4}\log x - \frac{x^4}{16} + C$$

$$y = \frac{x^2}{4}\log x - \frac{x^2}{16} + \frac{C}{x^2}$$

Final Answer

(i) $y = \frac{x}{2}\log x - \frac{x}{4} + \frac{C}{x}$

(ii) $y = \frac{x^2}{4}\log x - \frac{x^2}{16} + \frac{C}{x^2}$

Question 14

Solve the following differential equations:

(i) $\frac{dy}{dx} + y\cot x = 2\cos x$

(ii) $\frac{dy}{dx} - y\tan x = 2\sin x$

TYPE 98 General Solution of First-Order Linear Differential Equation with Trigonometric Coefficients

Solution:

(i) $\frac{dy}{dx} + y\cot x = 2\cos x$

Integrating Factor:

$$\text{I.F.} = e^{\int \cot x\,dx} = e^{\log|\sin x|} = \sin x$$

The general solution is:

$$y\cdot\sin x = \int 2\cos x\cdot\sin x\,dx + C$$
$$y\sin x = \int \sin 2x\,dx + C$$
$$y\sin x = -\frac{\cos 2x}{2} + C$$
$$y = -\frac{\cos 2x}{2\sin x} + C\csc x$$

(ii) $\frac{dy}{dx} - y\tan x = 2\sin x$

Integrating Factor:

$$\text{I.F.} = e^{\int -\tan x\,dx} = e^{\log|\cos x|} = \cos x$$

The general solution is:

$$y\cdot\cos x = \int 2\sin x\cdot\cos x\,dx + C$$
$$y\cos x = \int \sin 2x\,dx + C$$
$$y\cos x = -\frac{\cos 2x}{2} + C$$
$$y = -\frac{\cos 2x}{2\cos x} + C\sec x$$

Final Answer

(i) $y = -\frac{\cos 2x}{2\sin x} + C\csc x$

(ii) $y = -\frac{\cos 2x}{2\cos x} + C\sec x$

Question 15

Solve the following differential equations:
(i) $(\tan x)\frac{dy}{dx} + 2y = \sec x$
(ii) $dy = x(x^2 - 2y)\,dx$

TYPE 99 **General Solution of First-Order Linear Differential Equation Requiring Substitution in Integration**

Solution:

(i) $(\tan x)\frac{dy}{dx} + 2y = \sec x$
Dividing by $\tan x$:

$$\frac{dy}{dx} + 2\cot x \cdot y = \csc x$$

Integrating Factor:

$$\text{I.F.} = e^{\int 2\cot x\,dx} = e^{2\log|\sin x|} = \sin^2 x$$

The general solution is:

$$\begin{aligned} y \cdot \sin^2 x &= \int \csc x \cdot \sin^2 x\,dx + C \\ y\sin^2 x &= \int \sin x\,dx + C \\ y\sin^2 x &= -\cos x + C \\ y &= -\cos x\csc^2 x + C\csc^2 x \end{aligned}$$

(ii) $dy = x(x^2 - 2y)\,dx$
Dividing by dx:

$$\frac{dy}{dx} = x^3 - 2xy \implies \frac{dy}{dx} + 2xy = x^3$$

Integrating Factor:

$$\text{I.F.} = e^{\int 2x\,dx} = e^{x^2}$$

The general solution is:

$$y \cdot e^{x^2} = \int x^3 e^{x^2}\,dx + C$$

Let $x^2 = t$. Differentiating: $2x\,dx = dt \implies x\,dx = \frac{1}{2}dt$. The integral becomes:

$$\int x^2 \cdot e^{x^2} \cdot x\,dx = \int te^t \cdot \frac{1}{2}\,dt = \frac{1}{2}(t-1)e^t$$

Back-substituting $t = x^2$:

$$\int x^3 e^{x^2}\,dx = \frac{1}{2}(x^2 - 1)e^{x^2}$$

Thus:

$$\begin{aligned} ye^{x^2} &= \frac{1}{2}(x^2 - 1)e^{x^2} + C \\ y &= \frac{1}{2}(x^2 - 1) + Ce^{-x^2} \end{aligned}$$

Final Answer

(i) $y = -\cos x \csc^2 x + C \csc^2 x$

(ii) $y = \frac{1}{2}(x^2 - 1) + Ce^{-x^2}$

Question 16

Solve the following differential equations:
(i) $\frac{dy}{dx} - y = \cos x$
(ii) $x \log x \frac{dy}{dx} + y = 2\log x, x > 1$

TYPE 100 **General Solution of First-Order Linear Differential Equation with Logarithmic or Trigonometric Non-Homoge**

Solution:

(i) $\frac{dy}{dx} - y = \cos x$
Integrating Factor:

$$\text{I.F.} = e^{\int -1\,dx} = e^{-x}$$

The general solution is:

$$y \cdot e^{-x} = \int e^{-x} \cos x \, dx + C$$

Let $I = \int e^{-x} \cos x \, dx$. Using integration by parts: Let $u = \cos x \implies du = -\sin x \, dx$, and $dv = e^{-x}\,dx \implies v = -e^{-x}$.

$$I = -e^{-x}\cos x - \int e^{-x} \sin x \, dx$$

For the second integral, let $u = \sin x \implies du = \cos x \, dx$, and $dv = e^{-x}\,dx \implies v = -e^{-x}$.

$$I = -e^{-x}\cos x - \left(-e^{-x}\sin x + I\right) = e^{-x}(\sin x - \cos x) - I$$

$$2I = e^{-x}(\sin x - \cos x) \implies I = \frac{e^{-x}}{2}(\sin x - \cos x)$$

Thus:

$$ye^{-x} = \frac{e^{-x}}{2}(\sin x - \cos x) + C$$
$$y = \frac{1}{2}(\sin x - \cos x) + Ce^{x}$$

(ii) $x \log x \frac{dy}{dx} + y = 2\log x$
Dividing by $x \log x$:

$$\frac{dy}{dx} + \frac{1}{x\log x} y = \frac{2}{x}$$

Integrating Factor:

$$\text{I.F.} = e^{\int \frac{1}{x\log x}\,dx} = e^{\log(\log x)} = \log x$$

The general solution is:

$$y \cdot \log x = \int \frac{2}{x} \cdot \log x \, dx + C$$

Let $\log x = t$. Differentiating: $\frac{1}{x}\,dx = dt$.

$$\begin{aligned} y \log x &= \int 2t\,dt + C \\ y \log x &= t^2 + C \\ y \log x &= (\log x)^2 + C \\ y &= \log x + \frac{C}{\log x} \end{aligned}$$

Final Answer

(i) $y = \frac{1}{2}(\sin x - \cos x) + Ce^x$
(ii) $y = \log x + \frac{C}{\log x}$

Question 17

Solve the following differential equations:

(i) $y\,dx + (x - y^2)\,dy = 0, y > 0$

(ii) $(x + 3y^2)\,dy = y\,dx$

TYPE 101 General Solution of First-Order Linear Differential Equation in x with Polynomial Terms

Solution:

(i) $y\,dx + (x - y^2)\,dy = 0$

Dividing by dy:

$$y\frac{dx}{dy} + x - y^2 = 0 \implies y\frac{dx}{dy} + x = y^2$$

Dividing by y:

$$\frac{dx}{dy} + \frac{1}{y}x = y$$

Integrating Factor:

$$\text{I.F.} = e^{\int \frac{1}{y}\,dy} = y$$

The general solution is:

$$x \cdot y = \int y \cdot y\,dy + C$$

$$xy = \int y^2\,dy + C$$

$$xy = \frac{y^3}{3} + C$$

$$x = \frac{y^2}{3} + \frac{C}{y}$$

(ii) $(x + 3y^2)\,dy = y\,dx$

Dividing by dy:

$$x + 3y^2 = y\frac{dx}{dy} \implies y\frac{dx}{dy} - x = 3y^2$$

Dividing by y:

$$\frac{dx}{dy} - \frac{1}{y}x = 3y$$

Integrating Factor:

$$\text{I.F.} = e^{\int -\frac{1}{y}\,dy} = \frac{1}{y}$$

The general solution is:

$$x \cdot \frac{1}{y} = \int 3y \cdot \frac{1}{y}\, dy + C$$
$$\frac{x}{y} = \int 3\, dy + C$$
$$\frac{x}{y} = 3y + C$$
$$x = 3y^2 + Cy$$

Final Answer

(i) $x = \frac{y^2}{3} + \frac{C}{y}$

(ii) $x = 3y^2 + Cy$

Question 18

Solve the following differential equations:

(i) $dx + x\,dy = e^{-y}\log y\,dy, y > 0$

(ii) $(x + 2y^3)\,dy = y\,dx, y > 0$

TYPE 102 **General Solution of First-Order Linear Differential Equation in x with Logarithmic or Polynomial Terms**

Solution:

(i) $dx + x\,dy = e^{-y}\log y\,dy$

Dividing by dy:

$$\frac{dx}{dy} + x = e^{-y}\log y$$

Integrating Factor:

$$\text{I.F.} = e^{\int 1\,dy} = e^y$$

The general solution is:

$$x \cdot e^y = \int e^{-y}\log y \cdot e^y\,dy + C = \int \log y\,dy + C$$

Using integration by parts for $\int \log y\,dy$:

$$xe^y = y\log y - y + C$$
$$x = (y\log y - y + C)e^{-y}$$

(ii) $(x + 2y^3)\,dy = y\,dx$

Dividing by dy:

$$x + 2y^3 = y\frac{dx}{dy} \implies y\frac{dx}{dy} - x = 2y^3$$

Dividing by y:

$$\frac{dx}{dy} - \frac{1}{y}x = 2y^2$$

Integrating Factor:

$$\text{I.F.} = e^{\int -\frac{1}{y}\,dy} = \frac{1}{y}$$

The general solution is:

$$x \cdot \frac{1}{y} = \int 2y^2 \cdot \frac{1}{y}\,dy + C$$
$$\frac{x}{y} = \int 2y\,dy + C$$
$$\frac{x}{y} = y^2 + C$$
$$x = y^3 + Cy$$

Final Answer

(i) $x = (y\log y - y + C)e^{-y}$

(ii) $x = y^3 + Cy$

Question 19

Solve the differential equation:

$$(1 + y^2)\, dx = (\tan^{-1} y - x)\, dy$$

TYPE 103 General Solution of First-Order Linear Differential Equation in x with Inverse Trigonometric Term

Solution:

Dividing by dy:

$$(1 + y^2)\frac{dx}{dy} = \tan^{-1} y - x \implies (1 + y^2)\frac{dx}{dy} + x = \tan^{-1} y$$

Dividing by $1 + y^2$:

$$\frac{dx}{dy} + \frac{1}{1 + y^2}x = \frac{\tan^{-1} y}{1 + y^2}$$

Integrating Factor:

$$\text{I.F.} = e^{\int \frac{1}{1+y^2}\, dy} = e^{\tan^{-1} y}$$

The general solution is:

$$x \cdot e^{\tan^{-1} y} = \int \frac{\tan^{-1} y}{1 + y^2} e^{\tan^{-1} y}\, dy + C$$

Let $\tan^{-1} y = t$. Differentiating: $\frac{1}{1+y^2}\, dy = dt$. The integral becomes:

$$\int te^t\, dt = (t - 1)e^t$$

Back-substituting $t = \tan^{-1} y$:

$$\int \frac{\tan^{-1} y}{1 + y^2} e^{\tan^{-1} y}\, dy = (\tan^{-1} y - 1)e^{\tan^{-1} y}$$

Thus:

$$xe^{\tan^{-1} y} = (\tan^{-1} y - 1)e^{\tan^{-1} y} + C$$
$$x = \tan^{-1} y - 1 + Ce^{-\tan^{-1} y}$$

Final Answer

$x = \tan^{-1} y - 1 + Ce^{-\tan^{-1} y}$

Question 20

Solve the differential equation $(1+x^2)\frac{dy}{dx}+2xy-4x^2=0$ subject to the initial condition $y(0)=0$.

TYPE 104 Particular Solution of First-Order Linear Differential Equation with Polynomial Terms

Solution:

Rearranging the equation:

$$(1+x^2)\frac{dy}{dx}+2xy=4x^2$$

Dividing by $1+x^2$:

$$\frac{dy}{dx}+\frac{2x}{1+x^2}y=\frac{4x^2}{1+x^2}$$

Integrating Factor:

$$\text{I.F.}=e^{\int\frac{2x}{1+x^2}\,dx}=e^{\log(1+x^2)}=1+x^2$$

The general solution is:

$$\begin{aligned}
y\cdot(1+x^2)&=\int\frac{4x^2}{1+x^2}\cdot(1+x^2)\,dx+C\\
y(1+x^2)&=\int 4x^2\,dx+C\\
y(1+x^2)&=\frac{4x^3}{3}+C
\end{aligned}$$

Applying the initial condition $y(0)=0$:

$$0\cdot(1+0)=\frac{4(0)^3}{3}+C\implies C=0$$

Thus, the particular solution is:

$$\begin{aligned}
y(1+x^2)&=\frac{4x^3}{3}\\
y&=\frac{4x^3}{3(1+x^2)}
\end{aligned}$$

Final Answer

$y=\frac{4x^3}{3(1+x^2)}$

Question 21

Solve the differential equation $\frac{dy}{dx} + y\cot x = 2x + x^2\cot x$, given that $y = 0$ when $x = \frac{\pi}{2}$.

TYPE 105 **Particular Solution of First-Order Linear Differential Equation with Product Derivative Pattern**

Solution:

Here, $P(x) = \cot x$ and $Q(x) = 2x + x^2\cot x$. Integrating Factor:

$$\text{I.F.} = e^{\int \cot x\,dx} = e^{\log|\sin x|} = \sin x$$

The general solution is:

$$y \cdot \sin x = \int (2x + x^2\cot x)\sin x\,dx + C = \int (2x\sin x + x^2\cos x)\,dx + C$$

Using the product rule of differentiation, we observe that:

$$\frac{d}{dx}(x^2\sin x) = 2x\sin x + x^2\cos x$$

Thus:

$$y\sin x = x^2\sin x + C$$

Applying the initial condition $y = 0$ when $x = \frac{\pi}{2}$:

$$0 \cdot \sin\frac{\pi}{2} = \left(\frac{\pi}{2}\right)^2 \sin\frac{\pi}{2} + C$$

$$0 = \frac{\pi^2}{4} \cdot 1 + C \implies C = -\frac{\pi^2}{4}$$

Thus, the particular solution is:

$$y\sin x = x^2\sin x - \frac{\pi^2}{4}$$

$$y = x^2 - \frac{\pi^2}{4}\csc x$$

Final Answer

$y = x^2 - \frac{\pi^2}{4}\csc x$

Question 22

If $\frac{dy}{dx} + 2xy = x$, then prove that $2y = 1 + e^{-x^2}$, given that $y = 1$ when $x = 0$.

TYPE 106 Particular Solution Proof of First-Order Linear Differential Equation with Exponential Term

To Prove: $2y = 1 + e^{-x^2}$

Solution:

Here, $P(x) = 2x$ and $Q(x) = x$. Integrating Factor:

$$\text{I.F.} = e^{\int 2x\,dx} = e^{x^2}$$

The general solution is:

$$y \cdot e^{x^2} = \int xe^{x^2}\,dx + C$$

Let $x^2 = t$. Differentiating: $2x\,dx = dt \implies x\,dx = \frac{1}{2}dt$. The integral becomes:

$$\int e^t \cdot \frac{1}{2}\,dt = \frac{1}{2}e^t$$

Back-substituting $t = x^2$:

$$ye^{x^2} = \frac{1}{2}e^{x^2} + C$$

Dividing by e^{x^2}:

$$y = \frac{1}{2} + Ce^{-x^2}$$

Applying the initial condition $y = 1$ when $x = 0$:

$$1 = \frac{1}{2} + Ce^0 \implies 1 = \frac{1}{2} + C \implies C = \frac{1}{2}$$

Substituting $C = \frac{1}{2}$ back into the solution:

$$y = \frac{1}{2} + \frac{1}{2}e^{-x^2}$$

Multiplying both sides by 2:

$$2y = 1 + e^{-x^2}$$

Hence Proved

$LHS = RHS, henceproved.$

Question 23

Solve $(x + 2y^2)\frac{dy}{dx} = y$, given that $x = 2$ when $y = 1$.

TYPE 107 **Particular Solution of First-Order Linear Differential Equation in x**

Solution:

Taking the reciprocal of both sides:

$$\frac{dx}{dy} = \frac{x + 2y^2}{y} = \frac{x}{y} + 2y \implies \frac{dx}{dy} - \frac{1}{y}x = 2y$$

This is a linear differential equation in x with $P(y) = -\frac{1}{y}$ and $Q(y) = 2y$. Integrating Factor:

$$\text{I.F.} = e^{\int -\frac{1}{y}\,dy} = e^{-\log y} = \frac{1}{y}$$

The general solution is:

$$\begin{aligned} x \cdot \frac{1}{y} &= \int 2y \cdot \frac{1}{y}\,dy + C \\ \frac{x}{y} &= \int 2\,dy + C \\ \frac{x}{y} &= 2y + C \\ x &= 2y^2 + Cy \end{aligned}$$

Applying the initial condition $x = 2$ when $y = 1$:

$$2 = 2(1)^2 + C(1) \implies 2 = 2 + C \implies C = 0$$

Thus, the particular solution is:

$$x = 2y^2$$

Final Answer

$x = 2y^2$

Question 24

Solve the differential equation: $\frac{dy}{dx} + y\sec^2 x = \tan x \sec^2 x; y(0) = 1$.

TYPE 108 **Particular Solution of First-Order Linear Differential Equation with Trigonometric Substitution Integration**

Solution:

Here, $P(x) = \sec^2 x$ and $Q(x) = \tan x \sec^2 x$. Integrating Factor:

$$\text{I.F.} = e^{\int \sec^2 x\, dx} = e^{\tan x}$$

The general solution is:

$$y \cdot e^{\tan x} = \int \tan x \sec^2 x \cdot e^{\tan x}\, dx + C$$

Let $\tan x = t$. Differentiating: $\sec^2 x\, dx = dt$. The integral becomes:

$$\int te^t\, dt = (t-1)e^t$$

Back-substituting $t = \tan x$:

$$ye^{\tan x} = (\tan x - 1)e^{\tan x} + C$$

Dividing by $e^{\tan x}$:

$$y = \tan x - 1 + Ce^{-\tan x}$$

Applying the initial condition $y(0) = 1$:

$$1 = \tan 0 - 1 + Ce^{-\tan 0} \implies 1 = -1 + C \implies C = 2$$

Thus, the particular solution is:

$$y = \tan x - 1 + 2e^{-\tan x}$$

Final Answer

$y = \tan x - 1 + 2e^{-\tan x}$

Question 25

Find the particular solutions of the following differential equations:
(i) $y' - y = e^x$, given that $y = 1$ when $x = 0$
(ii) $y' + y = e^x$, given that $y = \frac{1}{2}$ when $x = 0$
(iii) $xy' + y = x \log x$, given that $y = \frac{1}{4}$ when $x = 1$

TYPE 109 Particular Solution of First-Order Linear Differential Equation with Initial Boundary Conditions

Solution:

(i) $y' - y = e^x \implies \frac{dy}{dx} - y = e^x$
Integrating Factor:

$$\text{I.F.} = e^{\int -1\,dx} = e^{-x}$$

The general solution is:

$$\begin{aligned} y \cdot e^{-x} &= \int e^x \cdot e^{-x}\,dx + C \\ ye^{-x} &= \int 1\,dx + C \\ ye^{-x} &= x + C \\ y &= (x + C)e^x \end{aligned}$$

Applying the initial condition $y = 1$ when $x = 0$:

$$1 = (0 + C)e^0 \implies C = 1$$

Thus, the particular solution is:

$$y = (x + 1)e^x$$

(ii) $y' + y = e^x \implies \frac{dy}{dx} + y = e^x$
Integrating Factor:

$$\text{I.F.} = e^{\int 1\,dx} = e^x$$

The general solution is:

$$\begin{aligned} y \cdot e^x &= \int e^x \cdot e^x\,dx + C \\ ye^x &= \int e^{2x}\,dx + C \\ ye^x &= \frac{e^{2x}}{2} + C \\ y &= \frac{e^x}{2} + Ce^{-x} \end{aligned}$$

Applying the initial condition $y = \frac{1}{2}$ when $x = 0$:

$$\frac{1}{2} = \frac{e^0}{2} + Ce^0 \implies \frac{1}{2} = \frac{1}{2} + C \implies C = 0$$

Thus, the particular solution is:

$$y = \frac{e^x}{2}$$

(iii) $xy' + y = x\log x \implies x\frac{dy}{dx} + y = x\log x$
Dividing by x:

$$\frac{dy}{dx} + \frac{1}{x}y = \log x$$

Integrating Factor:

$$\text{I.F.} = e^{\int \frac{1}{x}\,dx} = x$$

The general solution is:

$$y \cdot x = \int x\log x\,dx + C$$

Using integration by parts for $\int x\log x\,dx$:

$$xy = \frac{x^2}{2}\log x - \frac{x^2}{4} + C$$

Applying the initial condition $y = \frac{1}{4}$ when $x = 1$:

$$\frac{1}{4} = \frac{1^2}{2}\log 1 - \frac{1^2}{4} + C \implies \frac{1}{4} = -\frac{1}{4} + C \implies C = \frac{1}{2}$$

Thus, the particular solution is:

$$xy = \frac{x^2}{2}\log x - \frac{x^2}{4} + \frac{1}{2}$$
$$y = \frac{x}{2}\log x - \frac{x}{4} + \frac{1}{2x}$$

Final Answer

(i) $y = (x+1)e^x$
(ii) $y = \frac{e^x}{2}$
(iii) $y = \frac{x}{2}\log x - \frac{x}{4} + \frac{1}{2x}$

Question 25 (iv)

Solve the differential equation:

$$xy' - y = \log x$$

given that $y = 0$ when $x = 1$.

TYPE 110 Particular Solution of First-Order Linear Differential Equation with Logarithmic Integration

Given: $xy' - y = \log x$, with $y = 0$ when $x = 1$

Concept / Formula Used: $\frac{dy}{dx} + Py = Q$ $\quad$ I.F. $= e^{\int P\,dx}$ $\quad$ $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$

Solution:

We rewrite the given differential equation in standard form:

$$x\frac{dy}{dx} - y = \log x$$
$$\frac{dy}{dx} - \frac{1}{x}y = \frac{\log x}{x}$$

This is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where:

$$P = -\frac{1}{x}$$
$$Q = \frac{\log x}{x}$$

Now, we find the Integrating Factor (I.F.):

$$\begin{aligned}\text{I.F.} &= e^{\int P\,dx} \\ &= e^{\int -\frac{1}{x}\,dx} \\ &= e^{-\log|x|} \\ &= e^{\log\left(x^{-1}\right)} \\ &= \frac{1}{x}\end{aligned}$$

The general solution of the differential equation is given by:

$$\begin{aligned}y \cdot (\text{I.F.}) &= \int Q \cdot (\text{I.F.})\,dx + C \\ y \cdot \frac{1}{x} &= \int \frac{\log x}{x} \cdot \frac{1}{x}\,dx + C \\ \frac{y}{x} &= \int \frac{\log x}{x^2}\,dx + C\end{aligned}$$

To evaluate the integral $I = \int \frac{\log x}{x^2}\,dx$, we use integration by parts: Let $u = \log x$ and $dv = x^{-2}\,dx$. Differentiating u:

$$du = \frac{1}{x}\,dx$$

Integrating dv:

$$v = -\frac{1}{x}$$

Using the integration by parts formula $\int u\,dv = uv - \int v\,du$:

$$\begin{aligned}\int \frac{\log x}{x^2}\,dx &= (\log x)\left(-\frac{1}{x}\right) - \int \left(-\frac{1}{x}\right)\frac{1}{x}\,dx \\ &= -\frac{\log x}{x} + \int x^{-2}\,dx \\ &= -\frac{\log x}{x} - \frac{1}{x}\end{aligned}$$

Substituting this back into the general solution:

$$\frac{y}{x} = -\frac{\log x}{x} - \frac{1}{x} + C$$
$$y = -\log x - 1 + Cx$$

We are given the initial condition $y = 0$ when $x = 1$:

$$0 = -\log(1) - 1 + C(1)$$
$$0 = 0 - 1 + C$$
$$C = 1$$

Substituting $C = 1$ back into the solution:

$$y = x - 1 - \log x$$

Final Answer

$y = x - 1 - \log x$

Question 25 (v)

Solve the differential equation:

$$(1 + x^2)\frac{dy}{dx} + 2xy = \frac{1}{1 + x^2}$$

given that $y = 0$ when $x = 0$.

TYPE 111 **Particular Solution of First-Order Linear Differential Equation Reducible to Inverse Tangent Integration**

★ Likely Exam Question

Given: $(1 + x^2)\frac{dy}{dx} + 2xy = \frac{1}{1+x^2}$, with $y = 0$ when $x = 0$

Concept / Formula Used: $\frac{dy}{dx} + Py = Q$ I.F. $= e^{\int P\,dx}$ $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\, dx + C$

Solution:

We rewrite the given differential equation in standard form by dividing both sides by $1 + x^2$:

$$\frac{dy}{dx} + \frac{2x}{1 + x^2}y = \frac{1}{(1 + x^2)^2}$$

This is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where:

$$P = \frac{2x}{1 + x^2}$$
$$Q = \frac{1}{(1 + x^2)^2}$$

Now, we find the Integrating Factor (I.F.):

$$\begin{aligned} \text{I.F.} &= e^{\int P\,dx} \\ &= e^{\int \frac{2x}{1+x^2}\,dx} \end{aligned}$$

To evaluate the integral in the exponent, we use substitution: Let $t = 1 + x^2$. Differentiating both sides:

$$dt = 2x\,dx$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{2x}{1+x^2}\,dx &= \int \frac{1}{t}\,dt \\ &= \log|t| \end{aligned}$$

Back-substituting $t = 1 + x^2$:

$$= \log(1 + x^2)$$

Thus, the Integrating Factor is:

$$\begin{aligned} \text{I.F.} &= e^{\log(1+x^2)} \\ &= 1 + x^2 \end{aligned}$$

The general solution of the differential equation is given by:

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q \cdot (\text{I.F.})\,dx + C \\ y(1+x^2) &= \int \frac{1}{(1+x^2)^2}(1+x^2)\,dx + C \\ y(1+x^2) &= \int \frac{1}{1+x^2}\,dx + C \end{aligned}$$

Using the standard integration result $\int \frac{1}{1+x^2}\,dx = \tan^{-1} x$:

$$y(1+x^2) = \tan^{-1} x + C$$

We are given the initial condition $y = 0$ when $x = 0$:

$$\begin{aligned} 0(1+0^2) &= \tan^{-1}(0) + C \\ 0 &= 0 + C \\ C &= 0 \end{aligned}$$

Substituting $C = 0$ back into the solution:

$$y(1+x^2) = \tan^{-1} x$$

Final Answer

$y(1+x^2) = \tan^{-1} x$

Question 25 (vi)

Solve the differential equation:

$$\frac{dy}{dx} + 2y\tan x = \sin x$$

given that $y = 0$ when $x = \frac{\pi}{3}$.

TYPE 112 Particular Solution of First-Order Linear Differential Equation with Trigonometric Integrals

Given: $\frac{dy}{dx} + 2y\tan x = \sin x$, with $y = 0$ when $x = \frac{\pi}{3}$

Concept / Formula Used: $\frac{dy}{dx} + Py = Q$ I.F. $= e^{\int P\,dx}$ $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$

Solution:

The given differential equation is already in standard form:

$$\frac{dy}{dx} + 2y\tan x = \sin x$$

This is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where:

$$P = 2\tan x$$
$$Q = \sin x$$

Now, we find the Integrating Factor (I.F.):

$$\begin{aligned}
\text{I.F.} &= e^{\int P\,dx} \\
&= e^{\int 2\tan x\,dx} \\
&= e^{2\log|\sec x|} \\
&= e^{\log(\sec^2 x)} \\
&= \sec^2 x
\end{aligned}$$

The general solution of the differential equation is given by:

$$\begin{aligned}
y \cdot (\text{I.F.}) &= \int Q \cdot (\text{I.F.})\,dx + C \\
y\sec^2 x &= \int \sin x\sec^2 x\,dx + C \\
y\sec^2 x &= \int \sin x \cdot \frac{1}{\cos^2 x}\,dx + C \\
y\sec^2 x &= \int \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x}\,dx + C \\
y\sec^2 x &= \int \sec x\tan x\,dx + C
\end{aligned}$$

Using the standard integration result $\int \sec x \tan x \, dx = \sec x$:

$$y \sec^2 x = \sec x + C$$

Multiplying both sides by $\cos^2 x$:

$$y = \sec x \cdot \cos^2 x + C \cos^2 x$$
$$y = \cos x + C \cos^2 x$$

We are given the initial condition $y = 0$ when $x = \frac{\pi}{3}$:

$$0 = \cos\left(\frac{\pi}{3}\right) + C \cos^2\left(\frac{\pi}{3}\right)$$
$$0 = \frac{1}{2} + C\left(\frac{1}{2}\right)^2$$
$$0 = \frac{1}{2} + \frac{C}{4}$$
$$\frac{C}{4} = -\frac{1}{2}$$
$$C = -2$$

Substituting $C = -2$ back into the solution:

$$y = \cos x - 2\cos^2 x$$

Final Answer

$y = \cos x - 2\cos^2 x$

Question 25 (vii)

Solve the differential equation:

$$y' + 2y = e^{-2x} \sin x$$

given that $y = 0$ when $x = 0$.

TYPE 113 **Particular Solution of First-Order Linear Differential Equation with Exponential-Trigonometric Source Term**

Given: $y' + 2y = e^{-2x} \sin x$, with $y = 0$ when $x = 0$

Concept / Formula Used: $\frac{dy}{dx} + Py = Q$ I.F. $= e^{\int P\,dx}$ $y \cdot$ I.F. $= \int Q \cdot$ I.F. $dx + C$

Solution:

The given differential equation is:

$$\frac{dy}{dx} + 2y = e^{-2x} \sin x$$

This is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where:

$$P = 2$$
$$Q = e^{-2x} \sin x$$

Now, we find the Integrating Factor (I.F.):

$$\begin{aligned} \text{I.F.} &= e^{\int 2\,dx} \\ &= e^{2x} \end{aligned}$$

The general solution of the differential equation is given by:

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q \cdot (\text{I.F.})\,dx + C \\ ye^{2x} &= \int e^{-2x} \sin x \cdot e^{2x}\,dx + C \\ ye^{2x} &= \int \sin x\,dx + C \end{aligned}$$

Using the standard integration result $\int \sin x\,dx = -\cos x$:

$$ye^{2x} = -\cos x + C$$

We are given the initial condition $y = 0$ when $x = 0$:

$$\begin{aligned} 0 \cdot e^{2(0)} &= -\cos(0) + C \\ 0 &= -1 + C \\ C &= 1 \end{aligned}$$

Substituting $C = 1$ back into the solution:

$$ye^{2x} = 1 - \cos x$$

Final Answer

$ye^{2x} = 1 - \cos x$

Question 25 (viii)

Solve the differential equation:

$$xy' + y = x\cos x + \sin x$$

given that $y = 1$ when $x = \frac{\pi}{2}$.

TYPE 114 **Particular Solution of First-Order Linear Differential Equation with Product Rule Derivative Integrand**

Given: $xy' + y = x\cos x + \sin x$, with $y = 1$ when $x = \frac{\pi}{2}$

Concept / Formula Used: $\frac{dy}{dx} + Py = Q$ I.F. $= e^{\int P\,dx}$ $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$

Solution:

We rewrite the given differential equation in standard form by dividing both sides by x:

$$\frac{dy}{dx} + \frac{1}{x}y = \cos x + \frac{\sin x}{x}$$

This is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where:

$$P = \frac{1}{x}$$
$$Q = \cos x + \frac{\sin x}{x}$$

Now, we find the Integrating Factor (I.F.):

$$\begin{aligned} \text{I.F.} &= e^{\int \frac{1}{x}\,dx} \\ &= e^{\log|x|} \\ &= x \end{aligned}$$

The general solution of the differential equation is given by:

$$\begin{aligned} y \cdot (\text{I.F.}) &= \int Q \cdot (\text{I.F.})\,dx + C \\ y \cdot x &= \int \left(\cos x + \frac{\sin x}{x}\right) x\,dx + C \\ xy &= \int (x\cos x + \sin x)\,dx + C \end{aligned}$$

To evaluate the integral, we recognize that the integrand is the derivative of $x\sin x$ using the product rule:

$$\frac{d}{dx}(x\sin x) = x\cos x + \sin x$$

Therefore:

$$\begin{aligned} xy &= x\sin x + C \\ y &= \sin x + \frac{C}{x} \end{aligned}$$

We are given the initial condition $y = 1$ when $x = \frac{\pi}{2}$:

$$\begin{aligned} 1 &= \sin\left(\frac{\pi}{2}\right) + \frac{C}{\frac{\pi}{2}} \\ 1 &= 1 + \frac{2C}{\pi} \\ \frac{2C}{\pi} &= 0 \\ C &= 0 \end{aligned}$$

Substituting $C = 0$ back into the solution:

$$y = \sin x$$

Final Answer

$y = \sin x$

Question 25 (ix)

Solve the differential equation:

$$x\frac{dy}{dx} + y + \frac{1}{1+x^2} = 0$$

given that $y(1) = 0$.

TYPE 115 **Particular Solution of First-Order Linear Differential Equation with Rational Inverse Trigonometric Integrati**

Given: $x\frac{dy}{dx} + y + \frac{1}{1+x^2} = 0$, with $y(1) = 0$

Concept / Formula Used: $\frac{dy}{dx} + Py = Q \quad \text{I.F.} = e^{\int P\,dx} \quad y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$

Solution:

We rewrite the given differential equation in standard form:

$$x\frac{dy}{dx} + y = -\frac{1}{1+x^2}$$
$$\frac{dy}{dx} + \frac{1}{x}y = -\frac{1}{x(1+x^2)}$$

This is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where:

$$P = \frac{1}{x}$$
$$Q = -\frac{1}{x(1+x^2)}$$

Now, we find the Integrating Factor (I.F.):

$$\begin{aligned}\text{I.F.} &= e^{\int \frac{1}{x}\,dx}\\ &= e^{\log|x|}\\ &= x\end{aligned}$$

The general solution of the differential equation is given by:

$$\begin{aligned}y \cdot (\text{I.F.}) &= \int Q \cdot (\text{I.F.})\,dx + C\\ y \cdot x &= \int \left(-\frac{1}{x(1+x^2)}\right) x\,dx + C\\ xy &= -\int \frac{1}{1+x^2}\,dx + C\end{aligned}$$

Using the standard integration result $\int \frac{1}{1+x^2}\,dx = \tan^{-1} x$:

$$xy = -\tan^{-1} x + C$$

We are given the initial condition $y(1) = 0$ (i.e., $y = 0$ when $x = 1$):

$$\begin{aligned}(1)(0) &= -\tan^{-1}(1) + C \\ 0 &= -\frac{\pi}{4} + C \\ C &= \frac{\pi}{4}\end{aligned}$$

Substituting $C = \frac{\pi}{4}$ back into the solution:

$$\begin{aligned}xy &= -\tan^{-1} x + \frac{\pi}{4} \\ yx + \tan^{-1} x &= \frac{\pi}{4}\end{aligned}$$

Final Answer

$yx + \tan^{-1} x = \frac{\pi}{4}$

Question 25 (x)

Solve the differential equation:

$$\frac{dy}{dx} - 2xy = 3x^2 e^{x^2}$$

given that $y(0) = 5$.

TYPE 116 Particular Solution of First-Order Linear Differential Equation with Exponential Integrand

Given: $\frac{dy}{dx} - 2xy = 3x^2 e^{x^2}$, with $y(0) = 5$

Concept / Formula Used: $\frac{dy}{dx} + Py = Q$ I.F. $= e^{\int P\,dx}$ $y \cdot \text{I.F.} = \int Q \cdot \text{I.F.}\,dx + C$

Solution:

The given differential equation is already in standard form:

$$\frac{dy}{dx} - 2xy = 3x^2 e^{x^2}$$

This is a linear differential equation of the form $\frac{dy}{dx} + Py = Q$, where:

$$\begin{aligned}P &= -2x \\ Q &= 3x^2 e^{x^2}\end{aligned}$$

Now, we find the Integrating Factor (I.F.):

$$\begin{aligned}\text{I.F.} &= e^{\int P\,dx} \\ &= e^{\int -2x\,dx} \\ &= e^{-x^2}\end{aligned}$$

The general solution of the differential equation is given by:

$$y \cdot (\text{I.F.}) = \int Q \cdot (\text{I.F.})\, dx + C$$
$$ye^{-x^2} = \int 3x^2 e^{x^2} \cdot e^{-x^2}\, dx + C$$
$$ye^{-x^2} = \int 3x^2\, dx + C$$
$$ye^{-x^2} = x^3 + C$$

We are given the initial condition $y(0) = 5$ (i.e., $y = 5$ when $x = 0$):

$$5 \cdot e^{-0^2} = 0^3 + C$$
$$5 \cdot 1 = 0 + C$$
$$C = 5$$

Substituting $C = 5$ back into the solution:

$$ye^{-x^2} = x^3 + 5$$

Final Answer

$ye^{-x^2} = x^3 + 5$

Multiple Choice Question 1

The degree of the differential equation $x^2\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^3$ is:

(a) 1
(b) 2
(c) 3
(d) 6

TYPE 1 Order and Degree of Polynomial Differential Equation

To Find: The degree of the given differential equation

Solution:

The given differential equation is:

$$x^2\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^3$$

1. The highest order derivative present in the equation is $\frac{d^2y}{dx^2}$, which means the order of the differential equation is 2. 2. The equation is already a polynomial in terms of its derivatives $\frac{d^2y}{dx^2}$ and $\frac{dy}{dx}$. 3. The exponent of the highest order derivative $\frac{d^2y}{dx^2}$ is 1.

Therefore, the degree of the differential equation is 1. This corresponds to option (a).

Final Answer

Order $= 2$, Degree $= 1 \implies$ Option (a)

Multiple Choice Question 2

The degree of the differential equation $1 + \left(\frac{dy}{dx}\right)^2 = x$ is:
(a) 0
(b) 1
(c) 2
(d) 3

TYPE 1 Order and Degree of Polynomial Differential Equation

To Find: The degree of the given differential equation

Solution:

The given differential equation is:

$$1 + \left(\frac{dy}{dx}\right)^2 = x$$

1. The highest order derivative present in the equation is $\frac{dy}{dx}$, which means the order of the differential equation is 1. 2. The equation is a polynomial in terms of its derivative $\frac{dy}{dx}$. 3. The exponent of the highest order derivative $\frac{dy}{dx}$ is 2.

Therefore, the degree of the differential equation is 2. This corresponds to option (c).

Final Answer

Degree $= 2 \implies$ Option (c)

Multiple Choice Question 3

The degree of the differential equation $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x^2 \log\left(\frac{d^2y}{dx^2}\right)$ is:
(a) 1
(b) 2
(c) 3
(d) not defined

TYPE 4 Order and Undefined Degree from Non-Polynomial Derivative Term

To Find: The degree of the given differential equation

Solution:

The given differential equation is:

$$\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x^2 \log\left(\frac{d^2y}{dx^2}\right)$$

1. The highest order derivative present in the equation is $\frac{d^2y}{dx^2}$, so the order of the differential equation is 2. 2. However, the term $\log\left(\frac{d^2y}{dx^2}\right)$ contains the highest order derivative inside

a logarithmic function. 3. Because of this, the differential equation cannot be expressed as a polynomial in its derivatives.

Therefore, the degree of the differential equation is not defined. This corresponds to option (d).

Final Answer

Degree is not defined $\implies$ Option (d)

Multiple Choice Question 4

The order and the degree of the differential equation $1 + \left(\frac{dy}{dx}\right)^2 = \frac{d^2y}{dx^2}$ are:
(a) 2 and $\frac{3}{2}$
(b) 2 and 3
(c) 3 and 4
(d) 2 and 1

TYPE 1 Order and Degree of Polynomial Differential Equation

★ **Likely Exam Question**

To Find: The order and degree of the given differential equation

Solution:

The given differential equation is:

$$1 + \left(\frac{dy}{dx}\right)^2 = \frac{d^2y}{dx^2}$$

1. The highest order derivative present in the equation is $\frac{d^2y}{dx^2}$, which means the order of the differential equation is 2. 2. The equation is a polynomial in terms of its derivatives $\frac{d^2y}{dx^2}$ and $\frac{dy}{dx}$. 3. The exponent of the highest order derivative $\frac{d^2y}{dx^2}$ is 1.

Therefore, the order is 2 and the degree is 1. This corresponds to option (d).

Final Answer

Order = 2, Degree = 1 $\implies$ Option (d)

Multiple Choice Question 5

The order and the degree of the differential equation $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 3$, respectively, are:
(a) 1, 3
(b) 3, 1
(c) 2, 1
(d) 2, 2

TYPE 1 Order and Degree of Polynomial Differential Equation

★ Likely Exam Question

To Find: The order and degree of the given differential equation

Solution:

The given differential equation is:

$$\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 3$$

1. The highest order derivative present in the equation is $\frac{d^3y}{dx^3}$, which means the order of the differential equation is 3.
2. The equation is a polynomial in terms of its derivatives.
3. The exponent of the highest order derivative $\frac{d^3y}{dx^3}$ is 1.

Therefore, the order is 3 and the degree is 1. This corresponds to option (b).

Final Answer

Order $= 3$, Degree $= 1 \implies$ Option (b)

Multiple Choice Question 6

The order and the degree of the differential equation $\left(3 - \left(\frac{dy}{dx}\right)^2\right)^{5/3} = x^3 \frac{d^2y}{dx^2}$, respectively, are:

(a) 2, 1
(b) 2, 2
(c) 2, 3
(d) 2, 5

TYPE 1 **Order and Degree of Polynomial Differential Equation**

To Find: The order and degree of the given differential equation

Solution:

The given differential equation is:

$$\left(3 - \left(\frac{dy}{dx}\right)^2\right)^{5/3} = x^3 \frac{d^2y}{dx^2}$$

To eliminate the fractional exponent $\frac{5}{3}$, we cube both sides of the equation:

$$\left[\left(3 - \left(\frac{dy}{dx}\right)^2\right)^{5/3}\right]^3 = \left(x^3 \frac{d^2y}{dx^2}\right)^3$$
$$\left(3 - \left(\frac{dy}{dx}\right)^2\right)^5 = x^9 \left(\frac{d^2y}{dx^2}\right)^3$$

Now, we analyze the equation:

1. The highest order derivative present in the equation is $\frac{d^2y}{dx^2}$, which means the order of the differential equation is 2.
2. The equation is now in polynomial form with respect to its derivatives.
3. The exponent of the highest order derivative $\frac{d^2y}{dx^2}$ is 3.

Therefore, the order is 2 and the degree is 3. This corresponds to option (c).

Final Answer

Order $= 2$, Degree $= 3 \implies$ Option (c)